

Chapter 1

Fraïssé Classes and Fraïssé Limits

This chapter develops the Fraïssé-theoretic background needed for the rest of the thesis. We introduce the basic notions of age, hereditary property, joint embedding property, and amalgamation property, and we explain how these conditions characterize classes of finite structures arising from countable ultrahomogeneous structures. We then state and prove Fraïssé's theorem, which yields the existence and uniqueness of the Fraïssé limit of a Fraïssé class. These constructions provide the model-theoretic foundation for the later study of automorphism groups, Ramsey properties, and universal minimal flows.

1.1 Preliminaries on structures and embeddings

Throughout this chapter, L denotes a signature. An L -structure A consists of a set $\text{dom}(A)$ together with interpretations of the constant, relation, and function symbols of L . If c is a constant symbol, R an n -ary relation symbol, and F an n -ary function symbol, their interpretations in A are denoted by

$$c^A \in \text{dom}(A), \quad R^A \subseteq \text{dom}(A)^n, \quad F^A : \text{dom}(A)^n \rightarrow \text{dom}(A).$$

We write $|A|$ for $|\text{dom}(A)|$. Constants may, when convenient, be regarded as 0-ary function symbols.

If A and B are L -structures, a map $f : \text{dom}(A) \rightarrow \text{dom}(B)$ is a *homomorphism* if

1. $f(c^A) = c^B$ for every constant symbol c ,
2. whenever $\bar{a} \in R^A$, one has $f(\bar{a}) \in R^B$,
3. $f(F^A(\bar{a})) = F^B(f(\bar{a}))$ for every tuple $\bar{a}$.

An *embedding* is an injective homomorphism which also reflects relations, that is,

$$\bar{a} \in R^A \iff f(\bar{a}) \in R^B$$

for every relation symbol R . A bijective embedding is an *isomorphism*. We write $A \cong B$ if A and B are isomorphic. An *automorphism* of A is an isomorphism from A onto itself.

Lemma 1.1.1. *Let A, B, C be L -structures.*

- (i) *The identity map $1_A: A \rightarrow A$ is an isomorphism.*
- (ii) *If $f: A \rightarrow B$ is an isomorphism, then $f^{-1}: B \rightarrow A$ is an isomorphism.*
- (iii) *If $f: A \rightarrow B$ and $g: B \rightarrow C$ are isomorphisms, then $g \circ f: A \rightarrow C$ is an isomorphism.*

Proof. Part (i) is immediate. For (ii), since f is a bijective embedding, its inverse is again a bijection preserving constants, relations, and functions, and hence is an isomorphism. For (iii), by composition of embeddings, $g \circ f$ is an embedding, and since it is also a composition of bijections, it is bijective. Therefore $g \circ f$ is an isomorphism. $\dashv$

Definition 1.1.2 (Substructure). Let A and B be L -structures with $\text{dom}(A) \subseteq \text{dom}(B)$. We say that A is a *substructure* of B , and write $A \subseteq B$, if the inclusion map is an embedding. Equivalently,

$$c^A = c^B, \quad R^A = R^B \cap \text{dom}(A)^n, \quad F^A = F^B \upharpoonright_{\text{dom}(A)^n}$$

for all constant, relation, and function symbols of L .

Lemma 1.1.3. *Let B be an L -structure and let $X \subseteq \text{dom}(B)$. The following are equivalent:*

- (i) *$X = \text{dom}(A)$ for some substructure $A \subseteq B$;*
- (ii) *X contains all constants of B and is closed under all functions of B .*

If these conditions hold, the substructure A is uniquely determined.

Proof. If $A \subseteq B$, then $\text{dom}(A)$ plainly contains the constants and is closed under the functions of B . Conversely, if $X \subseteq \text{dom}(B)$ satisfies (ii), define an L -structure A on X by

$$c^A = c^B, \quad R^A = R^B \cap X^n, \quad F^A = F^B \upharpoonright_{X^n}.$$

Then the inclusion $A \hookrightarrow B$ is an embedding, so $A \subseteq B$. Uniqueness is immediate from the definition of substructure. $\dashv$

Definition 1.1.4 (Generated substructure). Let B be an L -structure and $Y \subseteq \text{dom}(B)$. By Lemma 1.1.3, there is a unique smallest substructure of B whose domain contains Y . It is denoted by

$$\langle Y \rangle_B$$

and is called the *substructure generated by Y* . When the ambient structure is clear, we simply write $\langle Y \rangle$. A structure is *finitely generated* if it is generated by a finite subset of its domain.

The following observation is used repeatedly: any embedding may be viewed as an inclusion after replacing the codomain by an isomorphic copy.

Lemma 1.1.5. *Let $f: A \hookrightarrow C$ be an embedding of L -structures. Then there exist an L -structure B and an isomorphism $h: B \rightarrow C$ such that $A \subseteq B$ and*

$$f = h \upharpoonright_A.$$

Proof. Let

$$\text{dom}(B) := \text{dom}(A) \dot{\cup} (\text{dom}(C) \setminus f(\text{dom}(A))),$$

and define a bijection $\sigma : \text{dom}(B) \rightarrow \text{dom}(C)$ by

$$\sigma(a) = f(a) \quad (a \in A), \quad \sigma(x) = x \quad (x \in \text{dom}(C) \setminus f(\text{dom}(A))).$$

Transport the L -structure of C to $\text{dom}(B)$ via σ . Then σ becomes an isomorphism $h : B \rightarrow C$. Since f is an embedding, the induced structure on $\text{dom}(A)$ is exactly the original structure A , so $A \subseteq B$ and $f = h \upharpoonright_A$. $\dashv$

For the examples relevant to Fraïssé theory, the language is usually finite and relational. In that case finitely generated structures are automatically finite.

Proposition 1.1.6. *Let L be a finite relational signature.*

1. *Every finitely generated L -structure is finite.*
2. *For each $n \geq 1$, there are only finitely many L -structures of cardinality n , up to isomorphism.*

Proof. Since L has no function symbols and no constants, the substructure generated by a set is just that set itself. Hence every finitely generated L -structure is finite.

Now fix $n \geq 1$. Every L -structure of size n is isomorphic to one with domain $\{1, \dots, n\}$. If $R_1, \dots, R_k$ are the relation symbols of L , of arities $m_1, \dots, m_k$, then each R_i may be interpreted arbitrarily as a subset of $\{1, \dots, n\}^{m_i}$. There are only finitely many such choices, so only finitely many L -structures on $\{1, \dots, n\}$, and therefore only finitely many isomorphism types of size n . $\dashv$

1.2 Ages, HP, and JEP

We now introduce the first invariant associated with a structure.

Definition 1.2.1 (Age). Let L be a signature and let D be an L -structure. The *age* of D , denoted $\text{Age}(D)$, is the class of all finitely generated L -structures that embed into D . Equivalently, $\text{Age}(D)$ is the class of all finitely generated substructures of D , taken up to isomorphism.

In this thesis we regard ages up to isomorphism. Thus to say that D has countable age means that there are only countably many isomorphism types of finitely generated substructures of D .

The following two properties are immediate for every age.

Definition 1.2.2. Let $\mathcal{K}$ be a class of finitely generated L -structures.

(HP) $\mathcal{K}$ has the *hereditary property* if whenever $A \in \mathcal{K}$ and B is a finitely generated substructure of A , then $B \in \mathcal{K}$ up to isomorphism.

(JEP) $\mathcal{K}$ has the *joint embedding property* if for all $A, B \in \mathcal{K}$, there exists $C \in \mathcal{K}$ such that both A and B embed into C .

Proposition 1.2.3. *Let D be an L -structure. Then $\text{Age}(D)$ is nonempty and satisfies (HP) and (JEP).*

Proof. Choose any $a \in D$. Then the substructure $\langle a \rangle$ generated by a is finitely generated and embeds into D , so $\text{Age}(D) \neq \emptyset$.

To prove (HP), let $A \in \text{Age}(D)$, and let $B \subseteq A$ be finitely generated. Since A embeds into D , the restriction of any embedding $A \hookrightarrow D$ gives an embedding $B \hookrightarrow D$. Hence $B \in \text{Age}(D)$.

To prove (JEP), let $A, B \in \text{Age}(D)$. Choose embeddings $e_A : A \hookrightarrow D$ and $e_B : B \hookrightarrow D$. Let

$$C := \langle e_A(A) \cup e_B(B) \rangle \subseteq D.$$

Since A and B are finitely generated, the structure C is finitely generated. Thus $C \in \text{Age}(D)$, and both A and B embed into C . $\dashv$

Fraïssé proved a converse: under mild countability assumptions, these two properties already characterize the ages of countable structures.

Proposition 1.2.4. *Let L be a signature, and let $\mathcal{K}$ be a nonempty class of finitely generated L -structures, countable up to isomorphism. Suppose that $\mathcal{K}$ satisfies (HP) and (JEP). Then there exists a countable L -structure C such that*

$$\mathcal{K} = \text{Age}(C).$$

Proof. Choose a sequence $(A_i)_{i < \omega}$ listing all isomorphism types in $\mathcal{K}$, with each type appearing at least once. We construct a chain

$$B_0 \subseteq B_1 \subseteq B_2 \subseteq \cdots$$

such that each $B_i \in \mathcal{K}$ up to isomorphism and A_i embeds into B_{i+1} .

Set $B_0 = A_0$. Suppose B_i has been constructed. By (JEP), there is some $C_i \in \mathcal{K}$ into which both B_i and A_{i+1} embed. By Lemma 1.1.5, after replacing C_i by an isomorphic copy, we may assume that $B_i \subseteq C_i$. Set $B_{i+1} := C_i$. This completes the construction.

Now let

$$C := \bigcup_{i < \omega} B_i.$$

Since C is a union of countably many finitely generated structures, it is countable.

By construction, every member of $\mathcal{K}$ embeds into C , so $\mathcal{K} \subseteq \text{Age}(C)$. Conversely, let A be a finitely generated substructure of C . A finite generating set for A is contained in some B_i , hence $A \subseteq B_i$. Since $B_i \in \mathcal{K}$ up to isomorphism and $\mathcal{K}$ satisfies (HP), it follows that $A \in \mathcal{K}$ up to isomorphism. Therefore $\text{Age}(C) \subseteq \mathcal{K}$.

Hence $\text{Age}(C) = \mathcal{K}$. $\dashv$

Remark 1.2.5. If L is a finite relational signature, then every finitely generated L -structure is finite. In that case the countability assumption on $\mathcal{K}$ is often easy to verify by Proposition 1.1.6.

1.3 A motivating example: finite linear orders

The preceding discussion shows that every age satisfies *(HP)* and *(JEP)*. A natural question is whether these conditions are strong enough to determine a distinguished countable structure. The example of linear orders shows that they are not.

Proposition 1.3.1. *Let $L = \{<\}$, and let $(M, <)$ be an infinite linear order. Then*

$$\text{Age}(M) = \{\text{finite linear orders}\}.$$

In particular, any two infinite linear orders have the same age.

Proof. Since the language $L = \{<\}$ is purely relational, every finitely generated substructure of M is simply the induced substructure on a finite subset of M . The restriction of $<$ to a finite subset is again a finite linear order. Thus every member of $\text{Age}(M)$ is a finite linear order.

Conversely, let $(F, <)$ be a finite linear order, say with

$$F = \{a_1, \dots, a_n\}, \quad a_1 < \dots < a_n.$$

Since M is infinite, we may choose distinct elements $b_1, \dots, b_n \in M$, and after reordering assume that

$$b_1 < \dots < b_n.$$

Then the map $e: F \rightarrow M$ given by $e(a_i) = b_i$ is an embedding. Hence every finite linear order belongs to $\text{Age}(M)$, and therefore

$$\text{Age}(M) = \{\text{finite linear orders}\}.$$

–

Let $\mathcal{K}_<$ denote the class of all finite linear orders, up to isomorphism. By Proposition 1.3.1,

$$\text{Age}(\mathbb{Q}, <) = \text{Age}(\mathbb{Z}, <) = \mathcal{K}_<.$$

Moreover, $\mathcal{K}_<$ satisfies *(HP)* and *(JEP)*: *(HP)* is immediate, and *(JEP)* holds because any two finite linear orders embed into a sufficiently long finite linear order.

Thus age, together with *(HP)* and *(JEP)*, does not distinguish the countable dense linear order without endpoints from the order of the integers. In particular, these conditions do not yet explain why the finite linear orders “tend” to $(\mathbb{Q}, <)$ rather than to $(\mathbb{Z}, <)$.

This shows that one further condition is needed. The next section introduces the *amalgamation property*, which is precisely the additional condition singled out by Fraïssé.

1.4 The amalgamation property and Fraïssé classes

The previous section shows that age, together with *(HP)* and *(JEP)*, does not suffice to distinguish $(\mathbb{Q}, <)$ from $(\mathbb{Z}, <)$. Fraïssé’s additional requirement is the amalgamation property.

Definition 1.4.1 (Amalgamation property). Let $\mathcal{K}$ be a class of finitely generated L -structures. We say that $\mathcal{K}$ has the *amalgamation property*, or *(AP)*, if whenever $A, B, C \in \mathcal{K}$ and

$$e: A \hookrightarrow B, \quad f: A \hookrightarrow C$$

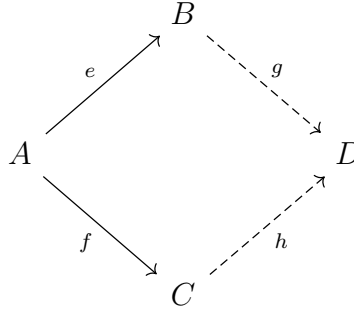
are embeddings, there exist $D \in \mathcal{K}$ and embeddings

$$g: B \hookrightarrow D, \quad h: C \hookrightarrow D$$

such that

$$g \circ e = h \circ f.$$

Thus *(AP)* says that any two structures in $\mathcal{K}$ which share a common substructure can be embedded into a larger structure in $\mathcal{K}$ so that the two copies of the common part coincide.



We now return to the class of finite linear orders.

Proposition 1.4.2. *The class of finite linear orders has the amalgamation property.*

Proof. We first treat the special case in which $A \subseteq B$ and $A \subseteq C$, the maps e and f are the inclusion maps, and

$$A = B \cap C.$$

Write

$$A = \{a_1 < \cdots < a_n\}.$$

For convenience, introduce formal symbols

$$a_0 = -\infty, \quad a_{n+1} = +\infty.$$

For each $0 \leq i \leq n$, let

$$B_i := \{b \in B \setminus A : a_i < b < a_{i+1}\}, \quad C_i := \{c \in C \setminus A : a_i < c < a_{i+1}\},$$

where the inequalities involving a_0 and a_{n+1} are interpreted in the obvious way. Since A is a substructure of both B and C , every element of $B \setminus A$ belongs to exactly one B_i , and every element of $C \setminus A$ belongs to exactly one C_i .

Let

$$D := B \cup C$$

as a set. We define a linear order $<_D$ on D as follows:

- $<_D$ agrees with the given orders on B and on C ;
- for each i , every element of B_i is declared smaller than every element of C_i ;
- if $x \in B_i \cup C_i$ and $y \in B_j \cup C_j$ with $i < j$, then $x <_D y$;
- if $x \in B_i \cup C_i$, then

$$a_i <_D x <_D a_{i+1},$$

with the obvious interpretation when $i = 0$ or $i = n$.

In other words, inside each interval determined by the elements of A , we place first the elements coming from B , in their original order, and then the elements coming from C , in their original order.

It is straightforward to verify that $<_D$ is a linear order on D . By construction, the inclusion maps

$$j_B: B \hookrightarrow D, \quad j_C: C \hookrightarrow D$$

are embeddings of linear orders, and they agree on A . Thus D is an amalgam of B and C over A . We now consider the general case. Let

$$e: A \hookrightarrow B, \quad f: A \hookrightarrow C$$

be embeddings of finite linear orders. By Lemma 1.1.5, there exist finite linear orders B_1 and C_1 , together with isomorphisms

$$\beta: B_1 \rightarrow B, \quad \gamma: C_1 \rightarrow C,$$

such that $A \subseteq B_1$, $A \subseteq C_1$, and

$$\beta \upharpoonright_A = e, \quad \gamma \upharpoonright_A = f.$$

Replacing B_1 and C_1 by isomorphic copies if necessary, we may moreover assume that

$$B_1 \cap C_1 = A.$$

By the special case already proved, there exist a finite linear order D_1 and embeddings

$$j_B: B_1 \hookrightarrow D_1, \quad j_C: C_1 \hookrightarrow D_1$$

such that

$$j_B \upharpoonright_A = j_C \upharpoonright_A.$$

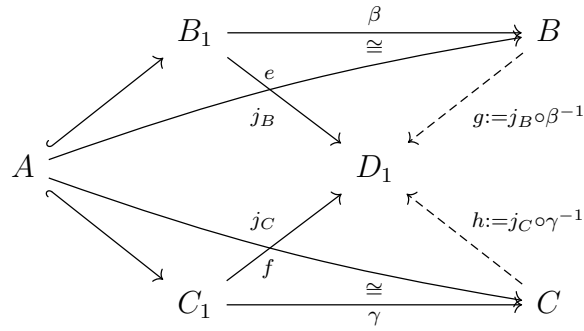
Define

$$g := j_B \circ \beta^{-1}: B \hookrightarrow D_1, \quad h := j_C \circ \gamma^{-1}: C \hookrightarrow D_1.$$

Then

$$g \circ e = j_B \circ \beta^{-1} \circ e = j_B \upharpoonright_A = j_C \upharpoonright_A = j_C \circ \gamma^{-1} \circ f = h \circ f,$$

so D_1 is an amalgam of e and f .



Hence the class of finite linear orders has (AP) .

+

This example leads to the central class of finite structures considered in Fraïssé theory.

Definition 1.4.3 (Fraïssé class). Let L be a countable signature. A class $\mathcal{K}$ of finitely generated L -structures is called a *Fraïssé class* if it satisfies the following conditions:

1. $\mathcal{K}$ is closed under isomorphism;
2. $\mathcal{K}$ has countably many isomorphism types;
3. $\mathcal{K}$ has the hereditary property (HP) ;
4. $\mathcal{K}$ has the joint embedding property (JEP) ;
5. $\mathcal{K}$ has the amalgamation property (AP) .

1.5 Fraïssé limits

We now arrive at the central notion of the chapter.

Definition 1.5.1 (Ultrahomogeneous structure). Let D be an L -structure. We say that D is *ultrahomogeneous* if every isomorphism between finitely generated substructures of D extends to an automorphism of D .

Fraïssé's theorem shows that every Fraïssé class determines, up to isomorphism, a unique countable ultrahomogeneous structure whose age is exactly that class.

Theorem 1.5.2 (Fraïssé's theorem). *Let $\mathcal{K}$ be a Fraïssé class. Then there exists a countable ultrahomogeneous L -structure D such that*

$$\text{Age}(D) = \mathcal{K}.$$

Moreover, D is unique up to isomorphism.

Definition 1.5.3 (Fraïssé limit). Let $\mathcal{K}$ be a Fraïssé class. The unique countable ultrahomogeneous structure D satisfying

$$\text{Age}(D) = \mathcal{K}$$

is called the *Fraïssé limit* of $\mathcal{K}$.

The next two sections are devoted to the proof of Theorem 1.5.2: first the uniqueness of the Fraïssé limit, and then its existence.

1.6 Uniqueness of the Fraïssé limit

We begin with a weaker extension property which is sufficient, in the countable case, to determine a structure uniquely from its age.

Definition 1.6.1 (Weak homogeneity). Let D be an L -structure. We say that D is *weakly homogeneous* if whenever $A \subseteq B$ are finitely generated substructures of D and

$$f: A \hookrightarrow D$$

is an embedding, there exists an embedding

$$g: B \hookrightarrow D$$

extending f .

$$\begin{array}{ccc} A & \xrightarrow{f} & D \\ \downarrow & \nearrow g & \\ B & & \end{array}$$

Weak homogeneity is strictly weaker than ultrahomogeneity as a formal condition, but for finite or countable structures the two notions coincide.

Example 1.6.2. The ordered set $(\mathbb{Z}, <)$ is not weakly homogeneous.

Proof. Let

$$A = \{0, 2\} \subseteq \mathbb{Z}, \quad B = \{0, 1, 2\} \subseteq \mathbb{Z},$$

with the induced order, and define $f: A \rightarrow \mathbb{Z}$ by

$$f(0) = 0, \quad f(2) = 1.$$

Then f is an embedding. If $g: B \rightarrow \mathbb{Z}$ were an embedding extending f , we would have

$$0 = g(0) < g(1) < g(2) = 1,$$

which is impossible in $\mathbb{Z}$. Hence no such extension exists. $\dashv$

Lemma 1.6.3. *Every ultrahomogeneous structure is weakly homogeneous.*

Proof. Let $A \subseteq B$ be finitely generated substructures of an ultrahomogeneous structure D , and let $f: A \hookrightarrow D$ be an embedding. Then $f: A \rightarrow f(A)$ is an isomorphism between finitely generated substructures of D . By ultrahomogeneity, f extends to an automorphism α of D . The restriction

$$g := \alpha \upharpoonright_B: B \hookrightarrow D$$

is then an embedding extending f . $\dashv$

The next lemma shows that a countable weakly homogeneous structure is universal for its age.

Lemma 1.6.4. *Let C and D be at most countable L -structures. Suppose that*

$$\text{Age}(C) \subseteq \text{Age}(D)$$

and that D is weakly homogeneous. Then every embedding of a finitely generated substructure of C into D extends to an embedding of C into D .

Proof. Let $f_0: A_0 \hookrightarrow D$ be an embedding, where $A_0 \subseteq C$ is finitely generated. Choose an increasing chain of finitely generated substructures

$$A_0 \subseteq A_1 \subseteq A_2 \subseteq \cdots$$

with

$$C = \bigcup_{n < \omega} A_n.$$

For example, if $C \setminus A_0 = \{c_0, c_1, \dots\}$, we may take

$$A_{n+1} := \langle A_n \cup \{c_n\} \rangle.$$

We construct inductively embeddings

$$f_n: A_n \hookrightarrow D \quad (n < \omega)$$

such that $f_n \subseteq f_{n+1}$. Suppose f_n has been constructed. Since A_{n+1} is finitely generated and $\text{Age}(C) \subseteq \text{Age}(D)$, there exist a finitely generated substructure $B \subseteq D$ and an isomorphism

$$u: A_{n+1} \xrightarrow{\cong} B.$$

Then

$$f_n \circ (u \upharpoonright_{A_n})^{-1}: u(A_n) \hookrightarrow D$$

is an embedding. Since $u(A_n) \subseteq B$ and D is weakly homogeneous, this embedding extends to an embedding

$$h: B \hookrightarrow D.$$

Define

$$f_{n+1} := h \circ u: A_{n+1} \hookrightarrow D.$$

For $a \in A_n$,

$$f_{n+1}(a) = h(u(a)) = (f_n \circ (u \upharpoonright_{A_n})^{-1})(u(a)) = f_n(a),$$

so f_{n+1} extends f_n .

Now let

$$f := \bigcup_{n < \omega} f_n: C \rightarrow D.$$

This is well defined because the maps f_n are compatible. Since every finite tuple from C is contained in some A_n , and each f_n is an embedding, it follows that f is an embedding of C into D . By construction, f extends f_0 . $\dashv$

We now obtain the uniqueness statement by the usual back-and-forth argument.

Lemma 1.6.5. *Let C and D be at most countable weakly homogeneous L -structures with*

$$\text{Age}(C) = \text{Age}(D).$$

Then $C \cong D$. More precisely, every isomorphism between finitely generated substructures of C and D extends to an isomorphism from C onto D .

Proof. Let

$$f_0: A_0 \xrightarrow{\cong} B_0$$

be an isomorphism between finitely generated substructures $A_0 \subseteq C$ and $B_0 \subseteq D$. Choose increasing chains of finitely generated substructures

$$A_0 \subseteq A_1 \subseteq A_2 \subseteq \cdots, \quad B_0 \subseteq B_1 \subseteq B_2 \subseteq \cdots$$

such that

$$C = \bigcup_{n < \omega} A_n, \quad D = \bigcup_{n < \omega} B_n.$$

We construct inductively a chain of partial isomorphisms

$$f_0 \subseteq f_1 \subseteq f_2 \subseteq \cdots$$

between finitely generated substructures of C and D such that for every n ,

$$A_n \subseteq \text{dom}(f_{2n}), \quad B_n \subseteq \text{im}(f_{2n+1}).$$

Suppose f_{2n} has been constructed. Since D is weakly homogeneous and $\text{Age}(C) \subseteq \text{Age}(D)$, Lemma 1.6.4 extends f_{2n} to an embedding

$$g: C \hookrightarrow D.$$

Let

$$f_{2n+1} := g \upharpoonright_{\langle \text{dom}(f_{2n}) \cup A_{n+1} \rangle}.$$

Then f_{2n+1} extends f_{2n} , and its domain contains A_{n+1} .

Now f_{2n+1}^{-1} is an isomorphism from the finitely generated substructure $\text{im}(f_{2n+1}) \subseteq D$ into C . Since C is weakly homogeneous and $\text{Age}(D) \subseteq \text{Age}(C)$, Lemma 1.6.4 extends f_{2n+1}^{-1} to an embedding

$$h: D \hookrightarrow C.$$

Define

$$f_{2n+2} := (h \upharpoonright_{\langle \text{im}(f_{2n+1}) \cup B_{n+1} \rangle})^{-1}.$$

Then f_{2n+2} extends f_{2n+1} , and its image contains B_{n+1} .

Thus the construction continues indefinitely. Let

$$f := \bigcup_{n < \omega} f_n.$$

Because the maps f_n are compatible, f is a well-defined embedding $C \rightarrow D$. The domain condition implies that every element of C lies in $\text{dom}(f)$, and the image condition implies that every element of D lies in $\text{im}(f)$. Hence f is an isomorphism from C onto D . By construction, it extends f_0 . $\dashv$

Corollary 1.6.6. *For a finite or countable L -structure D , the following are equivalent:*

1. D is ultrahomogeneous;
2. D is weakly homogeneous.

Proof. By Lemma 1.6.3, every ultrahomogeneous structure is weakly homogeneous.

Conversely, suppose that D is finite or countable and weakly homogeneous. Let

$$f: A \xrightarrow{\cong} B$$

be an isomorphism between finitely generated substructures of D . Applying Lemma 1.6.5 with $C = D$, $D = D$, and the given partial isomorphism f , we obtain an automorphism of D extending f . Hence D is ultrahomogeneous. $\dashv$

In particular, if C and D are countable ultrahomogeneous structures with the same age, then $C \cong D$. This proves the uniqueness part of Fraïssé's theorem.

1.7 Existence of the Fraïssé limit

We now prove the existence part of Fraïssé's theorem. The idea is to construct a countable structure D with age $\mathcal{K}$ and with the extension property that characterizes weak homogeneity.

We first record a simple lemma on increasing unions.

Lemma 1.7.1. *Let*

$$D_0 \subseteq D_1 \subseteq D_2 \subseteq \cdots$$

be an increasing chain of L -structures, and let

$$D := \bigcup_{i < \omega} D_i.$$

Let $\mathcal{K}$ be a class of finitely generated L -structures.

(i) *If $\text{Age}(D_i) \subseteq \mathcal{K}$ for all $i < \omega$, then*

$$\text{Age}(D) \subseteq \mathcal{K}.$$

(ii) *If, in addition, every $A \in \mathcal{K}$ embeds into some D_i , then*

$$\text{Age}(D) = \mathcal{K}.$$

Proof. Let A be a finitely generated substructure of D , generated by a finite tuple $\bar{a}$. Since the chain is increasing, all entries of $\bar{a}$ lie in some D_i . Hence

$$A = \langle \bar{a} \rangle_D = \langle \bar{a} \rangle_{D_i},$$

so $A \in \text{Age}(D_i) \subseteq \mathcal{K}$. This proves (i).

For (ii), part (i) already gives $\text{Age}(D) \subseteq \mathcal{K}$. Conversely, let $A \in \mathcal{K}$. By assumption, A embeds into some D_i , and therefore also into D . Thus $A \in \text{Age}(D)$, so $\mathcal{K} \subseteq \text{Age}(D)$. $\dashv$

Proof of Fraïssé's theorem. Let $\mathcal{K}$ be a Fraïssé class. Since only isomorphism types matter, we may assume that $\mathcal{K}$ is closed under isomorphism.

We shall construct an increasing chain

$$D_0 \subseteq D_1 \subseteq D_2 \subseteq \cdots$$

of members of $\mathcal{K}$ such that the following extension property holds:

whenever $A, B \in \mathcal{K}$ with $A \subseteq B$, and $f: A \hookrightarrow D_i$ is an embedding, there exists $j > i$ and an embedding

$$g: B \hookrightarrow D_j$$

extending f .

Assume for the moment that such a chain has been constructed, and let

$$D := \bigcup_{i < \omega} D_i.$$

Since $\mathcal{K}$ has countably many isomorphism types and each member of $\mathcal{K}$ is finitely generated in a countable signature, each D_i is countable. Hence D is countable.

Because each D_i belongs to $\mathcal{K}$, Lemma 1.7.1(i) gives

$$\text{Age}(D) \subseteq \mathcal{K}.$$

We claim that in fact $\text{Age}(D) = \mathcal{K}$. Let $A \in \mathcal{K}$. By (JEP), there exist $B \in \mathcal{K}$ and embeddings

$$u: D_0 \hookrightarrow B, \quad v: A \hookrightarrow B.$$

By Lemma 1.1.5, after replacing B by an isomorphic copy, we may assume that

$$D_0 \subseteq B.$$

Now apply the extension property to the identity embedding $1_{D_0}: D_0 \hookrightarrow D_0$. Then for some $j > 0$, the inclusion $D_0 \subseteq B$ extends to an embedding

$$g: B \hookrightarrow D_j.$$

Composing with v , we obtain an embedding of A into D_j . Thus every $A \in \mathcal{K}$ embeds into some D_j , and Lemma 1.7.1(ii) yields

$$\text{Age}(D) = \mathcal{K}.$$

We next show that D is weakly homogeneous. Let $A \subseteq B$ be finitely generated substructures of D , and let

$$f: A \hookrightarrow D$$

be an embedding. Since A is finitely generated and $f(A)$ is finite or countable, there exists $i < \omega$ such that

$$f(A) \subseteq D_i.$$

Because $\text{Age}(D) = \mathcal{K}$, both A and B belong to $\mathcal{K}$. Applying the extension property to the embedding

$$f: A \hookrightarrow D_i,$$

we obtain $j > i$ and an embedding

$$g: B \hookrightarrow D_j \subseteq D$$

extending f . Hence D is weakly homogeneous.

Since D is countable, Corollary 1.6.6 implies that D is ultrahomogeneous. Therefore D is a countable ultrahomogeneous structure with age $\mathcal{K}$, as required.

It remains to construct the chain $(D_i)_{i < \omega}$.

Let

$$\mathcal{P} = \{(A_n, B_n) : n < \omega\}$$

be a countable set of representatives of the isomorphism types of pairs (A, B) such that $A, B \in \mathcal{K}$ and $A \subseteq B$. Fix a bijection

$$\pi: \omega \times \omega \rightarrow \omega$$

such that $\pi(i, j) \geq i$ for all $i, j < \omega$. Choose any $D_0 \in \mathcal{K}$.

Suppose D_i has been constructed. Since D_i is countable and each A_n is finitely generated, there are only countably many embeddings of the various A_n into D_i . Enumerate all triples

$$(f_{ij}, A_{ij}, B_{ij}) \quad (j < \omega),$$

where $A_{ij} \subseteq B_{ij}$ is one of the pairs from $\mathcal{P}$ and

$$f_{ij}: A_{ij} \hookrightarrow D_i$$

is an embedding.

Now suppose D_k has been defined, and write

$$k = \pi(i, j).$$

Then

$$f_{ij}: A_{ij} \hookrightarrow D_i \subseteq D_k$$

is an embedding. Apply (AP) to the inclusion

$$A_{ij} \hookrightarrow B_{ij}$$

and the embedding

$$f_{ij}: A_{ij} \hookrightarrow D_k.$$

We obtain some $E \in \mathcal{K}$ and embeddings

$$r: B_{ij} \hookrightarrow E, \quad s: D_k \hookrightarrow E$$

such that

$$r \upharpoonright_{A_{ij}} = s \circ f_{ij}.$$

By Lemma 1.1.5, after replacing E by an isomorphic copy, we may assume that

$$D_k \subseteq E$$

and that s is the inclusion map. Set

$$D_{k+1} := E.$$

Then the embedding $r: B_{ij} \hookrightarrow D_{k+1}$ extends f_{ij} .

This completes the construction of the chain. Because every embedding problem over every stage D_i appears as some triple (f_{ij}, A_{ij}, B_{ij}) , and because $\pi(i, j) \geq i$, each such problem is eventually handled at a later stage. Hence the required extension property holds. $\dashv$

1.8 The converse direction

We now show that the conditions in Fraïssé's theorem are also necessary.

Theorem 1.8.1. *Let L be a countable signature, and let D be a finite or countable ultrahomogeneous L -structure. Then $\text{Age}(D)$ is a Fraïssé class. More explicitly, $\text{Age}(D)$ is nonempty, has countably many isomorphism types, and satisfies (HP), (JEP), and (AP).*

Proof. Set

$$\mathcal{K} := \text{Age}(D).$$

By Proposition 1.2.3, the class $\mathcal{K}$ is nonempty and satisfies (HP) and (JEP).

It remains to verify that $\mathcal{K}$ has countably many isomorphism types and satisfies (AP).

Since D is finite or countable, there are only countably many finite tuples from D . Every member of $\mathcal{K}$ is finitely generated, hence is generated by some finite tuple from D . Therefore $\mathcal{K}$ has at most countably many isomorphism types.

To prove (AP), let $A, B, C \in \mathcal{K}$, and let

$$e: A \hookrightarrow B, \quad f: A \hookrightarrow C$$

be embeddings. Since $A \in \mathcal{K} = \text{Age}(D)$, there exists an embedding

$$u: A \hookrightarrow D.$$

By Corollary 1.6.6, the structure D is weakly homogeneous. Now

$$u \circ e^{-1}: e(A) \hookrightarrow D$$

is an embedding of the finitely generated substructure $e(A) \subseteq B$ into D . Since $B \in \mathcal{K}$, weak homogeneity yields an embedding

$$j_B: B \hookrightarrow D$$

extending $u \circ e^{-1}$. Thus

$$j_B \circ e = u.$$

Similarly, the embedding

$$u \circ f^{-1}: f(A) \hookrightarrow D$$

extends to an embedding

$$j_C: C \hookrightarrow D$$

with

$$j_C \circ f = u.$$

Hence

$$j_B \circ e = j_C \circ f.$$

Let

$$E := \langle j_B(B) \cup j_C(C) \rangle \subseteq D.$$

Since B and C are finitely generated, the structure E is finitely generated, so $E \in \mathcal{K}$. The maps $j_B: B \hookrightarrow E$ and $j_C: C \hookrightarrow E$ are embeddings and satisfy

$$j_B \circ e = j_C \circ f,$$

so E is an amalgam of B and C over A . Therefore $\mathcal{K}$ satisfies (AP) .

Thus $\mathcal{K}$ is a Fraïssé class. +

Chapter 2

Automorphism Groups

Fraïssé theory leads naturally from countable structures to their automorphism groups. In this chapter we study automorphism groups of first-order structures as permutation groups on the underlying domain and equip them with the topology of pointwise convergence. We begin with the basic notions of stabilizers, orbits, transitivity, and rigidity, and then introduce the permutation topology on $\text{Sym}(\Omega)$ and on its subgroups.

We next characterize those subgroups of $\text{Sym}(\Omega)$ which arise as automorphism groups of structures on Ω , in terms of closed subgroups and canonical structures. In the countable case, this shows that automorphism groups are closed subgroups of S_∞ , and hence naturally carry the structure of Polish groups. These results provide the topological framework needed in the following chapters, where we study continuous actions of such groups on compact spaces.

2.1 Automorphism Groups as Permutation Groups

Let A be an L -structure with domain $\Omega = \text{dom}(A)$. Every automorphism of A is a permutation of Ω , and the set of all automorphisms of A forms a group under composition. We denote this group by $\text{Aut}(A)$ and regard it as a subgroup of the symmetric group $\text{Sym}(\Omega)$.

More generally, if Ω is any set, we write $\text{Sym}(\Omega)$ for the group of all permutations of Ω . Thus every automorphism group is naturally a permutation group. Our aim in this chapter is to study those subgroups of $\text{Sym}(\Omega)$ which arise as automorphism groups of structures on Ω , and to equip them with the topology of pointwise convergence.

Let $G \leq \text{Sym}(\Omega)$. For a subset $X \subseteq \Omega$, the *pointwise stabilizer* of X in G is

$$G_{(X)} = \{g \in G : g(a) = a \text{ for all } a \in X\},$$

and the *setwise stabilizer* of X in G is

$$G_{\{X\}} = \{g \in G : g(X) = X\}.$$

Thus $G_{(X)} \leq G_{\{X\}} \leq G$. If $\bar{a} = (a_1, \dots, a_n)$ is a finite tuple from Ω , we also write $G_{(\bar{a})}$ for the pointwise stabilizer of the set $\{a_1, \dots, a_n\}$.

For $a \in \Omega$, the *orbit* of a under G is the set

$$G \cdot a = \{g(a) : g \in G\}.$$

More generally, if $\bar{a} = (a_1, \dots, a_n) \in \Omega^n$, the *orbit* of $\bar{a}$ under G is

$$G \cdot \bar{a} = \{g\bar{a} : g \in G\},$$

where

$$g\bar{a} = (g(a_1), \dots, g(a_n)).$$

The orbits of G on Ω partition Ω . We say that G is *transitive* if it has exactly one orbit on Ω .

Accordingly, a structure A is called *transitive* if $\text{Aut}(A)$ acts transitively on $\text{dom}(A)$. At the opposite extreme, A is called *rigid* if $\text{Aut}(A) = \{1_\Omega\}$.

Example 2.1.1. The following examples will be useful throughout the thesis.

- (i) If A is a pure set with no non-logical symbols, then every permutation of $\text{dom}(A)$ is an automorphism, so $\text{Aut}(A) = \text{Sym}(\text{dom}(A))$.
- (ii) Every finite linear order is rigid. Indeed, if $A = (\{a_1, \dots, a_n\}, <)$ with $a_1 < \dots < a_n$, then any automorphism must preserve the least element, then the next least element, and so on; hence it fixes every point.
- (iii) The ordered set $(\mathbb{Q}, <)$ is transitive. Given $q, r \in \mathbb{Q}$, there exists an order automorphism of $(\mathbb{Q}, <)$ sending q to r , so all elements lie in a single orbit under $\text{Aut}(\mathbb{Q}, <)$.

These notions will reappear throughout the chapter. In particular, stabilizers of finite tuples will provide the basic open neighbourhoods in the permutation topology, while orbits on finite powers Ω^n will play a central role in the description of closed and dense subgroups.

2.2 The Permutation Topology

We now introduce the natural topology on permutation groups. Let Ω be a set. For $n < \omega$ and tuples $\bar{a}, \bar{b} \in \Omega^n$, set

$$S(\bar{a}, \bar{b}) = \{g \in \text{Sym}(\Omega) : g\bar{a} = \bar{b}\}.$$

A subset of $\text{Sym}(\Omega)$ is called *basic open* if it is of the form $S(\bar{a}, \bar{b})$ for some finite tuples $\bar{a}, \bar{b}$ of the same length. An *open* subset of $\text{Sym}(\Omega)$ is any union of basic open sets. This is the *permutation topology*, or equivalently the topology of pointwise convergence.

If $G \leq \text{Sym}(\Omega)$, we always equip G with the subspace topology induced from $\text{Sym}(\Omega)$. Thus the basic open subsets of G are precisely the sets of the form

$$G \cap S(\bar{a}, \bar{b}),$$

where $\bar{a}, \bar{b}$ are finite tuples from Ω of the same length.

Proposition 2.2.1. *Let $G \leq \text{Sym}(\Omega)$.*

- (a) *The sets $S(\bar{a}, \bar{b})$ form a basis for a topology on $\text{Sym}(\Omega)$. With this topology, $\text{Sym}(\Omega)$ is a topological group, and hence so is every subgroup $G \leq \text{Sym}(\Omega)$ with the induced topology.*

- (b) A subgroup $H \leq G$ is open if and only if it contains $G_{(\bar{a})}$ for some finite tuple $\bar{a}$ from Ω .
- (c) A subset $F \subseteq G$ is closed if and only if whenever $g \in G$ has the property that every basic open neighbourhood of g meets F , then $g \in F$.
- (d) A subgroup $H \leq G$ is dense in G if and only if H and G have the same orbits on Ω^n for every positive integer n .

Proof. For (a), note first that

$$\text{Sym}(\Omega) = S(\emptyset, \emptyset).$$

Moreover, if $g \in S(\bar{a}, \bar{b}) \cap S(\bar{c}, \bar{d})$, then

$$g \in S(\bar{a}\bar{c}, \bar{b}\bar{d}) \subseteq S(\bar{a}, \bar{b}) \cap S(\bar{c}, \bar{d}),$$

where $\bar{a}\bar{c}$ and $\bar{b}\bar{d}$ denote concatenation of tuples. Thus the sets $S(\bar{a}, \bar{b})$ form a basis for a topology on $\text{Sym}(\Omega)$.

To prove continuity of inversion, observe that

$$g \in S(\bar{a}, \bar{b}) \iff g^{-1} \in S(\bar{b}, \bar{a}).$$

For continuity of multiplication, let $(g, h) \in \text{Sym}(\Omega) \times \text{Sym}(\Omega)$ and suppose that

$$gh \in S(\bar{a}, \bar{b}).$$

Set $\bar{c} = h\bar{a}$. Then

$$h \in S(\bar{a}, \bar{c}) \quad \text{and} \quad g \in S(\bar{c}, \bar{b}),$$

and whenever $g' \in S(\bar{c}, \bar{b})$ and $h' \in S(\bar{a}, \bar{c})$, we have

$$g'h'\bar{a} = g'\bar{c} = \bar{b},$$

so that

$$g'h' \in S(\bar{a}, \bar{b}).$$

Hence

$$S(\bar{c}, \bar{b}) \cdot S(\bar{a}, \bar{c}) \subseteq S(\bar{a}, \bar{b}),$$

which proves continuity of multiplication. Therefore $\text{Sym}(\Omega)$ is a topological group. Since subgroups inherit the subspace topology, every subgroup $G \leq \text{Sym}(\Omega)$ is also a topological group.

For (b), if $\bar{a}$ is a finite tuple from Ω , then

$$G_{(\bar{a})} = G \cap S(\bar{a}, \bar{a}),$$

so $G_{(\bar{a})}$ is open in G . If $H \leq G$ contains $G_{(\bar{a})}$, then

$$H = \bigcup_{h \in H} hG_{(\bar{a})},$$

and each left coset $hG_{(\bar{a})}$ is open, since left translation is a homeomorphism of G . Thus H is open.

Conversely, suppose that $H \leq G$ is open. Since $1_G \in H$, there exists a basic open neighbourhood of 1_G contained in H . Such a neighbourhood has the form

$$G \cap S(\bar{a}, \bar{a}) = G_{(\bar{a})}$$

for some finite tuple $\bar{a}$ from Ω . Hence $G_{(\bar{a})} \subseteq H$.

Statement (c) is just the usual characterization of closed sets in terms of neighbourhoods: a subset $F \subseteq G$ is closed if and only if every point outside F has an open neighbourhood disjoint from F , equivalently, whenever every basic open neighbourhood of g meets F , one has $g \in F$.

For (d), first suppose that H is dense in G , and let $\bar{a}, \bar{b} \in \Omega^n$ lie in the same G -orbit. Choose $g \in G$ with $g\bar{a} = \bar{b}$. Then $S(\bar{a}, \bar{b})$ is a basic open neighbourhood of g , so by density it meets H . Hence there exists $h \in H$ such that $h\bar{a} = \bar{b}$. Thus $\bar{a}$ and $\bar{b}$ lie in the same H -orbit. It follows that every G -orbit on Ω^n is also an H -orbit, and therefore H and G have the same orbits on Ω^n .

Conversely, suppose that H and G have the same orbits on Ω^n for every $n \geq 1$. To show that H is dense in G , it suffices by part (c) to show that every nonempty basic open subset of G meets H . Let

$$G \cap S(\bar{a}, \bar{b})$$

be a nonempty basic open subset of G . Then there exists $g \in G$ such that $g\bar{a} = \bar{b}$, so $\bar{a}$ and $\bar{b}$ lie in the same G -orbit on Ω^n . By assumption they also lie in the same H -orbit, so there exists $h \in H$ such that $h\bar{a} = \bar{b}$. Hence

$$h \in H \cap S(\bar{a}, \bar{b}) \subseteq H \cap G \cap S(\bar{a}, \bar{b}),$$

and therefore $G \cap S(\bar{a}, \bar{b})$ meets H . This proves that H is dense in G . ◊

In particular, the stabilizers of finite tuples form a neighbourhood basis at the identity in any subgroup of $\text{Sym}(\Omega)$, and orbit structure on finite powers Ω^n detects density. These two observations will be used repeatedly in the next section.

2.3 Closed Subgroups and Canonical Structures

We now characterize those subgroups of $\text{Sym}(\Omega)$ which arise as automorphism groups of structures on Ω . In view of the preceding section, the relevant notion of closedness is topological closedness in the permutation topology.

Theorem 2.3.1. *Let Ω be a set, let $G \leq \text{Sym}(\Omega)$, and let $H \leq G$. Then the following are equivalent.*

- (a) H is closed in G .
- (b) There exists a structure A with domain Ω such that

$$H = G \cap \text{Aut}(A).$$

In particular, a subgroup $H \leq \text{Sym}(\Omega)$ is of the form $\text{Aut}(A)$ for some structure A with domain Ω if and only if H is closed in $\text{Sym}(\Omega)$.

Proof. Assume first that H is closed in G . For each $n < \omega$ and each orbit Δ of H on Ω^n , introduce an n -ary relation symbol R_Δ . Let L be the language consisting of all these relation symbols, and define an L -structure A with domain Ω by setting

$$R_\Delta^A = \Delta$$

for every orbit Δ of H on Ω^n .

Every element of H preserves each relation R_Δ^A , since each R_Δ^A is an H -orbit. Hence

$$H \subseteq G \cap \text{Aut}(A).$$

Conversely, let $g \in G \cap \text{Aut}(A)$. We claim that every basic open neighbourhood of g in G meets H . Let $\bar{a} \in \Omega^n$, and let Δ be the H -orbit of $\bar{a}$. Since $\bar{a} \in \Delta = R_\Delta^A$ and g is an automorphism of A , we have

$$g\bar{a} \in R_\Delta^A = \Delta.$$

Thus there exists $h \in H$ such that

$$h\bar{a} = g\bar{a}.$$

It follows that

$$h \in H \cap G \cap S(\bar{a}, g\bar{a}),$$

so the basic open neighbourhood $G \cap S(\bar{a}, g\bar{a})$ of g meets H . Since H is closed in G , we conclude that $g \in H$. Therefore

$$G \cap \text{Aut}(A) \subseteq H,$$

and hence

$$H = G \cap \text{Aut}(A).$$

Now assume that there exists a structure A with domain Ω such that

$$H = G \cap \text{Aut}(A).$$

To show that H is closed in G , let $g \in G$ and suppose that every basic open neighbourhood of g in G meets H . We show that $g \in \text{Aut}(A)$.

Let $\varphi(\bar{x})$ be an atomic formula in the language of A , and let $\bar{a}$ be a tuple from Ω of the same length as $\bar{x}$. By assumption, the basic open neighbourhood

$$G \cap S(\bar{a}, g\bar{a})$$

meets H . Choose

$$h \in H \cap G \cap S(\bar{a}, g\bar{a}).$$

Then $h\bar{a} = g\bar{a}$. Since $h \in H \subseteq \text{Aut}(A)$, we have

$$A \models \varphi(\bar{a}) \iff A \models \varphi(h\bar{a}).$$

Since $h\bar{a} = g\bar{a}$, it follows that

$$A \models \varphi(\bar{a}) \iff A \models \varphi(g\bar{a}).$$

Thus g preserves all atomic formulas, and since g is bijective it follows that $g \in \text{Aut}(A)$. Therefore

$$g \in G \cap \text{Aut}(A) = H.$$

Hence H is closed in G . ⊢

Corollary 2.3.2. *Let A be a structure with domain Ω . Then $\text{Aut}(A)$ is a closed subgroup of $\text{Sym}(\Omega)$.*

Proof. Apply Theorem 2.3.1 with $G = \text{Sym}(\Omega)$ and $H = \text{Aut}(A)$. ⊢

When H is a closed subgroup of $\text{Sym}(\Omega)$, the structure constructed in the proof of Theorem 2.3.1 is called the *canonical structure* of H . Its domain is Ω , and for each $n < \omega$ its basic n -ary relations are precisely the H -orbits on Ω^n . By construction,

$$H = \text{Aut}(A).$$

Thus the canonical structure records exactly the orbit structure of H on finite powers of Ω . In particular, closed subgroups of $\text{Sym}(\Omega)$ and automorphism groups of structures on Ω may be regarded as two equivalent ways of encoding the same data.

2.4 Countable Automorphism Groups as Polish Groups

We now specialize to the countable case, which is the one relevant for Fraïssé limits. Throughout this section, let Ω be a countable set, and fix an enumeration

$$\Omega = \{\omega_1, \omega_2, \dots\}.$$

Our goal is to show that automorphism groups of countable structures are Polish. Since $\text{Sym}(\Omega)$ is already a topological group by Proposition 2.2.1, it remains to verify in this setting separability, complete metrizability, and the fact that closed subgroups of Polish groups are again Polish.

Proposition 2.4.1. *The permutation topology on $\text{Sym}(\Omega)$ is second countable. Consequently, every subgroup of $\text{Sym}(\Omega)$, endowed with the induced topology, is also second countable.*

Proof. For each $n < \omega$ and each pair of tuples $\bar{a}, \bar{b} \in \Omega^n$, the set $S(\bar{a}, \bar{b})$ is basic open. Since Ω is countable, each Ω^n is countable, and hence

$$\bigcup_{n < \omega} \Omega^n \times \Omega^n$$

is countable. Therefore there are only countably many basic open sets of the form $S(\bar{a}, \bar{b})$, so $\text{Sym}(\Omega)$ has a countable basis. Since second countability passes to subspaces, every subgroup of $\text{Sym}(\Omega)$ is also second countable. ⊢

Corollary 2.4.2. *The space $\text{Sym}(\Omega)$ is separable. Consequently, every subgroup of $\text{Sym}(\Omega)$ is separable.*

Proof. Every second countable space is separable, so the conclusion follows from Proposition 2.4.1. $\dashv$

We now describe an explicit compatible complete metric. First define a metric d on $\text{Sym}(\Omega)$ by

$$d(g, h) = \begin{cases} 0, & \text{if } g = h, \\ \frac{1}{n}, & \text{if } g \neq h, \text{ where } n = \min\{m \geq 1 : g(\omega_m) \neq h(\omega_m)\}. \end{cases}$$

Thus $d(g, h)$ records the first point of the enumeration on which g and h disagree.

Proposition 2.4.3. *The function d is a metric on $\text{Sym}(\Omega)$, and it induces the permutation topology.*

Proof. The only nontrivial point is the triangle inequality. Let $g, h, k \in \text{Sym}(\Omega)$, and suppose that

$$d(g, k) = \frac{1}{a}, \quad d(k, h) = \frac{1}{b}.$$

Then $g(\omega_m) = k(\omega_m) = h(\omega_m)$ for every $m < \min(a, b)$. Hence

$$d(g, h) \leq \frac{1}{\min(a, b)} = \max\{d(g, k), d(k, h)\},$$

so d is in fact an ultrametric.

For $g \in \text{Sym}(\Omega)$ and $n \geq 1$, the open ball of radius $1/(n+1)$ about g is

$$B_d\left(g, \frac{1}{n+1}\right) = \{h \in \text{Sym}(\Omega) : h(\omega_m) = g(\omega_m) \text{ for all } m \leq n\}.$$

Thus the d -balls are exactly the sets of permutations agreeing with g on a finite initial segment of the enumeration. These form a neighbourhood basis at g for the permutation topology, so d induces that topology. $\dashv$

The metric d is not complete, since a sequence of longer and longer finite cycles can be Cauchy for d while converging pointwise to an injective non-surjective map. To obtain a complete compatible metric, define

$$d'(g, h) = d(g, h) + d(g^{-1}, h^{-1}).$$

Proposition 2.4.4. *The function d' is a complete metric on $\text{Sym}(\Omega)$, and it induces the permutation topology.*

Proof. Since d is a metric, so is d' . Moreover,

$$d(g, h) \leq d'(g, h),$$

so every d' -open set is d -open. Conversely, if h agrees with g on $\omega_1, \dots, \omega_n$, then every permutation sufficiently close to h in the metric d' also agrees with g on those points. Hence every basic d -open neighbourhood is d' -open, and the two metrics induce the same topology.

It remains to prove completeness. Let $F = \Omega^\Omega$, and define a metric δ on F by

$$\delta(f_1, f_2) = \begin{cases} 0, & \text{if } f_1 = f_2, \\ \frac{1}{n}, & \text{if } f_1 \neq f_2, \text{ where } n = \min\{m \geq 1 : f_1(\omega_m) \neq f_2(\omega_m)\}. \end{cases}$$

Exactly as above, (F, δ) is complete. Equip $F \times F$ with the product metric

$$\rho((f_1, h_1), (f_2, h_2)) = \delta(f_1, f_2) + \delta(h_1, h_2).$$

Then $(F \times F, \rho)$ is complete.

Now define

$$\Phi : \text{Sym}(\Omega) \rightarrow F \times F, \quad \Phi(g) = (g, g^{-1}).$$

For $g, h \in \text{Sym}(\Omega)$,

$$\rho(\Phi(g), \Phi(h)) = \delta(g, h) + \delta(g^{-1}, h^{-1}) = d'(g, h),$$

so Φ is an isometric embedding.

Its image consists exactly of the pairs $(f, h) \in F \times F$ satisfying

$$f \circ h = \text{id}_\Omega \quad \text{and} \quad h \circ f = \text{id}_\Omega.$$

This is a closed subset of $F \times F$, since for each $\omega \in \Omega$ the conditions

$$f(h(\omega)) = \omega \quad \text{and} \quad h(f(\omega)) = \omega$$

define closed sets. Therefore $\Phi(\text{Sym}(\Omega))$ is complete, and hence $(\text{Sym}(\Omega), d')$ is complete. $\dashv$

Definition 2.4.5. A *Polish group* is a topological group whose topology is separable and completely metrizable.

Proposition 2.4.6. Let G be a Polish group and let $H \leq G$ be a closed subgroup. Then H , equipped with the subspace topology, is again a Polish group.

Proof. Since multiplication and inversion on H are restrictions of the corresponding continuous maps on G , the group H is a topological group with the subspace topology.

Let d be a compatible complete metric on G . Then the restriction $d|_{H \times H}$ is a compatible metric on H , and it is complete because H is closed in the complete metric space (G, d) .

Since G is Polish, it is in particular separable and metrizable, hence second countable. Therefore the subspace H is also second countable, and thus separable. It follows that H is separable and completely metrizable, so H is Polish. $\dashv$

Theorem 2.4.7. *Let A be a structure with countable domain Ω . Then $\text{Aut}(A)$, endowed with the permutation topology, is a Polish group.*

Proof. By Corollary 2.3.2, the group $\text{Aut}(A)$ is a closed subgroup of $\text{Sym}(\Omega)$. By Proposition 2.4.4, the group $\text{Sym}(\Omega)$ is completely metrizable, and by Corollary 2.4.2 it is separable. Since $\text{Sym}(\Omega)$ is a topological group by Proposition 2.2.1, it is therefore a Polish group. Now Proposition 2.4.6 shows that $\text{Aut}(A)$ is Polish. $\dashv$

Proposition 2.4.8. *Let $G \leq \text{Sym}(\Omega)$. Then the family*

$$\{G_{(F)} : F \subseteq \Omega \text{ finite}\}$$

is a countable neighbourhood basis at the identity consisting of open subgroups.

Proof. Each $G_{(F)}$ is a subgroup of G , and by Proposition 2.4.3 it is open. By Proposition 2.4.1, the finite subsets of Ω form a countable family. Finally, every basic open neighbourhood of the identity in G contains $G_{(F)}$ for some finite set $F \subseteq \Omega$, so these subgroups form a neighbourhood basis at the identity. $\dashv$

In particular, automorphism groups of countable structures are non-archimedean Polish groups: they admit a countable local basis at the identity consisting of open subgroups. This is the topological setting in which we will study flows, minimality, and universal minimal flows in the following chapters.

Chapter 3

Topological Groups

This chapter collects the general facts on topological groups that will be used later in the thesis. Chapter 2 showed that automorphism groups of countable structures, equipped with the permutation topology, are Polish groups. We now develop the broader topological-group background needed for their actions on compact spaces. We begin with the definition of a topological group and some basic examples, then study neighborhoods of the identity, subgroups, coset spaces, and quotient groups, and finally recall the notions of metrizability and Polish groups.

3.1 Topological groups and basic examples

In this section we introduce the basic notion of a topological group and record several standard examples. The structural facts established here will be used later in the study of flows and universal minimal flows.

Definition 3.1.1. Let G be a group. A topology on G is said to be *compatible* with the group structure if the maps

$$m: G \times G \rightarrow G, \quad m(x, y) = xy,$$

and

$$i: G \rightarrow G, \quad i(x) = x^{-1},$$

are continuous, where $G \times G$ is equipped with the product topology. A *topological group* is a group together with a compatible topology.

Thus a topological group is a group for which the algebraic operations vary continuously with the topology. In particular, a topological group is simultaneously an algebraic and a topological object, and the interaction between these two structures is the source of its rigidity.

Example 3.1.2. Every group equipped with the discrete topology is a topological group.

Example 3.1.3. Let G be a topological group and let $H \leq G$ be a subgroup. Then H , equipped with the subspace topology inherited from G , is again a topological group.

Proof. The multiplication map on H is the restriction of the multiplication map on G to $H \times H$, and the inversion map on H is the restriction of the inversion map on G to H . Since restrictions of continuous maps are continuous, both operations are continuous on H . $\dashv$

Example 3.1.4. The additive groups $(\mathbb{R}, +)$, $(\mathbb{Q}, +)$, and $(\mathbb{Z}, +)$, equipped with their usual topologies, are topological groups.

Proof. Addition and additive inversion are continuous on $\mathbb{R}$. Hence $(\mathbb{R}, +)$ is a topological group. Since $\mathbb{Q}$ and $\mathbb{Z}$ are subgroups of $(\mathbb{R}, +)$, the previous example shows that they are topological groups with the induced topologies. $\dashv$

Example 3.1.5. The multiplicative group

$$\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\},$$

equipped with the subspace topology inherited from $\mathbb{C}$, is a topological group. It is called the *circle group*.

Proof. The set $\mathbb{T}$ is a subgroup of the multiplicative group $\mathbb{C} \setminus \{0\}$, and multiplication and inversion are continuous on $\mathbb{C} \setminus \{0\}$. Hence $\mathbb{T}$ is a topological group with the relative topology. $\dashv$

Example 3.1.6. If $(G_\alpha)_{\alpha \in A}$ is a family of topological groups, then the product

$$\prod_{\alpha \in A} G_\alpha,$$

equipped with the product topology and coordinatewise group operations, is a topological group.

Proof. Multiplication and inversion are defined coordinatewise, so they are continuous by the universal property of the product topology. $\dashv$

The next proposition records the basic symmetries of a topological group. It will be used repeatedly.

Proposition 3.1.7. *Let G be a topological group and let $a, b \in G$. Then the following maps are homeomorphisms of G :*

$$x \mapsto x^{-1}, \quad x \mapsto ax, \quad x \mapsto xb, \quad x \mapsto axb, \quad x \mapsto axa^{-1}.$$

In particular, every left translation and every right translation is a homeomorphism.

Proof. The inversion map $x \mapsto x^{-1}$ is continuous by definition of a topological group, and it is its own inverse, hence a homeomorphism.

The map $x \mapsto ax$ is continuous because it is obtained by composing the continuous map

$$x \mapsto (a, x)$$

from G to $G \times G$ with the multiplication map. Its inverse is the map $x \mapsto a^{-1}x$, which is continuous by the same argument. Thus $x \mapsto ax$ is a homeomorphism. Similarly, $x \mapsto xb$ is a homeomorphism.

The map $x \mapsto axb$ is the composition of the two homeomorphisms $x \mapsto ax$ and $x \mapsto xb$, so it is again a homeomorphism. Finally, $x \mapsto axa^{-1}$ is a special case of the previous map. $\dashv$

Proposition 3.1.7 shows that a topological group is homogeneous as a topological space: any point can be moved to any other point by a homeomorphism. Consequently, much of the topology of G is already determined by the neighborhoods of the identity element. This observation will be developed in the next section.

3.2 Neighborhoods of the identity

By Proposition 3.1.7, every left and right translation in a topological group is a homeomorphism. As a result, the topology of a topological group is determined by the neighborhoods of the identity element. We record the basic properties of this neighborhood system and the converse construction.

Theorem 3.2.1. *Let G be a topological group, and let $\mathcal{V}$ denote the collection of all neighborhoods of the identity element e . Then $\mathcal{V}$ has the following properties:*

1. $e \in V$ for every $V \in \mathcal{V}$;
2. if $V, W \in \mathcal{V}$, then $V \cap W \in \mathcal{V}$;
3. if $V \in \mathcal{V}$, then there exists $W \in \mathcal{V}$ such that $WW \subseteq V$;
4. if $V \in \mathcal{V}$, then $V^{-1} \in \mathcal{V}$;
5. if $V \in \mathcal{V}$ and $a \in G$, then $aVa^{-1} \in \mathcal{V}$;
6. if $V \in \mathcal{V}$ and $V \subseteq W \subseteq G$, then $W \in \mathcal{V}$.

Proof. Properties (1), (2), and (6) are true for the neighborhood filter of any point in a topological space.

For (3), let $V \in \mathcal{V}$. Since the multiplication map

$$m: G \times G \rightarrow G, \quad m(x, y) = xy,$$

is continuous at (e, e) and $m(e, e) = e$, there exist neighborhoods W_1 and W_2 of e such that

$$W_1 W_2 \subseteq V.$$

Setting $W = W_1 \cap W_2$, we still have $W \in \mathcal{V}$, and then

$$WW \subseteq W_1 W_2 \subseteq V.$$

Property (4) follows because inversion is a homeomorphism of G fixing e .

Property (5) follows because the map

$$x \mapsto axa^{-1}$$

is a homeomorphism of G fixing e . ◻

The preceding theorem shows that the neighborhoods of e satisfy strong algebraic compatibility conditions. Conversely, such a family determines a unique group topology.

Theorem 3.2.2. *Let G be an abstract group with identity element e , and let $\mathcal{V}$ be a nonempty collection of subsets of G satisfying (1)–(6) of Theorem 3.2.1. Then there is a unique topology on G such that:*

1. G becomes a topological group;
2. $\mathcal{V}$ is exactly the collection of neighborhoods of e .

Proof. For each $a \in G$, define

$$\mathcal{V}_a = \{aV : V \in \mathcal{V}\}.$$

We claim that the family $(\mathcal{V}_a)_{a \in G}$ is the neighborhood system of a topology on G .

First, every set in $\mathcal{V}_a$ contains a , since $e \in V$ for every $V \in \mathcal{V}$. Also, $\mathcal{V}_a$ is closed under finite intersections and upward inclusion, because $\mathcal{V}$ has the corresponding properties.

It remains to verify the neighborhood axiom. Let $N \in \mathcal{V}_a$, say $N = aV$ with $V \in \mathcal{V}$. By (3), choose $W \in \mathcal{V}$ such that $WW \subseteq V$, and set $M = aW$. Then $M \in \mathcal{V}_a$. If $y \in M$, write $y = aw$ with $w \in W$. We claim that $N \in \mathcal{V}_y$. Indeed,

$$yW = (aw)W = a(wW) \subseteq aWW \subseteq aV = N,$$

and $yW \in \mathcal{V}_y$. Thus every point of M has a neighborhood contained in N , so the family $(\mathcal{V}_a)_{a \in G}$ defines a topology on G .

By construction, $\mathcal{V}$ is the collection of neighborhoods of e . We now show that the group operations are continuous.

Let $a, b \in G$, and let abV be a neighborhood of ab , where $V \in \mathcal{V}$. By (3), choose $W \in \mathcal{V}$ such that $WW \subseteq V$. By (5), the set bWb^{-1} belongs to $\mathcal{V}$. Hence

$$U := W \cap bWb^{-1}$$

also belongs to $\mathcal{V}$. We show that

$$(aU)(bU) \subseteq abV.$$

Indeed, if $u_1, u_2 \in U$, then

$$(au_1)(bu_2) = ab(b^{-1}u_1b)u_2.$$

Since $u_1 \in U \subseteq bWb^{-1}$, we have $b^{-1}u_1b \in W$, and since $u_2 \in U \subseteq W$, it follows that

$$(b^{-1}u_1b)u_2 \in WW \subseteq V.$$

Therefore $(au_1)(bu_2) \in abV$, proving continuity of multiplication.

To prove continuity of inversion, let $a \in G$ and let $a^{-1}V$ be a neighborhood of a^{-1} , where $V \in \mathcal{V}$. By (4) and (5), the set

$$U := a^{-1}V^{-1}a$$

belongs to $\mathcal{V}$. If $x \in aU$, then $x = au$ for some $u \in U$, and hence

$$x^{-1} = u^{-1}a^{-1}.$$

Since $u \in a^{-1}V^{-1}a$, we have $u^{-1} \in a^{-1}Va$, so

$$x^{-1} = u^{-1}a^{-1} \in a^{-1}V.$$

Thus

$$(aU)^{-1} \subseteq a^{-1}V,$$

and inversion is continuous at a . Hence G is a topological group.

Finally, uniqueness is immediate: in any topological group, the neighborhoods of an arbitrary point a are exactly the left translates aV of neighborhoods V of e . Therefore the topology is completely determined by the neighborhood system at the identity. $\dashv$

In practice one usually works with a neighborhood basis at the identity rather than with the full neighborhood filter.

Corollary 3.2.3. *Let G be an abstract group, and let $\mathcal{B}$ be a nonempty collection of subsets of G such that:*

1. $e \in B$ for every $B \in \mathcal{B}$;
2. for all $B_1, B_2 \in \mathcal{B}$, there exists $B_3 \in \mathcal{B}$ with $B_3 \subseteq B_1 \cap B_2$;
3. for every $B \in \mathcal{B}$, there exists $C \in \mathcal{B}$ such that $CC \subseteq B$;
4. for every $B \in \mathcal{B}$, there exists $C \in \mathcal{B}$ such that $C \subseteq B^{-1}$;
5. for every $B \in \mathcal{B}$ and every $a \in G$, there exists $C \in \mathcal{B}$ such that $C \subseteq aBa^{-1}$.

Then there exists a unique topology on G making G a topological group and making $\mathcal{B}$ a neighborhood basis at e .

Proof. Let $\mathcal{V}$ be the collection of all subsets $V \subseteq G$ containing some $B \in \mathcal{B}$. Then $\mathcal{V}$ satisfies the hypotheses of Theorem 3.2.2, so the result follows. $\dashv$

The importance of these results is that they reduce many topological questions about a group to the behavior of a suitably chosen family of neighborhoods of the identity. This will be used repeatedly in the metrizable case.

3.3 Subgroups, coset spaces, and quotient groups

We now consider the basic constructions obtained from a topological group by passing to subgroups and quotients. These constructions will be used later both for coset spaces of open subgroups and for quotient groups by closed normal subgroups.

Proposition 3.3.1. *Let G be a topological group and let $H \leq G$. Then H , equipped with the subspace topology inherited from G , is a topological group.*

Proof. The multiplication map on H is the restriction of the multiplication map on G to $H \times H$, and the inversion map on H is the restriction of the inversion map on G to H . Since restrictions of continuous maps are continuous, both operations are continuous. $\dashv$

Proposition 3.3.2. *Let G be a topological group and let $H \leq G$. Then $\overline{H}$ is a subgroup of G . If H is normal, then $\overline{H}$ is normal.*

Proof. Let $x, y \in \overline{H}$. Since multiplication is continuous, we have

$$xy^{-1} \in \overline{HH^{-1}}.$$

As H is a subgroup, $HH^{-1} = H$, so $xy^{-1} \in \overline{H}$. Hence $\overline{H}$ is a subgroup.

Now assume that H is normal. Fix $g \in G$. Since the conjugation map

$$c_g: G \rightarrow G, \quad c_g(x) = gxg^{-1},$$

is a homeomorphism, we obtain

$$g\overline{H}g^{-1} = c_g(\overline{H}) = \overline{c_g(H)} = \overline{gHg^{-1}} = \overline{H}.$$

Thus $\overline{H}$ is normal. ◻

We next turn to coset spaces.

Definition 3.3.3. Let G be a topological group and let $H \leq G$. The set of left cosets of H in G is denoted by

$$G/H = \{gH : g \in G\}.$$

Let

$$\pi_H: G \rightarrow G/H, \quad \pi_H(g) = gH,$$

be the canonical map. We equip G/H with the quotient topology induced by π_H .

Proposition 3.3.4. *Let G be a topological group and let $H \leq G$. Then the quotient map*

$$\pi_H: G \rightarrow G/H$$

is continuous, surjective, and open.

Proof. Continuity and surjectivity are immediate from the definition of the quotient topology.

Let $U \subseteq G$ be open. We show that $\pi_H(U)$ is open in G/H . By definition of the quotient topology, it suffices to show that

$$\pi_H^{-1}(\pi_H(U))$$

is open in G . But

$$\pi_H^{-1}(\pi_H(U)) = UH = \bigcup_{h \in H} Uh,$$

and each right translate Uh is open by Proposition 3.1.7. Hence UH is open, so $\pi_H(U)$ is open. ◻

The case of open subgroups is especially important.

Corollary 3.3.5. *Let G be a topological group and let $H \leq G$ be open. Then G/H is discrete.*

Proof. Since H is open in G , Proposition 3.3.4 implies that

$$\{\pi_H(e)\} = \pi_H(H) = \{H\}$$

is open in G/H . Every singleton in G/H is a left translate of $\{H\}$, hence is open. Therefore G/H is discrete. $\dashv$

We now consider quotient groups. For this, normality is needed in order that multiplication of cosets be well defined.

Proposition 3.3.6. *Let G be a topological group and let $N \trianglelefteq G$ be a normal subgroup. Then G/N , equipped with the quotient topology, is a topological group under the operations*

$$(gN)(hN) = ghN \quad \text{and} \quad (gN)^{-1} = g^{-1}N.$$

Proof. Since N is normal, the operations are well defined algebraically.

Let

$$q: G \rightarrow G/N$$

denote the quotient map. To prove continuity of multiplication on G/N , consider the diagram

$$\begin{array}{ccc} G \times G & \xrightarrow{m} & G \\ q \times q \downarrow & & \downarrow q \\ G/N \times G/N & \xrightarrow{\tilde{m}} & G/N \end{array}$$

where $m(g, h) = gh$. Since $q(gh) = q(g)q(h)$, the map $\tilde{m}$ is well defined and satisfies

$$\tilde{m} \circ (q \times q) = q \circ m.$$

The map $q \times q$ is a quotient map because q is open and surjective. Since $q \circ m$ is continuous, it follows from the universal property of quotient maps that $\tilde{m}$ is continuous.

Similarly, inversion on G/N is induced by inversion on G . Indeed, if $i(g) = g^{-1}$, then the map

$$\tilde{i}: G/N \rightarrow G/N, \quad \tilde{i}(gN) = g^{-1}N,$$

satisfies

$$\tilde{i} \circ q = q \circ i.$$

Since q is a quotient map and $q \circ i$ is continuous, $\tilde{i}$ is continuous. Thus G/N is a topological group. $\dashv$

In many applications one wants the quotient group to remain Hausdorff. This is guaranteed by closedness of the normal subgroup.

Proposition 3.3.7. *Let G be a Hausdorff topological group and let $N \trianglelefteq G$ be a closed normal subgroup. Then G/N is Hausdorff.*

Proof. Let $gN, hN \in G/N$ with $gN \neq hN$. Then $h^{-1}g \notin N$. Since N is closed and G is Hausdorff, there exists an open neighborhood U of $h^{-1}g$ such that

$$U \cap N = \emptyset.$$

By continuity of multiplication, there exist neighborhoods V of g and W of h such that

$$W^{-1}V \subseteq U.$$

We claim that $\pi_N(V)$ and $\pi_N(W)$ are disjoint neighborhoods of gN and hN , respectively. Suppose not. Then there exist $v \in V$, $w \in W$, and $n_1, n_2 \in N$ such that

$$vn_1 = wn_2.$$

Hence

$$w^{-1}v = n_2n_1^{-1} \in N.$$

But $w^{-1}v \in W^{-1}V \subseteq U$, contradicting $U \cap N = \emptyset$. Therefore $\pi_N(V)$ and $\pi_N(W)$ are disjoint, and G/N is Hausdorff. $\dashv$

We shall later apply these facts in two different ways. For open subgroups $V \leq G$, the coset space G/V is discrete by Corollary 3.3.5. For closed normal subgroups $N \trianglelefteq G$, the quotient G/N is again a Hausdorff topological group by Propositions 3.3.6 and 3.3.7.

3.4 Metrizable and Polish groups

For the groups that arise in this thesis, the most important topological groups are the metrizable ones, and especially the Polish groups. We therefore record the basic definitions and the standard metrization theorem for topological groups.

Definition 3.4.1. Let G be a topological group.

1. A metric d on G is said to be *compatible* if it induces the given topology on G .
2. A metric d on G is *left-invariant* if

$$d(gx, gy) = d(x, y)$$

for all $g, x, y \in G$.

3. A topological group is *metrizable* if its topology is induced by some metric.
4. A topological group is *Polish* if its topology is separable and completely metrizable.

The key fact is that, for Hausdorff topological groups, first countability already implies metrizability.

Theorem 3.4.2 (Birkhoff–Kakutani). *Let G be a Hausdorff topological group. Then the following are equivalent:*

1. G is first countable;
2. G is metrizable;
3. G admits a compatible left-invariant metric.

Proof. The implication (3) $\Rightarrow$ (2) is immediate, and every metric space is first countable, so (2) $\Rightarrow$ (1).

It remains to prove (1) $\Rightarrow$ (3). Assume that G is first countable. Since G is Hausdorff, we have

$$\bigcap \mathcal{V}_e = \{e\},$$

where $\mathcal{V}_e$ denotes the collection of neighborhoods of e . Choose a countable neighborhood basis $(U_n)_{n \geq 1}$ at e . By Corollary 3.2.3, we may recursively choose a sequence $(V_n)_{n \geq 1}$ of neighborhoods of e such that

$$V_1 = G, \quad V_{n+1} = V_{n+1}^{-1}, \quad V_{n+1}^3 \subseteq V_n \cap U_n$$

for every $n \geq 1$. In particular,

$$V_{n+1} \subseteq V_n \quad \text{and} \quad \bigcap_{n \geq 1} V_n = \{e\}.$$

Define a function $\rho: G \times G \rightarrow [0, \infty)$ by

$$\rho(x, y) = \inf \{2^{-n} : x^{-1}y \in V_n\}.$$

Then ρ is left-invariant, symmetric, and satisfies $\rho(x, y) = 0$ if and only if $x = y$.

Now define

$$d(x, y) = \inf \left\{ \sum_{i=0}^{m-1} \rho(z_i, z_{i+1}) : m \geq 1, z_0 = x, z_m = y, z_1, \dots, z_{m-1} \in G \right\}.$$

Since ρ is left-invariant and symmetric, the same is true of d . Also, by construction, d satisfies the triangle inequality.

We claim that d is a metric. It only remains to show that $d(x, y) = 0$ implies $x = y$. Suppose $x \neq y$. Then $x^{-1}y \neq e$, so there exists n such that $x^{-1}y \notin V_n$. By the standard Birkhoff–Kakutani argument using the condition $V_{n+1}^3 \subseteq V_n$, any chain from x to y has total ρ -length at least 2^{-n-1} . Hence $d(x, y) > 0$, as required.

Finally, the metric d induces the original topology on G . Indeed, one shows again from the construction that for each $n \geq 1$,

$$B_d(e, 2^{-n-1}) \subseteq V_n \subseteq B_d(e, 2^{1-n}),$$

so the d -balls about e form a neighborhood basis at the identity. By translation, d induces the given topology on all of G . ◻

For later use we isolate the most important consequence.

Corollary 3.4.3. *Every Hausdorff first countable topological group is metrizable. More precisely, it admits a compatible left-invariant metric.*

Proof. This is immediate from Theorem 3.4.2. ◻

We now turn to Polish groups.

Definition 3.4.4. A topological space is called *Polish* if it is separable and completely metrizable. A *Polish group* is a topological group whose underlying topological space is Polish.

Polish groups form the natural ambient class for the automorphism groups considered in this thesis. In Chapter 2 we used the fact that closed subgroups of Polish groups are again Polish in order to show that automorphism groups of countable structures are Polish. In the next chapter these groups will act on compact spaces, and that dynamical viewpoint will become central.

The main point of this chapter is that the topological structure of a group is largely controlled by the neighborhoods of the identity, and that in the metrizable setting one obtains a particularly tractable class of groups, namely Polish groups. In the next chapter we turn from the internal structure of topological groups to their actions on compact spaces, which will provide the dynamical framework needed for the Kechris–Pestov–Todorcevic correspondence.

Chapter 4

Topological Dynamics

In the previous chapter we studied automorphism groups of countable structures as topological groups, with particular emphasis on closed subgroups of S_∞ and the Polish group structure they inherit. The next step in the thesis is to study continuous actions of such groups on compact spaces. This is the setting of topological dynamics, and it is here that the fixed-point properties and universal objects appearing in the Kechris–Pestov–Todorćević correspondence naturally arise.

The main notions introduced in this chapter are those of a G -flow, a minimal flow, the universal minimal flow, extreme amenability, and a G -ambit. Among these, the universal minimal flow and extreme amenability are especially important for the later chapters. The universal minimal flow serves as a canonical dynamical invariant of the group, while extreme amenability expresses the strongest possible fixed-point property for continuous actions on compact spaces.

Our treatment will be limited to the parts of topological dynamics that are needed for the rest of the thesis. In particular, we do not attempt to develop the general theory in full. Instead, we focus on the basic definitions, the existence of minimal subflows, the existence and uniqueness of the universal minimal flow, and the notion of the greatest ambit. These concepts will later be related to structural Ramsey theory and to ordered expansions of Fraïssé structures, leading to concrete descriptions of universal minimal flows for automorphism groups of Fraïssé limits.

4.1 Flows, invariant sets, and subflows

We begin with the basic objects of topological dynamics.

Definition 4.1.1. Let G be a topological group. A G -flow is a compact Hausdorff space X together with a continuous action

$$G \times X \rightarrow X, \quad (g, x) \mapsto g \cdot x.$$

Thus a G -flow is simply a continuous action of G on a compact Hausdorff space. In the sequel, whenever X is a G -flow and $x \in X$, we write

$$G \cdot x = \{g \cdot x : g \in G\}$$

for the orbit of x .

Definition 4.1.2. Let X be a G -flow.

1. A subset $Y \subseteq X$ is called G -invariant if

$$g \cdot Y \subseteq Y \quad \text{for all } g \in G.$$

2. A *subflow* of X is a nonempty closed G -invariant subset of X .

If $Y \subseteq X$ is G -invariant, then in fact $g \cdot Y = Y$ for every $g \in G$, since

$$Y = g^{-1} \cdot (g \cdot Y) \subseteq g^{-1} \cdot Y.$$

Applying g to both sides gives $g \cdot Y = Y$.

The simplest source of subflows is given by orbit closures.

Proposition 4.1.3. Let X be a G -flow and let $x \in X$. Then the closure $\overline{G \cdot x}$ of the orbit of x is a subflow of X .

Proof. The set $\overline{G \cdot x}$ is nonempty and closed. It remains to show that it is G -invariant.

Fix $g \in G$. Since the map

$$X \rightarrow X, \quad y \mapsto g \cdot y$$

is continuous, we have

$$g \cdot \overline{G \cdot x} \subseteq \overline{g \cdot (G \cdot x)}.$$

Now

$$g \cdot (G \cdot x) = \{g \cdot (h \cdot x) : h \in G\} = \{(gh) \cdot x : h \in G\} = G \cdot x,$$

so

$$g \cdot \overline{G \cdot x} \subseteq \overline{G \cdot x}.$$

Hence $\overline{G \cdot x}$ is G -invariant, and therefore a subflow of X . ◻

We shall repeatedly use the fact that closed invariant subsets of a G -flow are again G -flows by restriction of the action. In particular, the study of a flow naturally leads to the study of its subflows. The most important of these are the minimal subflows, to which we now turn.

4.2 Minimal flows

Minimality is the basic irreducibility notion in topological dynamics.

Definition 4.2.1. A G -flow X is said to be *minimal* if it has no proper subflow; that is, if the only nonempty closed G -invariant subset of X is X itself.

Thus a minimal flow is one in which the action cannot be restricted to a smaller nontrivial closed invariant set. The following characterization is fundamental.

Proposition 4.2.2. *Let X be a G -flow. Then the following are equivalent:*

1. X is minimal.
2. For every $x \in X$, the orbit $G \cdot x$ is dense in X .

Proof. Assume first that X is minimal, and let $x \in X$. By Proposition 4.1.3, the orbit closure $\overline{G \cdot x}$ is a subflow of X . Since X is minimal, it follows that

$$\overline{G \cdot x} = X.$$

Hence $G \cdot x$ is dense in X .

Conversely, assume that every orbit is dense in X , and let $Y \subseteq X$ be a nonempty closed G -invariant subset. Choose $y \in Y$. Since Y is G -invariant, we have

$$G \cdot y \subseteq Y,$$

and since Y is closed,

$$\overline{G \cdot y} \subseteq Y.$$

By assumption, $\overline{G \cdot y} = X$, so $X \subseteq Y$. Therefore $Y = X$, and hence X is minimal. $\dashv$

In particular, every point of a minimal flow has dense orbit. Minimality therefore expresses a strong form of topological transitivity.

The next result shows that minimal flows always exist inside arbitrary flows.

Proposition 4.2.3. *Every G -flow contains a minimal subflow.*

Proof. Let X be a G -flow, and let

$$\mathcal{S} = \{Y \subseteq X : Y \neq \emptyset, Y \text{ is closed and } G\text{-invariant}\}.$$

Since $X \in \mathcal{S}$, the set $\mathcal{S}$ is nonempty. We partially order $\mathcal{S}$ by reverse inclusion:

$$Y_1 \preceq Y_2 \iff Y_1 \supseteq Y_2.$$

Let $\mathcal{C} \subseteq \mathcal{S}$ be a chain. We claim that $\mathcal{C}$ has an upper bound in $\mathcal{S}$. Put

$$Y = \bigcap_{Z \in \mathcal{C}} Z.$$

Each $Z \in \mathcal{C}$ is closed, so Y is closed; and since each Z is G -invariant, so is Y .

It remains to show that $Y \neq \emptyset$. Since X is compact and the members of $\mathcal{C}$ are closed, it suffices to show that the family $\mathcal{C}$ has the finite intersection property. But this is immediate, because $\mathcal{C}$ is a chain under reverse inclusion: given finitely many sets $Z_1, \dots, Z_n \in \mathcal{C}$, one of them is contained in all the others, and hence

$$Z_1 \cap \dots \cap Z_n \neq \emptyset.$$

Thus $Y \in \mathcal{S}$, and Y is an upper bound for $\mathcal{C}$.

By Zorn's lemma, $\mathcal{S}$ has a maximal element with respect to $\preceq$, that is, a minimal element with respect to inclusion. Let $M \in \mathcal{S}$ be such an element. Then M is a nonempty closed G -invariant subset of X , and it contains no proper nonempty closed G -invariant subset. Hence M is a minimal subflow of X . $\dashv$

Minimal subflows will play a central role in what follows. In particular, they allow one to define a universal minimal flow, which is the basic dynamical invariant attached to a topological group.

4.3 Universal minimal flows and extreme amenability

To compare G -flows, one uses morphisms that respect the action.

Definition 4.3.1. Let X and Y be G -flows. A *homomorphism* from X to Y is a continuous map

$$\pi: X \rightarrow Y$$

such that

$$\pi(g \cdot x) = g \cdot \pi(x) \quad \text{for all } g \in G, x \in X.$$

Homomorphisms onto minimal flows are automatically surjective.

Proposition 4.3.2. *Let X and Y be G -flows, and suppose that Y is minimal. If*

$$\pi: X \rightarrow Y$$

is a homomorphism, then π is surjective.

Proof. The image $\pi(X)$ is compact, hence closed in Y , and it is nonempty. Moreover, since π is a G -map, the set $\pi(X)$ is G -invariant. Thus $\pi(X)$ is a nonempty closed G -invariant subset of Y . Since Y is minimal, it follows that $\pi(X) = Y$. $\dashv$

This leads to the basic universal object in topological dynamics.

Definition 4.3.3. A G -flow M is called a *universal minimal flow* of G if

1. M is minimal, and
2. for every minimal G -flow X , there exists a homomorphism

$$\pi: M \rightarrow X.$$

By Proposition 4.3.2, the map π in the definition is necessarily surjective.

Theorem 4.3.4. *For every topological group G , there exists a universal minimal flow. It is unique up to isomorphism.*

Sketch of proof. The existence is obtained by a standard product argument. Choose one representative from each isomorphism class of minimal G -flows, say $\{X_i : i \in I\}$, and consider the product

$$X = \prod_{i \in I} X_i$$

with the coordinatewise action of G . Since each X_i is compact Hausdorff, so is X , and hence X is a G -flow. By Proposition 4.2.3, X contains a minimal subflow M .

For each $i \in I$, let

$$\pi_i: M \rightarrow X_i$$

be the restriction of the i -th coordinate projection. Then π_i is a homomorphism. Since X_i is minimal, Proposition 4.3.2 shows that π_i is surjective. Therefore M maps onto every minimal G -flow, so M is universal.

The uniqueness up to isomorphism is standard. We therefore write $M(G)$ for the universal minimal flow of G . $\dashv$

The universal minimal flow is a canonical dynamical invariant of the group. It is the largest minimal G -flow in the sense that every other minimal G -flow is its factor.

We now introduce the fixed-point property that will later be related to structural Ramsey theory.

Definition 4.3.5. Let X be a G -flow. A point $x \in X$ is called a *fixed point* if

$$g \cdot x = x \quad \text{for all } g \in G.$$

The topological group G is said to be *extremely amenable* if every G -flow has a fixed point.

Since every G -flow contains a minimal subflow, extreme amenability may equivalently be expressed by saying that every minimal G -flow is a singleton. In terms of the universal minimal flow, this becomes particularly simple.

Proposition 4.3.6. *A topological group G is extremely amenable if and only if $M(G)$ is a singleton.*

Proof. Assume first that G is extremely amenable. Then the G -flow $M(G)$ has a fixed point x . Since $\{x\}$ is a nonempty closed G -invariant subset of $M(G)$ and $M(G)$ is minimal, it follows that

$$M(G) = \{x\}.$$

Conversely, assume that $M(G)$ is a singleton, and let X be any G -flow. By Proposition 4.2.3, X contains a minimal subflow Y . Since $M(G)$ is universal, there is a surjective homomorphism

$$\pi: M(G) \rightarrow Y.$$

But $M(G)$ has only one point, so Y also has only one point. Hence Y is a fixed point of X . Therefore every G -flow has a fixed point, and G is extremely amenable. $\dashv$

Extreme amenability is one of the central dynamical properties studied in this thesis. In later chapters it will be characterized, for automorphism groups of Fraïssé limits, by Ramsey properties of suitable classes of finite structures.

4.4 Ambits and the greatest ambit

A distinguished role is played by flows generated by a single point with dense orbit.

Definition 4.4.1. A G -ambit is a pair (X, x_0) , where X is a G -flow and $x_0 \in X$ is such that the orbit

$$G \cdot x_0$$

is dense in X .

Thus a G -ambit is a point-transitive G -flow together with a specified transitive point. The distinguished point is part of the data.

Definition 4.4.2. Let (X, x_0) and (Y, y_0) be G -ambits. A *homomorphism* from (X, x_0) to (Y, y_0) is a homomorphism of G -flows

$$\pi: X \rightarrow Y$$

such that

$$\pi(x_0) = y_0.$$

Because the distinguished point has dense orbit, a homomorphism of ambits is determined uniquely by its value at that point.

Proposition 4.4.3. *Let (X, x_0) and (Y, y_0) be G -ambits. Then there is at most one homomorphism*

$$\pi: (X, x_0) \rightarrow (Y, y_0).$$

Proof. Suppose that

$$\pi_1, \pi_2: X \rightarrow Y$$

are homomorphisms of ambits. For every $g \in G$, we have

$$\pi_1(g \cdot x_0) = g \cdot \pi_1(x_0) = g \cdot y_0$$

and similarly

$$\pi_2(g \cdot x_0) = g \cdot \pi_2(x_0) = g \cdot y_0.$$

Hence π_1 and π_2 agree on the orbit $G \cdot x_0$. Since $G \cdot x_0$ is dense in X and both maps are continuous, it follows that $\pi_1 = \pi_2$. $\dashv$

The category of G -ambits has a universal object.

Definition 4.4.4. A G -ambit $(S(G), s_0)$ is called the *greatest ambit* of G if for every G -ambit (X, x_0) there exists a homomorphism

$$\pi: (S(G), s_0) \rightarrow (X, x_0).$$

By Proposition 4.4.3, such a homomorphism is automatically unique whenever it exists.

Theorem 4.4.5. *For every topological group G , there exists a greatest ambit. It is unique up to isomorphism.*

Comment. The existence and uniqueness of the greatest ambit are standard. One may construct it by a universal compactness argument, and in the general theory it also arises canonically from a compactification of G . Since we will not need the construction itself in the sequel, we record only the statement. $\dashv$

The greatest ambit is useful because it contains, in a universal form, all point-transitive G -flows. In particular, one may recover the universal minimal flow by passing to a minimal subflow of the greatest ambit.

The notions introduced in this chapter provide the dynamical framework for the rest of the thesis. In particular, the universal minimal flow and the property of extreme amenability attach to a topological group canonical dynamical invariants and fixed-point properties. In the chapters that follow, these concepts will be related to structural Ramsey theory and to ordered expansions of Fraïssé classes, leading to concrete descriptions of universal minimal flows for automorphism groups of Fraïssé limits.

Chapter 5

Extreme Amenability and the Ramsey Property

The aim of this chapter is to connect the dynamical notion of extreme amenability introduced in the previous chapter with the Ramsey property for classes of finite structures. This is the first decisive step in the Kechris–Pestov–Todorcevic correspondence. For closed subgroups of S_∞ , the topology is controlled by stabilizers of finite sets, and this makes it possible to reformulate a fixed-point property for flows in terms of finite colorings of finite configurations. The result is a bridge from topological dynamics to structural Ramsey theory.

We begin by recalling the basic notions of structural Ramsey theory and some standard examples. We then prove a characterization of extremely amenable closed subgroups of S_∞ in terms of a Ramsey property for the corresponding finite types. Finally, we translate this characterization into Fraïssé-theoretic language and obtain the fundamental criterion that if F is the Fraïssé limit of a Fraïssé order class $\mathcal{K}$, then $\text{Aut}(F)$ is extremely amenable if and only if $\mathcal{K}$ has the Ramsey property. This criterion will be the main tool in the next chapter, where we construct concrete examples of extremely amenable automorphism groups.

5.1 Structural Ramsey theory

We begin by recalling the basic notions of structural Ramsey theory that will be used in the sequel. The point of this section is to formulate the combinatorial property that later characterizes extreme amenability for automorphism groups of Fraïssé limits.

Let A and B be L -structures. We write

$$A \leq B$$

if A embeds into B . If $A \leq B$, we let

$$\binom{B}{A} = \{A_0 \subseteq B : A_0 \cong A\},$$

the set of all substructures of B isomorphic to A .

If $A \leq B \leq C$ and $k \geq 2$, we write

$$C \rightarrow (B)_k^A$$

to mean that for every coloring

$$c: \binom{C}{A} \rightarrow \{1, \dots, k\},$$

there is some $B_0 \in \binom{C}{B}$ such that

$$c \upharpoonright \binom{B_0}{A}$$

is constant. In this situation we say that B_0 is *homogeneous* for the coloring c .

Definition 5.1.1. Let $\mathcal{K}$ be a class of finite L -structures. We say that $\mathcal{K}$ has the *Ramsey property* if $\mathcal{K}$ is hereditary and for every $A, B \in \mathcal{K}$ with $A \leq B$ and every integer $k \geq 2$, there exists $C \in \mathcal{K}$ such that

$$C \rightarrow (B)_k^A.$$

As usual, it is enough to verify the Ramsey property for two colors.

Proposition 5.1.2. *A hereditary class $\mathcal{K}$ of finite structures has the Ramsey property if and only if for every $A, B \in \mathcal{K}$ with $A \leq B$, there exists $C \in \mathcal{K}$ such that*

$$C \rightarrow (B)_2^A.$$

Proof. Only one implication needs proof. Assume the two-color condition holds. We show by induction on $k \geq 2$ that for all $A \leq B$ in $\mathcal{K}$ there exists $C \in \mathcal{K}$ with

$$C \rightarrow (B)_k^A.$$

The case $k = 2$ is our assumption. Suppose the statement has been proved for $k - 1$, where $k \geq 3$, and fix $A \leq B$ in $\mathcal{K}$. Choose $D \in \mathcal{K}$ such that

$$D \rightarrow (B)_{k-1}^A,$$

and then choose $C \in \mathcal{K}$ such that

$$C \rightarrow (D)_2^A.$$

Now let

$$c: \binom{C}{A} \rightarrow \{1, \dots, k\}$$

be a k -coloring. Define a two-coloring

$$c': \binom{C}{A} \rightarrow \{0, 1\}$$

by

$$c'(A_0) = \begin{cases} 0, & c(A_0) = k, \\ 1, & c(A_0) \in \{1, \dots, k-1\}. \end{cases}$$

Since $C \rightarrow (D)_2^A$, there is $D_0 \in \binom{C}{D}$ homogeneous for c' . If c' is constantly 0 on $\binom{D_0}{A}$, then every copy of A in D_0 has color k , so D_0 contains a homogeneous copy of B . If c' is constantly 1, then the original coloring c uses only the colors $1, \dots, k-1$ on $\binom{D_0}{A}$. Since $D_0 \cong D$ and $D \rightarrow (B)_{k-1}^A$, there is a copy of B inside D_0 homogeneous for c .
 Thus $C \rightarrow (B)_k^A$, as required. $\dashv$

We now record several standard examples.

Example 5.1.3. The class $\mathcal{LO}$ of finite linear orders has the Ramsey property.

Proof. Two finite linear orders are isomorphic if and only if they have the same cardinality. Thus, for finite linear orders A , B , and C , the set

$$\binom{C}{A}$$

may be identified with the family of $|A|$ -element subsets of the underlying set of C . Under this identification, the statement

$$C \rightarrow (B)_k^A$$

is exactly the classical finite Ramsey theorem. $\dashv$

Example 5.1.4. Let $L = \{<, E\}$, where $<$ is a binary relation symbol and E is a binary relation symbol interpreted as a symmetric irreflexive graph relation. Then the class $\mathcal{OG}$ of finite ordered graphs has the Ramsey property.

This example matters because it shows a basic phenomenon that will reappear throughout the thesis: adding a linear order can restore the Ramsey property to a class that does not have it in the unordered form.

Here $\mathcal{OG}$ consists of finite structures $(A, <^A, E^A)$ such that $<^A$ is a linear order on A and E^A is a graph relation on A . So a copy of A inside some ordered graph C is not just a graph copy: it must preserve both the graph relation and the order. In other words, when we color elements of

$$\binom{C}{A},$$

we are coloring ordered copies of A inside C .

Why is the order important? The class of finite graphs by itself does not have the Ramsey property. The reason is that finite graphs usually have nontrivial automorphisms and too much symmetry. Once a linear order is added, each finite structure becomes rigid, and the combinatorics become much better behaved. The theorem of Nešetřil and Rödl says that this extra order is enough: for every ordered graphs $A \leq B$ and every $k \geq 2$, there is an ordered graph C such that every coloring of the ordered copies of A in C contains a copy of B on which all copies of A have the same color.

So the slogan is: unordered graphs are not Ramsey, but ordered graphs are. This is one of the main reasons ordered expansions play such a central role in the Kechris–

Pestov–Todorćevic correspondence.

This is the theorem of Nešetřil and Rödl [56, 58].

Example 5.1.5. Let F be a finite field, and let $\mathcal{V}_F$ be the class of finite-dimensional vector spaces over F , regarded as structures in a language coding addition and scalar multiplication. Then $\mathcal{V}_F$ has the Ramsey property.

This example is algebraic rather than relational, and it is worth pausing to see what the Ramsey statement means concretely.

If $U \leq V \leq W$ are finite-dimensional vector spaces over F , then

$$\binom{W}{U}$$

is the set of subspaces of W that are isomorphic to U . Since finite-dimensional vector spaces over a fixed field are classified up to isomorphism by dimension, this is just the set of subspaces of W of dimension $\dim U$. So a coloring

$$c: \binom{W}{U} \rightarrow \{1, \dots, k\}$$

is simply a coloring of the $\dim U$ -dimensional subspaces of W .

Then the statement

$$W \rightarrow (V)_k^U$$

means that there is a copy $V_0 \leq W$ of V such that all $\dim U$ -dimensional subspaces of V_0 receive the same color. In dimension language: for every pair of integers $d \leq m$ and every k , there exists n such that whenever the d -dimensional subspaces of F^n are colored with k colors, one can find an m -dimensional subspace all of whose d -dimensional subspaces have the same color.

So this is a vector-space analogue of the classical Ramsey theorem. Instead of coloring subsets of a finite set, we color subspaces of a finite vector space. The fact that such a theorem holds is highly nontrivial; it is the Graham–Leeb–Rothschild theorem. It is a good example to keep in mind because it shows that the Ramsey phenomenon is not tied only to ordered relational structures.

This is the Graham–Leeb–Rothschild theorem [32].

Example 5.1.6. Let $\mathcal{BA}$ be the class of finite Boolean algebras in the usual algebraic language. Then $\mathcal{BA}$ has the Ramsey property.

This example is a little less transparent at first sight, because Boolean algebras are algebraic objects rather than combinatorial ones. For finite Boolean algebras, however, there is a very concrete picture.

Every finite Boolean algebra is isomorphic to the power-set algebra $\mathcal{P}(n)$ for some

finite set $n = \{1, \dots, n\}$. Equivalently, every finite Boolean algebra is generated by its atoms, and the number of atoms determines it up to isomorphism. Thus copies of a finite Boolean algebra A inside a larger finite Boolean algebra C can be understood in terms of partitions of the atoms of C into blocks corresponding to the atoms of A .

From this point of view, the Ramsey statement for $\mathcal{BA}$ says that if one colors all copies of a smaller finite Boolean algebra inside a sufficiently large one, then one can find a copy of a prescribed larger Boolean algebra inside which all copies of the smaller one have the same color. This turns out to be equivalent to the Dual Ramsey Theorem.

The word “dual” appears because the underlying combinatorics are naturally phrased in terms of partitions or quotient-type data rather than subsets. Ordinary Ramsey theory colors subsets; dual Ramsey theory colors partitions. Finite Boolean algebras provide an algebraic language in which that dual phenomenon is expressed very naturally.

This is an equivalent formulation of the Dual Ramsey Theorem of Graham and Rothschild; see, for example, [51, §4.13].

Definition 5.1.7. A structure A is said to be *rigid* if every automorphism of A is the identity. Equivalently,

$$\text{Aut}(A) = \{\text{id}_A\}.$$

A class of finite structures is called rigid if every member of the class is rigid.

This notion is important here because finite linearly ordered structures are automatically rigid: an automorphism must preserve the linear order, and therefore it must fix the least element, then the next one, and so on, hence every element.

For our purposes, ordered classes are especially important, since finite linearly ordered structures are rigid. This rigidity ties the Ramsey property to amalgamation.

Proposition 5.1.8. *Let $\mathcal{K}$ be a hereditary class of finite rigid L -structures. If $\mathcal{K}$ has the joint embedding property and the Ramsey property, then $\mathcal{K}$ has the amalgamation property.*

Proof. Let $A, B, C \in \mathcal{K}$ and let

$$f: A \rightarrow B, \quad g: A \rightarrow C$$

be embeddings. Since $\mathcal{K}$ has the joint embedding property, there exists $E \in \mathcal{K}$ into which both B and C embed. Fix embeddings

$$u: B \rightarrow E, \quad v: C \rightarrow E.$$

By the Ramsey property, there is $D \in \mathcal{K}$ such that

$$D \rightarrow (E)_4^A.$$

We define a coloring of $\binom{D}{A}$ with values in the four-element set $\mathcal{P}(\{B, C\})$. For $A_0 \in \binom{D}{A}$, put

$$B \in c(A_0) \iff \text{there exists an embedding } r: B \rightarrow D \text{ with } r(f(A)) = A_0,$$

and define similarly

$$C \in c(A_0) \iff \text{there exists an embedding } s: C \rightarrow D \text{ with } s(g(A)) = A_0.$$

Choose $E_0 \in \binom{D}{E}$ homogeneous for this coloring. Since $E_0 \cong E$ contains copies of both B and C , there is some copy of A inside E_0 whose color contains B , and also some copy of A inside E_0 whose color contains C . Because E_0 is homogeneous, the common color of all members of

$$\binom{E_0}{A}$$

must therefore be exactly $\{B, C\}$.

Now choose any $A_0 \in \binom{E_0}{A}$. Since $c(A_0) = \{B, C\}$, there are embeddings

$$r: B \rightarrow D, \quad s: C \rightarrow D$$

such that

$$r(f(A)) = A_0 = s(g(A)).$$

Hence both $r \circ f$ and $s \circ g$ are isomorphisms from A onto A_0 . Since A is rigid, this isomorphism is unique, so

$$r \circ f = s \circ g.$$

Thus D , together with the embeddings r and s , is an amalgam of B and C over A . Therefore $\mathcal{K}$ has the amalgamation property. $\dashv$

No. This proposition does not explain the vector-spaces example.

The proposition says:

$$\text{Ramsey property} + \text{rigidity} + \text{JEP} \implies \text{AP}.$$

It is a way of deducing amalgamation once one already knows the Ramsey property, provided the structures are rigid.

But finite-dimensional vector spaces over a finite field are *not* rigid. Even a one-dimensional space has nontrivial automorphisms, and in higher dimensions there are many linear automorphisms. So the hypothesis of rigidity fails, and Proposition 5.1.8 does not apply to $\mathcal{V}_F$.

Also, the point of the vector-spaces example is different: it is there to show that there are important nontrivial classes with the Ramsey property outside the ordered relational setting. Its proof comes from the Graham–Leeb–Rothschild theorem, not from the proposition above.

What *is* true is that the class of finite-dimensional vector spaces has amalgamation for a completely separate algebraic reason: if

$$U \hookrightarrow V, \quad U \hookrightarrow W,$$

then one can amalgamate V and W over U by taking a pushout, or more concretely by identifying the two copies of U inside a larger vector space. So AP is easy there, but it does not come from rigidity, and it does not explain the Ramsey property.

In particular, if $\mathcal{K}$ is a non-empty countable hereditary class of finite rigid structures, has the joint embedding property, contains structures of arbitrarily large cardinality, and has the Ramsey property, then $\mathcal{K}$ is a Fraïssé class.

The importance of these observations for us is that many of the classes arising in the Kechris–Pestov–Todorcevic correspondence are order classes. For such classes, rigidity is automatic, and Proposition 5.1.8 shows that once the Ramsey property is established, the amalgamation property follows under mild additional hypotheses.

5.2 A characterization of extreme amenability for closed subgroups of S_∞

We now reformulate extreme amenability for closed subgroups of S_∞ in combinatorial terms. The basic idea is that for such groups, open subgroups are determined by finite information, and this allows one to express a fixed-point property for flows as a Ramsey-type statement about finite configurations.

We begin with a general criterion for the existence of a fixed point.

Lemma 5.2.1. *Let G be a topological group and let X be a G -flow. Then the following are equivalent.*

- (i) X has a fixed point.
- (ii) For every $n \geq 1$, every continuous map $f: X \rightarrow \mathbb{R}^n$, every $\epsilon > 0$, and every finite set $F \subseteq G$, there exists $x \in X$ such that

$$|f(x) - f(g \cdot x)| \leq \epsilon \quad \text{for all } g \in F,$$

where $|\cdot|$ denotes the Euclidean norm.

Proof. The implication (i) $\Rightarrow$ (ii) is immediate.

For (ii) $\Rightarrow$ (i), define, for each continuous $f: X \rightarrow \mathbb{R}^n$, each $\epsilon > 0$, and each finite $F \subseteq G$,

$$A_{f,\epsilon,F} = \{x \in X : |f(x) - f(g \cdot x)| \leq \epsilon \text{ for all } g \in F\}.$$

Each set $A_{f,\epsilon,F}$ is closed.

The argument is just that $A_{f,\epsilon,F}$ is a finite intersection of inverse images of closed sets under continuous maps.

For each fixed $g \in F$, define

$$\varphi_g(x) = |f(x) - f(g \cdot x)|.$$

Since the action map $x \mapsto g \cdot x$ is continuous and f is continuous, the map $\varphi_g: X \rightarrow \mathbb{R}$

is continuous. Then

$$\{x \in X : \varphi_g(x) \leq \epsilon\} = \varphi_g^{-1}((-\infty, \epsilon])$$

is closed, because $(-\infty, \epsilon]$ is closed in $\mathbb{R}$.

Therefore

$$A_{f,\epsilon,F} = \bigcap_{g \in F} \{x \in X : \varphi_g(x) \leq \epsilon\}$$

is closed as a finite intersection of closed sets.

We claim that the family

$$\{A_{f,\epsilon,F}\}$$

has the finite intersection property. Indeed, given finitely many triples (f_j, ϵ_j, F_j) , $j = 1, \dots, m$, let

$$\bar{F} = F_1 \cup \dots \cup F_m, \quad \bar{\epsilon} = \min\{\epsilon_1, \dots, \epsilon_m\},$$

and define

$$\bar{f} = (f_1, \dots, f_m): X \rightarrow \mathbb{R}^{n_1 + \dots + n_m},$$

where $f_j: X \rightarrow \mathbb{R}^{n_j}$. Then

$$A_{\bar{f}, \bar{\epsilon}, \bar{F}} \subseteq \bigcap_{j=1}^m A_{f_j, \epsilon_j, F_j},$$

and $A_{\bar{f}, \bar{\epsilon}, \bar{F}}$ is non-empty by (ii). Since X is compact, it follows that

$$\bigcap_{f,\epsilon,F} A_{f,\epsilon,F} \neq \emptyset.$$

Choose

$$x \in \bigcap_{f,\epsilon,F} A_{f,\epsilon,F}.$$

We show that x is fixed. Suppose not. Then there exists $g \in G$ such that $g \cdot x \neq x$. By complete regularity of the compact Hausdorff space X , there is a continuous function $f: X \rightarrow \mathbb{R}$ such that

$$f(x) = 0, \quad f(g \cdot x) = 1.$$

The point is simply that compact Hausdorff spaces are completely regular.

Since $x \neq g \cdot x$, the singleton $\{g \cdot x\}$ is closed. Complete regularity says that if $p \in X$ and $C \subseteq X$ is closed with $p \notin C$, then there exists a continuous map

$$f: X \rightarrow [0, 1]$$

such that

$$f(p) = 0 \quad \text{and} \quad f \upharpoonright C \equiv 1.$$

Apply this with

$$p = x, \quad C = \{g \cdot x\}.$$

Then we get a continuous function f with

$$f(x) = 0, \quad f(g \cdot x) = 1.$$

If you prefer, one can also appeal to Urysohn's lemma, since compact Hausdorff spaces are normal.

But then

$$x \notin A_{f, 1/2, \{g\}},$$

a contradiction. Hence x is a fixed point. $\dashv$

We now specialize to closed subgroups of S_∞ .

Proposition 5.2.2. *Let $G \leq S_\infty$ be a closed subgroup. Then the following are equivalent.*

- (i) G is extremely amenable.
- (ii) For every open subgroup $V \leq G$, every finite coloring

$$c: G/V \rightarrow \{1, \dots, k\},$$

and every finite set $A \subseteq G/V$, there exist $g \in G$ and $i \in \{1, \dots, k\}$ such that

$$c(g \cdot a) = i \quad \text{for all } a \in A,$$

where G acts on G/V by left translation.

Proof. Assume first that G is extremely amenable. Let $V \leq G$ be open, let

$$c: G/V \rightarrow \{1, \dots, k\}$$

be a coloring, and let $A \subseteq G/V$ be finite. Consider the compact space

$$Y = \{1, \dots, k\}^{G/V}$$

with the product topology, endowed with the shift action

$$(g \cdot p)(x) = p(g^{-1} \cdot x) \quad (g \in G, p \in Y, x \in G/V).$$

Let $X = \overline{G \cdot c} \subseteq Y$. Since G is extremely amenable, X has a fixed point γ . Because the action of G on G/V is transitive,

Yes. You already defined transitivity in Chapter 2.

There you said that an action is *transitive* if it has exactly one orbit. Equivalently, an action of G on a set Ω is transitive if for every $x, y \in \Omega$ there exists $g \in G$ such that

$$g \cdot x = y.$$

The action of G on G/V by left translation is transitive for a very simple reason: given any two cosets

$$aV, bV \in G/V,$$

take

$$g = ba^{-1}.$$

Then

$$g \cdot (aV) = (ba^{-1})aV = bV.$$

So every coset can be moved to every other coset by some element of G .

every fixed point of Y is constant.

This means the following. A point of

$$Y = \{1, \dots, k\}^{G/V}$$

is just a function

$$p: G/V \rightarrow \{1, \dots, k\}.$$

Saying that p is a fixed point means

$$g \cdot p = p \quad \text{for every } g \in G.$$

By the definition of the action, this says

$$p(g^{-1} \cdot x) = p(x) \quad \text{for all } g \in G, x \in G/V.$$

Now take any two points $x, y \in G/V$. Since the action on G/V is transitive, there exists $g \in G$ such that

$$g^{-1} \cdot x = y.$$

Plugging this into the fixed-point identity gives

$$p(y) = p(g^{-1} \cdot x) = p(x).$$

Since this holds for all x, y , the function p has the same value everywhere. In other words, p is constant.

Thus there is $i \in \{1, \dots, k\}$ such that

$$\gamma(x) = i \quad \text{for all } x \in G/V.$$

Since $\gamma \in \overline{G \cdot c}$, there is $g \in G$ such that

$$(g^{-1} \cdot c)(a) = \gamma(a) = i \quad \text{for all } a \in A.$$

why? where did A come from all the sudden?

Equivalently,

$$c(g \cdot a) = i \quad \text{for all } a \in A.$$

Conversely, assume (ii), and let X be a G -flow. By Lemma 5.2.1, it is enough to verify condition (ii) of that lemma. So fix a continuous map

$$f: X \rightarrow \mathbb{R}^n,$$

an $\epsilon > 0$, and a finite set $F \subseteq G$.

By continuity of the action and compactness of X , there exists an open neighbourhood U of the identity in G such that

$$|f(x) - f(u \cdot x)| \leq \epsilon/3 \quad \text{for all } x \in X, u \in U.$$

The clean way to see this is to look at the map

$$\Phi: G \times X \rightarrow \mathbb{R}, \quad \Phi(u, x) = |f(x) - f(u \cdot x)|.$$

This map is continuous, because the action

$$G \times X \rightarrow X, \quad (u, x) \mapsto u \cdot x$$

is continuous, f is continuous, and the norm is continuous.

For every $x \in X$ we have

$$\Phi(1_G, x) = |f(x) - f(x)| = 0.$$

Since Φ is continuous at $(1_G, x)$, there exist an open neighbourhood U_x of 1_G in G and an open neighbourhood O_x of x in X such that

$$\Phi(u, y) < \epsilon/3 \quad \text{whenever } u \in U_x, y \in O_x.$$

The sets O_x cover X . Since X is compact, choose finitely many points $x_1, \dots, x_m$ such that

$$X = O_{x_1} \cup \dots \cup O_{x_m}.$$

Now let

$$U = U_{x_1} \cap \dots \cap U_{x_m}.$$

Then U is still an open neighbourhood of 1_G .

If $u \in U$ and $y \in X$, choose j with $y \in O_{x_j}$. Since $u \in U \subseteq U_{x_j}$, we get

$$|f(y) - f(u \cdot y)| = \Phi(u, y) < \epsilon/3.$$

This is exactly the required uniform estimate.

Since G is a closed subgroup of S_∞ , it has a neighbourhood basis at the identity consisting of open subgroups.

This comes from the pointwise convergence topology on S_∞ .

In S_∞ , a basic open neighbourhood of the identity is obtained by fixing finitely many natural numbers. Concretely, for a finite set $E \subseteq \mathbb{N}$, put

$$S_{\infty, (E)} = \{\pi \in S_\infty : \pi(n) = n \text{ for all } n \in E\}.$$

Each $S_{\infty, (E)}$ is an open subgroup of S_∞ , and these sets form a neighbourhood basis at the identity.

Since G is a subgroup of S_∞ with the subspace topology, the sets

$$G \cap S_{\infty, (E)}$$

form a neighbourhood basis at the identity in G . But

$$G \cap S_{\infty, (E)} = \{g \in G : g(n) = n \text{ for all } n \in E\} = G_{(E)}.$$

So G has a neighbourhood basis at the identity consisting of open subgroups, namely the pointwise stabilizers of finite sets.

Choose an open subgroup

$$V \leq U.$$

Partition the compact set $f(X) \subseteq \mathbb{R}^n$ into finitely many sets

$$A_1, \dots, A_k$$

of diameter at most $\epsilon/3$. Fix $x_0 \in X$ and define

$$U_i = \{g \in G : f(g \cdot x_0) \in A_i\}.$$

Let

$$W_i = VU_i = \{vg : v \in V, g \in U_i\}.$$

Each W_i is a union of right cosets of V . Equivalently, each W_i^{-1} is a union of left cosets of V . Since

$$G = U_1 \cup \dots \cup U_k,$$

we also have

$$G = W_1 \cup \dots \cup W_k.$$

The first covering is immediate from the definition of the sets U_i .

Take any $g \in G$. Then

$$g \cdot x_0 \in X,$$

so

$$f(g \cdot x_0) \in f(X).$$

Since

$$f(X) \subseteq A_1 \cup \dots \cup A_k,$$

there is some i such that

$$f(g \cdot x_0) \in A_i.$$

By definition of U_i , this means

$$g \in U_i.$$

So indeed

$$G = U_1 \cup \cdots \cup U_k.$$

For the second covering, note that

$$W_i = VU_i$$

contains U_i , because the identity element belongs to V . So

$$U_i \subseteq W_i \quad \text{for every } i.$$

Therefore

$$G = U_1 \cup \cdots \cup U_k \subseteq W_1 \cup \cdots \cup W_k \subseteq G.$$

Hence

$$G = W_1 \cup \cdots \cup W_k.$$

Define a coloring

$$c: G/V \rightarrow \{1, \dots, k\}$$

so that

$$c(gV) = i \implies g^{-1} \in W_i.$$

This is possible because the sets W_i^{-1} cover G and each is a union of left cosets of V .

Apply (ii) to the finite subset

$$A' = \{hV : h \in F \cup \{1_G\}\} \subseteq G/V.$$

There exist $g \in G$ and $i \in \{1, \dots, k\}$ such that

$$c(ghV) = i \quad \text{for all } h \in F \cup \{1_G\}.$$

By the definition of c , this means that

$$h^{-1}g^{-1} \in W_i = VU_i \quad \text{for all } h \in F \cup \{1_G\},$$

or equivalently,

$$hg \in U_i^{-1}V.$$

Replacing U_i by U_i^{-1} in the construction if necessary, we may assume that for each $h \in F \cup \{1_G\}$ there exists $v_h \in V$ such that

$$v_h hg \in U_i.$$

Set

$$x = g \cdot x_0.$$

Then for each $h \in F \cup \{1_G\}$ we have

$$f(v_h h \cdot x) = f(v_h hg \cdot x_0) \in A_i.$$

Since $v_h \in V \subseteq U$, it follows that

$$|f(h \cdot x) - f(v_h h \cdot x)| \leq \epsilon/3.$$

Hence $f(h \cdot x)$ lies in the $\epsilon/3$ -neighbourhood of A_i for every $h \in F \cup \{1_G\}$. As the diameter of A_i is at most $\epsilon/3$, we obtain

$$|f(x) - f(h \cdot x)| \leq \epsilon \quad \text{for all } h \in F.$$

Thus X has a fixed point, and G is extremely amenable. $\dashv$

For closed subgroups of S_∞ , it is enough to consider pointwise stabilizers of finite sets. For each non-empty finite $F \subseteq \mathbb{N}$, let

$$G_{(F)} = \{g \in G : g(n) = n \text{ for all } n \in F\}.$$

These subgroups form a neighbourhood basis at the identity.

We now recast the preceding proposition in terms of finite subsets of $\mathbb{N}$. The action of G on $\mathbb{N}$ induces an action on the set of all finite subsets of $\mathbb{N}$ by

$$g \cdot F = \{g(n) : n \in F\}.$$

For a finite non-empty set $F \subseteq \mathbb{N}$, let

$$G_F = \{g \in G : g \cdot F = F\}$$

be its setwise stabilizer. The orbit

$$G \cdot F$$

will be called the G -type of F . If ρ and σ are G -types, we write

$$\rho \leq \sigma$$

if some, equivalently every, member of σ contains some member of ρ .

We also need the logic action on the space of structures on $\mathbb{N}$. If L is a countable signature, let X_L denote the space of all L -structures with universe $\mathbb{N}$. When L is relational, X_L is compact, and the natural action of S_∞ on X_L is continuous.

Here is the standard proof.

Assume first that L is relational, say

$$L = \{R_i : i \in I\},$$

where each R_i has finite arity n_i , and I is countable.

An L -structure M on universe $\mathbb{N}$ is completely determined by the truth values of all statements

$$R_i(a_1, \dots, a_{n_i}), \quad a_1, \dots, a_{n_i} \in \mathbb{N}.$$

So we can identify M with the tuple of characteristic functions of the relations R_i^M . Equivalently,

$$X_L \cong \prod_{i \in I} 2^{\mathbb{N}^{n_i}},$$

where each factor $2^{\mathbb{N}^{n_i}}$ is the set of all subsets of $\mathbb{N}^{n_i}$, viewed as a product of copies of the discrete two-point space $2 = \{0, 1\}$.

Now each space 2 is compact, so by Tychonoff's theorem each product

$$2^{\mathbb{N}^{n_i}}$$

is compact, and therefore the countable product

$$\prod_{i \in I} 2^{\mathbb{N}^{n_i}}$$

is compact as well. Hence X_L is compact.

Concretely, a basic clopen set in X_L is obtained by prescribing finitely many atomic facts and negated atomic facts, for example

$$R_i(\bar{a}) \text{ holds,} \quad R_j(\bar{b}) \text{ fails,}$$

for finitely many tuples $\bar{a}, \bar{b}$.

The action of S_∞ is defined by transport of structure:

$$R_i^{g \cdot M}(\bar{a}) \iff R_i^M(g^{-1}\bar{a}).$$

In words, $g \cdot M$ is the structure obtained by relabelling the underlying set by g .

Why is the action continuous? It is enough to check that the preimage of a basic clopen cylinder is open. So fix a basic condition such as

$$R_i(\bar{a}) \text{ holds in } g \cdot M.$$

By the definition of the action, this means

$$R_i^M(g^{-1}\bar{a}) \text{ holds in } M.$$

Thus whether (g, M) lands in that cylinder depends only on finitely many values of g and on finitely many coordinates of M . Therefore around any given pair (g_0, M_0) one can force the same condition by requiring:

- g to agree with g_0 on finitely many points, and
- M to agree with M_0 on finitely many relation-values.

That is exactly an open condition in

$$S_\infty \times X_L.$$

So the action is continuous.

This is why the relational assumption is convenient: an L -structure is then encoded by a product of discrete two-point spaces, and compactness becomes immediate.

In the special case $L = \{<\}$, let $\text{LO} \subseteq X_L$ be the compact space of all linear orderings on

$\mathbb{N}$. We say that G *preserves an ordering* if the G -flow LO has a fixed point, that is, if there is a linear order on $\mathbb{N}$ invariant under every element of G .

What KPT does here is slightly broader than what is needed in your current draft.

The general logic action is important because it gives a very concrete source of compact G -flows: if one starts with a countable structure on $\mathbb{N}$ and looks at expansions of it in a larger language, the orbit closure of one such expansion inside an appropriate space X_{L^*} is a compact G -flow. Later in KPT, this is exactly how one realizes universal minimal flows coming from ordered expansions.

In the present section, however, the only part needed immediately is the special case

$$L = \{<\},$$

because right now you only want to formulate the notion that G preserves a linear order on $\mathbb{N}$. For that purpose it is enough to introduce

$$\text{LO} \subseteq X_{\{<\}}$$

and say that a fixed point of this flow is precisely a G -invariant linear order.

So nothing essential is being hidden. The omitted material is the more general philosophy: logic actions let one turn spaces of expansions into compact G -flows, and those flows later encode the universal minimal flow. If you want a fuller conceptual bridge, then it would be reasonable to add a short explanatory paragraph here saying exactly that.

Proposition 5.2.3. *Let $G \leq S_\infty$ be a closed subgroup. Then the following are equivalent.*

(i) G is extremely amenable.

(ii) (a) For every finite non-empty $F \subseteq \mathbb{N}$,

$$G_{(F)} = G_F.$$

(b) Whenever $\rho \leq \sigma$ are G -types and

$$c: \rho \rightarrow \{1, \dots, k\}$$

is a finite coloring, there exist $i \in \{1, \dots, k\}$ and $F \in \sigma$ such that

$$c(F') = i \quad \text{for every } F' \subseteq F \text{ with } F' \in \rho.$$

(iii) (a') G preserves an ordering.

(b) The condition in (ii)(b) holds.

Proof. Here is the proof in full detail.

First assume that

$$G \text{ is extremely amenable.}$$

We want to prove (iii).

Since LO is a compact G -flow, extreme amenability implies that LO has a fixed point.

But a fixed point of LO is exactly a linear order on $\mathbb{N}$ preserved by every element of G . So G preserves an ordering. This proves (iii)(a').

Now we prove (iii)(b). Let

$$\rho \leq \sigma$$

be G -types, and let

$$c: \rho \rightarrow \{1, \dots, k\}$$

be a coloring.

Choose some finite set

$$F' \in \rho,$$

and let

$$V = G_{(F')}.$$

We claim that in fact

$$G_{(F')} = G_{F'}.$$

Why? Because G preserves a linear order on $\mathbb{N}$. If $g \in G_{F'}$, then g maps the finite set F' onto itself. Since g also preserves the global G -invariant linear order on $\mathbb{N}$, it preserves the induced linear order on the finite set F' . But a permutation of a finite linearly ordered set that preserves the order must be the identity. Therefore g fixes each element of F' pointwise, so

$$g \in G_{(F')}.$$

The reverse inclusion

$$G_{(F')} \subseteq G_{F'}$$

is obvious, so equality holds.

Because of this equality, the coset space

$$G/V = G/G_{(F')}$$

can be identified with the orbit

$$G \cdot F' = \rho.$$

The identification is

$$gV \mapsto g \cdot F'.$$

This is well defined because if

$$g_1V = g_2V,$$

then

$$g_2^{-1}g_1 \in V = G_{F'},$$

so

$$g_1 \cdot F' = g_2 \cdot F'.$$

It is surjective by definition of the orbit, and injective because the stabilizer of F' is exactly $G_{F'} = V$.

Next choose some

$$F_0 \in \sigma.$$

Consider the finite set

$$A = \{F'_0 \subseteq F_0 : F'_0 \in \rho\} \subseteq \rho.$$

This is finite because F_0 is finite.

Now apply Proposition 5.2.2 to the open subgroup V , the coloring c on $G/V \cong \rho$, and the finite subset A . Since G is extremely amenable, that proposition gives some

$$g \in G \quad \text{and} \quad i \in \{1, \dots, k\}$$

such that

$$c(g \cdot F'_0) = i \quad \text{for all } F'_0 \in A.$$

Let

$$F = g \cdot F_0.$$

Since $F_0 \in \sigma$ and σ is a G -orbit, we have

$$F \in \sigma.$$

We now show that F has the desired monochromatic property. So let

$$E \subseteq F \quad \text{with} \quad E \in \rho.$$

Apply g^{-1} to both sides of the inclusion $E \subseteq F = g \cdot F_0$. Then

$$g^{-1} \cdot E \subseteq F_0.$$

Also, since ρ is a G -orbit and $E \in \rho$, we have

$$g^{-1} \cdot E \in \rho.$$

Therefore

$$g^{-1} \cdot E \in A.$$

By the choice of g and i ,

$$c(E) = c(g \cdot (g^{-1} \cdot E)) = i.$$

Since this holds for every $E \subseteq F$ with $E \in \rho$, condition (iii)(b) follows.

This proves

$$(i) \implies (iii).$$

Next we prove

$$(iii) \implies (ii).$$

Here condition (b) is literally the same in (ii) and (iii), so only (a) needs explanation. Assume that G preserves a linear order on $\mathbb{N}$, and let $F \subseteq \mathbb{N}$ be finite and non-empty. We want to show

$$G_{(F)} = G_F.$$

The inclusion

$$G_{(F)} \subseteq G_F$$

is automatic. Conversely, let

$$g \in G_F.$$

Then g maps F onto itself. Since g preserves the invariant linear order on $\mathbb{N}$, it also preserves the induced order on the finite set F . But an order-preserving permutation of a finite linearly ordered set must be the identity. Hence g fixes every element of F , so

$$g \in G_{(F)}.$$

Therefore

$$G_F = G_{(F)}.$$

Thus (ii)(a) holds, and so (iii) implies (ii).

Finally we prove

$$(ii) \implies (i).$$

By Proposition 5.2.2, it is enough to verify the coloring condition there for open subgroups. Since the pointwise stabilizers of finite sets form a neighbourhood basis at the identity, it is enough to check that condition for subgroups of the form

$$G_{(F)}.$$

So fix a finite non-empty

$$F \subseteq \mathbb{N},$$

a coloring

$$c: G/G_{(F)} \rightarrow \{1, \dots, k\},$$

and a finite subset

$$A \subseteq G/G_{(F)}.$$

By assumption (ii)(a),

$$G_{(F)} = G_F.$$

Therefore the coset space can be identified with the orbit of F :

$$G/G_F \cong G \cdot F = \rho.$$

Under this identification, the coloring c becomes a coloring of the G -type ρ .

Now look at the finite family A of members of ρ , and let

$$F_0 = \bigcup A.$$

Since A is finite and each member of A is finite, the set F_0 is finite. Let σ be the G -type of F_0 . Every member of A is a member of ρ contained in F_0 , so certainly

$$\rho \leq \sigma.$$

Apply assumption (ii)(b) to the coloring $c: \rho \rightarrow \{1, \dots, k\}$. Then there exist

$$i \in \{1, \dots, k\} \quad \text{and} \quad E \in \sigma$$

such that

$$c(E') = i$$

for every

$$E' \subseteq E \quad \text{with} \quad E' \in \rho.$$

Because E and F_0 lie in the same G -type σ , there exists

$$g \in G$$

such that

$$E = g \cdot F_0.$$

Now take any

$$a \in A.$$

Since $a \subseteq F_0$ and $a \in \rho$, applying g gives

$$g \cdot a \subseteq g \cdot F_0 = E \quad \text{and} \quad g \cdot a \in \rho.$$

Therefore the monochromaticity property of E implies

$$c(g \cdot a) = i.$$

Since this holds for every $a \in A$, the condition in Proposition 5.2.2 is satisfied.

Hence G is extremely amenable.

So all three conditions are equivalent.

Assume first that G is extremely amenable. Then G preserves an ordering because LO is a compact G -flow. This gives (iii)(a').

To prove (iii)(b), let $\rho \leq \sigma$ be G -types and let

$$c: \rho \rightarrow \{1, \dots, k\}$$

be a coloring. Choose $F' \in \rho$ and let

$$V = G_{(F')}.$$

Since G preserves an ordering, every finite setwise stabilizer equals the corresponding pointwise stabilizer, so

$$G_{(F')} = G_{F'}.$$

Hence

$$G/V \cong G \cdot F' = \rho.$$

Now fix $F_0 \in \sigma$ and consider the finite subset

$$A = \{F'_0 \subseteq F_0 : F'_0 \in \rho\} \subseteq \rho.$$

Applying Proposition 5.2.2, we obtain $g \in G$ and $i \in \{1, \dots, k\}$ such that

$$c(g \cdot F'_0) = i \quad \text{for all } F'_0 \in A.$$

Let

$$F = g \cdot F_0 \in \sigma.$$

If $F' \subseteq F$ and $F' \in \rho$, then

$$g^{-1} \cdot F' \subseteq F_0 \quad \text{and} \quad g^{-1} \cdot F' \in \rho,$$

so

$$c(F') = i.$$

Thus (iii)(b) holds.

The implication (iii) $\Rightarrow$ (ii) is immediate, since an invariant linear order on $\mathbb{N}$ forces every finite setwise stabilizer to equal the corresponding pointwise stabilizer.

Finally, assume (ii). We verify condition (ii) of Proposition 5.2.2 for open subgroups of the form $G_{(F)}$. By (ii)(a), these equal the setwise stabilizers G_F . Fix a coloring

$$c: G/G_F \rightarrow \{1, \dots, k\}$$

and a finite subset

$$A \subseteq G/G_F.$$

Identify

$$G/G_F \cong G \cdot F = \rho.$$

Let

$$F_0 = \bigcup A$$

and let σ be the G -type of F_0 . Then $\rho \leq \sigma$. By (ii)(b), there exist $i \in \{1, \dots, k\}$ and $E \in \sigma$ such that

$$c(F') = i$$

for every $F' \subseteq E$ with $F' \in \rho$. Since E and F_0 lie in the same G -type, we may write

$$E = g \cdot F_0$$

for some $g \in G$. Every element of A is a member of ρ contained in F_0 , so after applying g it becomes a member of ρ contained in E . Hence

$$c(g \cdot a) = i \quad \text{for all } a \in A.$$

By Proposition 5.2.2, G is extremely amenable. +

We now isolate the corresponding Ramsey property for the group itself.

Definition 5.2.4. Let $G \leq S_\infty$. Suppose that $\rho \leq \sigma$ are G -types. For $F \in \sigma$, define

$$\binom{F}{\rho} = \{F' \subseteq F : F' \in \rho\}.$$

If $\rho \leq \sigma \leq \tau$ are G -types and $k \geq 2$, we write

$$\tau \rightarrow (\sigma)_k^\rho$$

if for every $F \in \tau$ and every coloring

$$c: \binom{F}{\rho} \rightarrow \{1, \dots, k\},$$

there exists

$$F_0 \in \binom{F}{\sigma}$$

such that

$$c \upharpoonright \binom{F_0}{\rho}$$

is constant.

We say that G has the *Ramsey property* if for all G -types $\rho \leq \sigma$ and all $k \geq 2$, there exists a G -type $\tau \geq \sigma$ such that

$$\tau \rightarrow (\sigma)_k^\rho.$$

Theorem 5.2.5. Let $G \leq S_\infty$ be a closed subgroup. Then the following are equivalent.

- (i) G is extremely amenable.
- (ii) G preserves an ordering and has the Ramsey property.

Proof. Here is the proof with all the main ideas spelled out.

We prove first that

$$(i) \implies (ii).$$

So assume that

G is extremely amenable.

By Proposition 5.2.3, extreme amenability already implies that G preserves an ordering. So only the Ramsey property remains to be proved.

As in the usual finite Ramsey theorem, it is enough to prove the two-color case. The reason is the same as before: once one can find monochromatic copies for two colors, one gets the general k -color version by induction on k . So suppose, towards a contradiction, that G does *not* have the Ramsey property for two colors.

Then there exist G -types

$$\rho \leq \sigma$$

such that for no G -type

$$\tau \geq \sigma$$

do we have

$$\tau \rightarrow (\sigma)_2^\rho.$$

Fix one particular

$$F_0 \in \sigma.$$

Now let E be any finite set with

$$F_0 \subseteq E.$$

Let τ be the G -type of E . Since E contains $F_0 \in \sigma$, we have

$$\tau \geq \sigma.$$

By the assumed failure of the Ramsey property, this type τ does not satisfy

$$\tau \rightarrow (\sigma)_2^\rho.$$

Therefore there exists a coloring

$$c_E: \binom{E}{\rho} \rightarrow \{1, 2\}$$

such that no member of

$$\binom{E}{\sigma}$$

is homogeneous for c_E .

So for every large enough finite set E containing F_0 , we have chosen a bad two-coloring c_E .

The next step is to combine all these bad colorings into one global coloring

$$c: \rho \rightarrow \{1, 2\}.$$

This is exactly where the ultrafilter is used.

Let I be the set of all finite non-empty subsets of $\mathbb{N}$. For every finite set $F \subseteq \mathbb{N}$, the collection

$$\{E \in I : F \subseteq E\}$$

is non-empty, and finite intersections of such sets are again of the same form. So these sets generate a filter on I . Extend that filter to an ultrafilter $\mathcal{U}$.

Now fix

$$D \in \rho.$$

Consider the two sets

$$S_D^1 = \{E \in I : D \cup F_0 \subseteq E \text{ and } c_E(D) = 1\},$$

$$S_D^2 = \{E \in I : D \cup F_0 \subseteq E \text{ and } c_E(D) = 2\}.$$

Their union is

$$\{E \in I : D \cup F_0 \subseteq E\},$$

which belongs to $\mathcal{U}$ by construction. Since $\mathcal{U}$ is an ultrafilter, exactly one of the two sets S_D^1, S_D^2 belongs to $\mathcal{U}$.

So we may define

$$c(D) = i$$

if

$$S_D^i \in \mathcal{U}.$$

This gives a well-defined coloring

$$c: \rho \rightarrow \{1, 2\}.$$

Now we use Proposition 5.2.3. Since G is extremely amenable, that proposition tells us that for the coloring c there exists some

$$F \in \sigma$$

such that all members of

$$\binom{F}{\rho}$$

receive the same color. Say that common color is

$$i \in \{1, 2\}.$$

For each

$$D \in \binom{F}{\rho},$$

define

$$A_D = \{E \in I : F \cup F_0 \subseteq E \text{ and } c_E(D) = i\}.$$

We claim that

$$A_D \in \mathcal{U}.$$

Indeed, since $c(D) = i$, the set

$$S_D^i = \{E \in I : D \cup F_0 \subseteq E \text{ and } c_E(D) = i\}$$

lies in $\mathcal{U}$. Also

$$\{E \in I : F \cup F_0 \subseteq E\} \in \mathcal{U}.$$

Because

$$D \subseteq F,$$

the intersection of these two sets is exactly A_D . Since ultrafilters are closed under finite intersections, $A_D \in \mathcal{U}$.

Now there are only finitely many elements of

$$\binom{F}{\rho},$$

because F is finite. Hence the intersection

$$\bigcap_{D \in \binom{F}{\rho}} A_D$$

still belongs to $\mathcal{U}$, and in particular is non-empty. Choose

$$E \in \bigcap_{D \in \binom{F}{\rho}} A_D.$$

Then

$$F \cup F_0 \subseteq E,$$

so $F \subseteq E$, and for every

$$D \in \binom{F}{\rho}$$

we have

$$c_E(D) = i.$$

In other words, the copy $F \in \sigma$ inside E is homogeneous for the coloring c_E .

But that is impossible, because c_E was chosen specifically so that no member of

$$\binom{E}{\sigma}$$

is homogeneous.

This contradiction shows that the assumed failure of the Ramsey property was impossible. Therefore G has the Ramsey property. This proves

$$(i) \implies (ii).$$

Now we prove the converse

$$(ii) \implies (i).$$

So assume that G preserves an ordering and has the Ramsey property. By Proposition 5.2.3, in order to prove extreme amenability it is enough to verify condition (ii)(b) of that proposition.

Accordingly, let

$$\rho \leq \sigma$$

be G -types, and let

$$c: \rho \rightarrow \{1, \dots, k\}$$

be any finite coloring.

Since G has the Ramsey property, there exists a G -type

$$\tau \geq \sigma$$

such that

$$\tau \rightarrow (\sigma)_k^\rho.$$

Choose some

$$F \in \tau.$$

Now restrict the coloring c to the finite set

$$\binom{F}{\rho}.$$

By the arrow relation

$$\tau \rightarrow (\sigma)_k^\rho,$$

there exists

$$F_0 \in \binom{F}{\sigma}$$

such that

$$c \upharpoonright \binom{F_0}{\rho}$$

is constant.

But this is exactly what Proposition 5.2.3(ii)(b) asks for: we have found a member

$$F_0 \in \sigma$$

such that all members of type ρ contained in F_0 have the same color.

Therefore condition (ii)(b) of Proposition 5.2.3 holds. Since G also preserves an ordering, that proposition tells us that G is extremely amenable.

So

$$(ii) \implies (i).$$

Hence the two conditions are equivalent.

Assume that G is extremely amenable. Then G preserves an ordering by Proposition 5.2.3.

We show that G has the Ramsey property. As in the usual Ramsey argument, it is enough to treat the case of two colors. Suppose instead that there exist G -types $\rho \leq \sigma$ such that no G -type $\tau \geq \sigma$ satisfies

$$\tau \rightarrow (\sigma)_2^\rho.$$

Fix $F_0 \in \sigma$. Then for every finite set $E \supseteq F_0$ there exists a coloring

$$c_E: \binom{E}{\rho} \rightarrow \{1, 2\}$$

with no homogeneous member of

$$\binom{E}{\sigma}.$$

Let I be the set of all finite non-empty subsets of $\mathbb{N}$, and choose an ultrafilter $\mathcal{U}$ on I containing every set of the form

$$\{E \in I : F \subseteq E\},$$

where $F \subseteq \mathbb{N}$ is finite. For each $D \in \rho$, exactly one of the sets

$$\{E \supseteq D \cup F_0 : c_E(D) = 1\}, \quad \{E \supseteq D \cup F_0 : c_E(D) = 2\}$$

belongs to $\mathcal{U}$. Define

$$c(D) = i$$

if the i th set belongs to $\mathcal{U}$. This gives a coloring

$$c: \rho \rightarrow \{1, 2\}.$$

By Proposition 5.2.3, there exists $F \in \sigma$ such that

$$c \upharpoonright \binom{F}{\rho}$$

is constant, say constantly equal to i . For each

$$D \in \binom{F}{\rho},$$

the set

$$A_D = \{E \supseteq F \cup F_0 : c_E(D) = i\}$$

belongs to $\mathcal{U}$. Since there are only finitely many such D , their intersection is non-empty. Choose

$$E \in \bigcap_{D \in \binom{F}{\rho}} A_D.$$

Then

$$E \supseteq F_0$$

and

$$c_E(D) = i \quad \text{for all } D \in \binom{F}{\rho}.$$

Thus F is a homogeneous member of

$$\binom{E}{\sigma}$$

for the coloring c_E , contradicting the choice of c_E . Therefore G has the Ramsey property.

Conversely, assume that G preserves an ordering and has the Ramsey property. To prove extreme amenability, by Proposition 5.2.3 it is enough to verify condition (ii)(b) there. Let $\rho \leq \sigma$ be G -types and let

$$c: \rho \rightarrow \{1, \dots, k\}$$

be a coloring. Since G has the Ramsey property, there exists a G -type $\tau \geq \sigma$ such that

$$\tau \rightarrow (\sigma)_k^p.$$

Choose $F \in \tau$. Then the restriction of c to

$$\begin{pmatrix} F \\ \rho \end{pmatrix}$$

has a homogeneous member of

$$\begin{pmatrix} F \\ \sigma \end{pmatrix},$$

which is exactly the statement required in Proposition 5.2.3. Hence G is extremely amenable. $\dashv$

Remark 5.2.6. A family T of G -types is called *cofinal* if for every G -type ρ there exists $\sigma \in T$ such that

$$\rho \leq \sigma.$$

It is enough in Definition 5.2.4 to consider ρ , σ , and τ ranging over any fixed cofinal family of G -types.

5.3 Fraïssé order classes and extremely amenable automorphism groups

We now translate the characterization obtained in the previous section into Fraïssé-theoretic language.

Let L be a signature containing a distinguished binary relation symbol $<$. An *order structure* for L is an L -structure A such that $<^A$ is a linear ordering on $\text{dom}(A)$. A class $\mathcal{K}$ of finite L -structures is called an *order class* if every member of $\mathcal{K}$ is an order structure.

The next theorem identifies the extremely amenable closed subgroups of S_∞ with automorphism groups of Fraïssé limits of Ramsey order classes.

Theorem 5.3.1. *Let $G \leq S_\infty$ be a closed subgroup. Then the following are equivalent.*

- (i) *G is extremely amenable.*
- (ii) *There is a Fraïssé order class $\mathcal{K}$ with the Ramsey property such that*

$$G = \text{Aut}(F),$$

where $F = \text{Flim}(\mathcal{K})$.

Proof. We prove the two implications separately.

Step 1: assume that G is extremely amenable. We want to build a Fraïssé order class whose limit has automorphism group G .

Start with the canonical structure A_G from Chapter 2. By construction, the relations of A_G record the G -orbits on tuples of natural numbers, and Chapter 2 showed that

$$\text{Aut}(A_G) = G.$$

It also showed that A_G is ultrahomogeneous and relational.

Since G is extremely amenable, the previous theorem says that G preserves a linear order on $\mathbb{N}$. So choose a linear order

$$\prec$$

on $\mathbb{N}$ such that every element of G preserves $\prec$.

Now expand A_G by adding this order. Call the new structure

$$A = \langle A_G, \prec \rangle.$$

This is now an order structure.

We claim that

$$\text{Aut}(A) = G.$$

First, if $g \in G = \text{Aut}(A_G)$, then g preserves the relations of A_G . Since $\prec$ was chosen to be G -invariant, g also preserves $\prec$. Therefore g is an automorphism of the expanded structure A . So

$$G \subseteq \text{Aut}(A).$$

Conversely, if $h \in \text{Aut}(A)$, then h certainly preserves all the old relations coming from A_G . So $h \in \text{Aut}(A_G) = G$. Hence

$$\text{Aut}(A) \subseteq G.$$

Therefore

$$\text{Aut}(A) = G.$$

Next we claim that A is ultrahomogeneous. Take finite substructures

$$B, C \leq A$$

and an isomorphism

$$\varphi: B \rightarrow C.$$

Because B and C are substructures of the expansion A , the map φ preserves in particular all the relations coming from A_G . So it is an isomorphism between finite substructures of A_G as well. Since A_G is ultrahomogeneous, φ extends to some

$$g \in \text{Aut}(A_G) = G.$$

But every element of G preserves the order $\prec$, so this same g is actually an automorphism of A . Thus every isomorphism between finite substructures of A extends to an automorphism of A , which is exactly ultrahomogeneity.

Since the language is relational, finitely generated substructures are just finite subsets with the induced structure. So every finitely generated substructure of A is finite. In other words, A is locally finite.

Now apply Fraïssé's theorem: because A is countable, relational, locally finite, and ultrahomogeneous, its age

$$\mathcal{K} := \text{Age}(A)$$

is a Fraïssé class, and A is its Fraïssé limit:

$$A = \text{Flim}(\mathcal{K}).$$

Since A carries a linear order, every member of $\mathcal{K}$ is an order structure, so $\mathcal{K}$ is in fact a Fraïssé order class.

It remains to prove that $\mathcal{K}$ has the Ramsey property.

So let

$$B_0, B_1 \in \mathcal{K} \quad \text{with} \quad B_0 \leq B_1,$$

and let $k \geq 2$. We must find some

$$C \in \mathcal{K}$$

such that

$$C \rightarrow (B_1)_k^{B_0}.$$

Since $B_0, B_1 \in \text{Age}(A)$, copies of them occur inside A . Let ρ_0 be the G -type consisting of all finite subsets of $\mathbb{N}$ whose induced structure in A is isomorphic to B_0 , and let ρ_1 be defined similarly for B_1 .

Because A is ultrahomogeneous, any two copies of B_0 in A lie in the same G -orbit, and similarly for B_1 . So ρ_0 and ρ_1 are well-defined G -types.

Since G is extremely amenable, the previous theorem tells us that G has the Ramsey property. Therefore there exists a G -type $\tau \geq \rho_1$ such that

$$\tau \rightarrow (\rho_1)_k^{\rho_0}.$$

Choose a finite subset of $\mathbb{N}$ belonging to τ , and let C be the induced substructure of A on that set. Then $C \in \mathcal{K}$ because $\mathcal{K} = \text{Age}(A)$.

Why does this C work? Because copies of B_0 inside C are exactly the members of type ρ_0 contained in the underlying set of C , and copies of B_1 inside C are exactly the members of type ρ_1 contained in that same set. So the statement

$$\tau \rightarrow (\rho_1)_k^{\rho_0}$$

says precisely that every coloring of copies of B_0 inside C has a copy of B_1 inside C on which all copies of B_0 have the same color. In other words,

$$C \rightarrow (B_1)_k^{B_0}.$$

This proves that $\mathcal{K}$ has the Ramsey property. So from an extremely amenable closed subgroup $G \leq S_\infty$ we have produced a Fraïssé order class $\mathcal{K}$ with the Ramsey property such that

$$G = \text{Aut}(A) = \text{Aut}(\text{Flim}(\mathcal{K})).$$

Step 2: assume now that there is a Fraïssé order class $\mathcal{K}$ with the Ramsey property, and let

$$F = \text{Flim}(\mathcal{K}), \quad G = \text{Aut}(F).$$

We want to prove that G is extremely amenable.

Since $\mathcal{K}$ is an order class, the Fraïssé limit F comes equipped with a distinguished linear order $<^F$. Every automorphism of F must preserve all relations of the language, including this order. Therefore G preserves an ordering.

By the theorem from the previous section, to prove extreme amenability it is enough to show that G has the Ramsey property.

By the remark on cofinal families of types, we do not need to check all possible G -types. It is enough to check the types that come from finite substructures of F , because every finite subset of $\mathbb{N}$ generates a finite induced substructure of F , so these types are cofinal.

So let

$$B_0, B_1 \in \text{Age}(F) = \mathcal{K} \quad \text{with} \quad B_0 \leq B_1,$$

and let ρ_0, ρ_1 be the corresponding G -types of copies of B_0 and B_1 inside F .

Because $\mathcal{K}$ has the Ramsey property, for every $k \geq 2$ there exists

$$C \in \mathcal{K}$$

such that

$$C \rightarrow (B_1)_k^{B_0}.$$

Since $C \in \text{Age}(F)$, the structure C appears inside F . Let τ be the G -type of copies of C in F .

Now interpret the Ramsey statement inside F . A member of τ is a copy of C , a member of ρ_1 is a copy of B_1 , and a member of ρ_0 is a copy of B_0 . Therefore

$$C \rightarrow (B_1)_k^{B_0}$$

is exactly the same combinatorial statement as

$$\tau \rightarrow (\rho_1)_k^{\rho_0}.$$

So for the cofinal family of types coming from finite substructures of F , the group G satisfies the Ramsey property. Hence G has the Ramsey property.

We have now proved two things:

- G preserves an ordering;
- G has the Ramsey property.

The theorem from the previous section then gives that G is extremely amenable.

This proves the converse implication, and therefore the two conditions are equivalent.

Assume first that G is extremely amenable. Let A_G be the canonical structure on $\mathbb{N}$ associated with G ; recall from Chapter 2 that its basic n -ary relations are the G -orbits on $\mathbb{N}^n$, and that

$$\text{Aut}(A_G) = G.$$

By construction, the structure A_G is ultrahomogeneous and has a relational signature.

Since G is extremely amenable, Theorem 5.2.5 shows that G preserves an ordering on $\mathbb{N}$. Let $\prec$ be a linear ordering of $\mathbb{N}$ fixed by every element of G . Expand A_G by a new binary relation symbol $<$ interpreted as $\prec$, and denote the resulting structure by

$$A = \langle A_G, \prec \rangle.$$

Then every element of G preserves $\prec$, so

$$G \leq \text{Aut}(A).$$

Conversely, every automorphism of A is in particular an automorphism of A_G , hence belongs to G . Therefore

$$\text{Aut}(A) = G.$$

Since A_G is ultrahomogeneous and $\prec$ is G -invariant, the expansion A is also ultrahomogeneous: if $\varphi: B \rightarrow C$ is an isomorphism between finite substructures of A , then φ is in particular an isomorphism between finite substructures of A_G , so it extends to some $g \in G = \text{Aut}(A_G)$; because g preserves $\prec$, it is an automorphism of A extending φ . Because the signature is relational, finitely generated substructures of A are finite, so A is locally finite. It follows that

$$\mathcal{K} := \text{Age}(A)$$

is a Fraïssé order class with Fraïssé limit A .

It remains to show that $\mathcal{K}$ has the Ramsey property. Let $B_0, B_1 \in \mathcal{K}$ with $B_0 \leq B_1$, and let $k \geq 2$. Consider the G -types ρ_0 and ρ_1 consisting of all subsets of $\mathbb{N}$ which, with the induced structure from A , are isomorphic to B_0 and B_1 , respectively. Since A is ultrahomogeneous, copies of a finite substructure in A form a single G -type. By Theorem 5.2.5, the group G has the Ramsey property. Hence there exists a G -type $\tau \geq \rho_1$ such that

$$\tau \rightarrow (\rho_1)_k^{\rho_0}.$$

Choose a finite substructure $C \leq A$ whose underlying set belongs to τ . Then $C \in \mathcal{K}$, and the statement

$$C \rightarrow (B_1)_k^{B_0}$$

is exactly the same as

$$\tau \rightarrow (\rho_1)_k^{\rho_0}.$$

Thus $\mathcal{K}$ has the Ramsey property.

This proves (i) $\Rightarrow$ (ii).

Now assume (ii). Let

$$F = \text{Flim}(\mathcal{K}) \quad \text{and} \quad G = \text{Aut}(F),$$

where $\mathcal{K}$ is a Fraïssé order class with the Ramsey property.

Since F is an order structure, the linear order $<^F$ on $\text{dom}(F)$ is preserved by every automorphism of F . Thus G preserves an ordering.

To show that G is extremely amenable, it is enough by Theorem 5.2.5 to verify that G has the Ramsey property. By Remark 5.2.6, it suffices to check this on any cofinal family of G -types. We take the family of G -types arising from finite substructures of F .

Let $B_0, B_1 \in \text{Age}(F) = \mathcal{K}$ with $B_0 \leq B_1$, and let ρ_0, ρ_1 be the corresponding G -types of copies of B_0 and B_1 in F . Since $\mathcal{K}$ has the Ramsey property, for every $k \geq 2$ there exists

$$C \in \mathcal{K}$$

such that

$$C \rightarrow (B_1)_k^{B_0}.$$

Let τ be the G -type of copies of C in F . Then

$$\tau \rightarrow (\rho_1)_k^{\rho_0}.$$

Therefore G has the Ramsey property on this cofinal family of types, hence G has the Ramsey property. Since G also preserves an ordering, Theorem 5.2.5 yields that G is extremely amenable. $\dashv$

The preceding theorem immediately gives the Fraïssé-theoretic form that will be used in the next chapter.

Corollary 5.3.2. *Let $\mathcal{K}$ be a Fraïssé order class and let*

$$F = \text{Flim}(\mathcal{K}).$$

Then the following are equivalent.

- (i) $\text{Aut}(F)$ is extremely amenable.
- (ii) $\mathcal{K}$ has the Ramsey property.

Proof. This is really just the theorem above applied to the special group

$$G = \text{Aut}(F).$$

Let us check both directions slowly.

Assume first that

$\text{Aut}(F)$ is extremely amenable.

Since F is the Fraïssé limit of the Fraïssé order class $\mathcal{K}$, the theorem above applies with

$$G = \text{Aut}(F).$$

That theorem says that if a closed subgroup of S_∞ is extremely amenable, then it comes from a Fraïssé order class with the Ramsey property. But here the relevant Fraïssé order class is already

$$\text{Age}(F) = \mathcal{K}.$$

Therefore $\mathcal{K}$ has the Ramsey property.

Conversely, assume that

$\mathcal{K}$ has the Ramsey property.

Again apply the theorem above with

$$G = \text{Aut}(F).$$

Since $F = \text{Flim}(\mathcal{K})$ and $\mathcal{K}$ is a Fraïssé order class with the Ramsey property, the theorem tells us directly that

$$\text{Aut}(F)$$

is extremely amenable.

So each condition implies the other, and the two are equivalent.

In short: for a Fraïssé order class, the theorem above says exactly that

$$\text{Aut}(\text{Flim}(\mathcal{K})) \text{ is extremely amenable} \iff \mathcal{K} \text{ has the Ramsey property.}$$

Apply Theorem 5.3.1 with $G = \text{Aut}(F)$. +

The preceding corollary is the form of the Kechris–Pestov–Todorcevic correspondence that will be used in the next chapter. It shows that, for a Fraïssé order class, the extreme amenability of the automorphism group of its Fraïssé limit is exactly equivalent to the Ramsey property of the class. In Chapter 6 we apply this criterion to concrete examples.

Chapter 6

Examples of Extremely Amenable Automorphism Groups

In Chapter 5 we proved that if $\mathcal{K}$ is a Fraïssé order class with Fraïssé limit F , then $\text{Aut}(F)$ is extremely amenable if and only if $\mathcal{K}$ has the Ramsey property. The aim of this chapter is to apply that criterion to several standard Fraïssé classes and obtain concrete examples of extremely amenable automorphism groups.

We begin with classes of finite ordered graphs and hypergraphs, where the required Ramsey property is provided by the theorems of Nešetřil and Rödl and their extensions. We then consider finite-dimensional vector spaces over a fixed finite field and finite Boolean algebras, which yield further examples of extremely amenable automorphism groups. Finally, we turn to finite metric spaces and the ordered rational Urysohn space, leading to Pestov's theorem on the extreme amenability of the isometry group of the Urysohn space.

The purpose of the chapter is not to give an exhaustive catalogue of Ramsey classes, but to show how the criterion from Chapter 5 works in a range of important settings. These examples will also prepare the ground for the later computation of universal minimal flows via ordered expansions.

6.1 Graphs and equivalence relations

We begin with finite ordered graphs and some closely related classes. Let

$$L_0 = \{E\} \quad \text{and} \quad L = \{E, <\},$$

where E is a binary relation symbol and $<$ is intended to denote a linear order. An L_0 -structure A_0 is called a *graph* if E^{A_0} is irreflexive and symmetric. An *ordered graph* is an L -structure

$$A = \langle A_0, <^A \rangle$$

such that A_0 is a graph and $<^A$ is a linear ordering of $\text{dom}(A)$.

The most important Fraïssé classes of finite graphs for our purposes are the class $\mathcal{GR}$ of all finite graphs and, for each $n \geq 3$, the class $\text{Forb}(K_n)$ of all finite K_n -free graphs. Their ordered expansions will give the first basic examples of extremely amenable automorphism

groups. We will also consider finite equivalence relations, which provide a useful contrast: some ordered expansions are not Ramsey, while a more carefully chosen ordered expansion does lead to extreme amenability.

6.1.1 Ordered graphs and K_n -free ordered graphs

Let

$$\vec{\mathcal{GR}}$$

denote the class of all finite ordered graphs, and for each $n \geq 3$ let

$$\vec{\text{Forb}}(K_n)$$

denote the class of all finite ordered K_n -free graphs.

Both classes are hereditary and clearly contain structures of arbitrarily large cardinality. Since $\vec{\mathcal{GR}}$ and $\vec{\text{Forb}}(K_n)$ have strong amalgamation and strong joint embedding, it follows that

$$\vec{\mathcal{GR}} \quad \text{and} \quad \vec{\text{Forb}}(K_n)$$

are Fraïssé order classes.

In any part of the thesis, were mentions of the strong amalgamation and strong joint embedding? should their defintion be included in chapter 1? plus showing why above examples have these properties in chapter 1? or even in this chapter?

The Ramsey property for these classes is due to Nešetřil and Rödl [56, 58]; see also [50]. Therefore Chapter 5 applies immediately.

Theorem 6.1.1. *Let*

$$OR = \text{Flim}(\vec{\mathcal{GR}})$$

be the random ordered graph. Then $\text{Aut}(OR)$ is extremely amenable.

why $OR = \text{Flim}(\vec{\mathcal{GR}})$? do you think the reason for it should be included in chapter 1? as an example? or even in this chapter?

Proof. The class $\vec{\mathcal{GR}}$ is a Fraïssé order class with the Ramsey property, so the conclusion follows from Corollary 5.3.2. +

Theorem 6.1.2. *For each $n \geq 3$, let*

$$OR_n = \text{Flim}(\vec{\text{Forb}}(K_n))$$

be the random ordered K_n -free graph. Then $\text{Aut}(OR_n)$ is extremely amenable.

why $OR_n = \text{Flim}(\vec{\text{Forb}}(K_n))$? do you think the reason for it should be included in chapter 1? as an example? or even in this chapter?

Proof. The class $\vec{\text{Forb}}(K_n)$ is a Fraïssé order class with the Ramsey property, so again the result follows from Corollary 5.3.2. $\dashv$

These are the basic graph-theoretic examples. In contrast, the unordered automorphism groups $\text{Aut}(\text{Flim}(\mathcal{GR}))$ and $\text{Aut}(\text{Flim}(\vec{\text{Forb}}(K_n)))$ are not extremely amenable, since they do not preserve a linear ordering.

why they don't? could a proof or even an explanation be included?

6.1.2 A failure of the Ramsey property: equivalence relations

We now turn to finite equivalence relations. Strictly speaking, an equivalence relation is not a graph, since it is reflexive. Nevertheless, it is convenient to treat it in parallel with the graph examples by suppressing the diagonal.

Let

$$\mathcal{EQ}$$

denote the class of finite equivalence relations, viewed as L_0 -structures, and let

$$\vec{\mathcal{EQ}}$$

be the class of all finite linearly ordered equivalence relations. This class is a Fraïssé order class.

why? should it be included in chapter 1 as an example? or even in this chapter?

Its Fraïssé limit is, up to isomorphism,

$$\langle \mathbb{Q}, E, < \rangle,$$

where $<$ is the usual order on $\mathbb{Q}$ and E is an equivalence relation with infinitely many classes, each of which is dense in $\mathbb{Q}$.

why? should it be included in chapter 1 as an example? or even in this chapter?

Unlike the graph classes above, $\vec{\mathcal{EQ}}$ does *not* have the Ramsey property.

Theorem 6.1.3. *The class $\vec{\mathcal{EQ}}$ does not have the Ramsey property. Consequently, the automorphism group of*

$$\langle \mathbb{Q}, E, < \rangle,$$

where E has infinitely many dense classes, is not extremely amenable.

Proof. Let A be the ordered structure with domain $\{a, b\}$ and

$$a <^A b,$$

and let B be the ordered structure with domain $\{a, b, c\}$ such that

$$a <^B b <^B c,$$

and the E^B -classes are $\{a, c\}$ and $\{b\}$. We claim that there is no

$$C \in \vec{\mathcal{Q}}$$

such that

$$C \rightarrow (B)_2^A.$$

Let C be any member of $\vec{\mathcal{Q}}$. Order the E^C -classes by the least element they contain. Color a copy $\{x, y\}$ of A in C red if the E^C -class of x is lower than the E^C -class of y , and color it blue otherwise.

Now consider any copy $\{x, y, z\}$ of B in C , with

$$x <^C y <^C z.$$

Since x and z lie in the same equivalence class while y lies in a different one, the pair $\{x, y\}$ and the pair $\{y, z\}$ must receive different colors. Hence no copy of B is monochromatic. Therefore $\vec{\mathcal{Q}}$ fails the Ramsey property.

The final statement follows from Corollary 5.3.2. ◊

This example is useful because it shows that simply adding a linear order to a Fraïssé class does not automatically produce a Ramsey class.

6.1.3 Convexly ordered equivalence relations

There is, however, a natural ordered expansion of finite equivalence relations that *does* have the Ramsey property.

Let $\mathcal{K}$ be the class of all finite L -structures

$$A = \langle A, E^A, <^A \rangle$$

such that E^A is an equivalence relation on A , $<^A$ is a linear ordering on A , and each E^A -class is convex with respect to $<^A$; that is, whenever

$$a <^A b <^A c \quad \text{and} \quad a E^A c,$$

we also have

$$a E^A b.$$

It is straightforward to check that $\mathcal{K}$ is a Fraïssé order class. Its Fraïssé limit is, up to isomorphism,

$$F = \langle \mathbb{Q}^2, E, <_\ell \rangle,$$

where $<_\ell$ is the lexicographic ordering on $\mathbb{Q}^2$ and

$$(r, s) E (r', s') \iff r = r'.$$

To prove that $\text{Aut}(F)$ is extremely amenable, we use the following standard closure properties.

Lemma 6.1.4.

- (i) *Let G be a topological group and let $\mathcal{H}$ be a directed family of extremely amenable subgroups of G . If $\bigcup \mathcal{H}$ is dense in G , then G is extremely amenable.*
- (ii) *Let N be a closed normal subgroup of a topological group G . If N and G/N are extremely amenable, then G is extremely amenable.*
- (iii) *An arbitrary direct product of extremely amenable groups is extremely amenable.*

Proof. what is the full, detailed proof?

For (i), let X be a compact G -flow. For each $H \in \mathcal{H}$, the set

$$X^H = \{x \in X : h \cdot x = x \text{ for all } h \in H\}$$

is non-empty and closed. Since $\mathcal{H}$ is directed, the family $\{X^H : H \in \mathcal{H}\}$ has the finite intersection property, so compactness gives

$$\bigcap_{H \in \mathcal{H}} X^H \neq \emptyset.$$

Any point in this intersection is fixed by $\bigcup \mathcal{H}$, hence by all of G because the action is continuous and $\bigcup \mathcal{H}$ is dense.

For (ii), let X be a compact G -flow. Since N is extremely amenable, the fixed-point set

$$X^N = \{x \in X : n \cdot x = x \text{ for all } n \in N\}$$

is non-empty. It is closed and G -invariant because N is normal. The induced action of G/N on X^N is continuous, and since G/N is extremely amenable, there is a point of X^N fixed by G/N , hence by G .

For (iii), every finite subproduct is extremely amenable by repeated use of (ii). These finite subproducts form a directed family whose union is dense in the full product, so (i) applies. ◻

Theorem 6.1.5. *Let*

$$F = \langle \mathbb{Q}^2, E, <_\ell \rangle$$

be as above. Then $\text{Aut}(F)$ is extremely amenable.

Proof. what is the full, detailed proof?

For each $r \in \mathbb{Q}$, let

$$I_r = \{r\} \times \mathbb{Q}.$$

Any automorphism $\pi \in \text{Aut}(F)$ permutes the fibers I_r , since these are exactly the E -classes. Thus π induces an order automorphism

$$f_\pi \in \text{Aut}(\langle \mathbb{Q}, < \rangle)$$

given by

$$\pi(I_r) = I_{f_\pi(r)}.$$

Moreover, for each $r \in \mathbb{Q}$, the restriction of π to the fiber I_r induces an order automorphism

$$(g_\pi)_r \in \text{Aut}(\langle \mathbb{Q}, < \rangle).$$

Therefore we obtain a homomorphism

$$\Theta: \text{Aut}(F) \rightarrow \text{Aut}(\langle \mathbb{Q}, < \rangle) \ltimes \text{Aut}(\langle \mathbb{Q}, < \rangle)^\mathbb{Q},$$

where the first factor acts on the second by coordinate shift.

It is straightforward to check that Θ is a topological group isomorphism. Hence it is enough to show that

$$\text{Aut}(\langle \mathbb{Q}, < \rangle) \ltimes \text{Aut}(\langle \mathbb{Q}, < \rangle)^\mathbb{Q}$$

is extremely amenable.

By Theorem ??, the group

$$\text{Aut}(\langle \mathbb{Q}, < \rangle)$$

is extremely amenable. By Lemma 6.1.4(iii), so is the product

$$\text{Aut}(\langle \mathbb{Q}, < \rangle)^\mathbb{Q}.$$

Applying Lemma 6.1.4(ii), we conclude that the semidirect product is extremely amenable, and therefore so is $\text{Aut}(F)$. $\dashv$

Corollary 6.1.6. *The class of finite convexly ordered equivalence relations has the Ramsey property.*

Proof. This follows from Theorem 6.1.5 and Corollary 5.3.2. $\dashv$

The contrast between Theorem 6.1.3 and Corollary 6.1.6 is instructive. In the first case, an arbitrary linear ordering does not interact well with the equivalence relation. In the second, convexity aligns the order with the equivalence classes, and the resulting ordered expansion fits the Ramsey framework.

6.2 Hypergraphs

The graph examples extend naturally to hypergraphs. Let

$$L_0 = \{R_i : i \in I\}$$

be a finite relational signature, where each R_i has arity $n_i \geq 2$. An L_0 -structure A_0 will be called a *hypergraph* of type L_0 if, for each $i \in I$, the relation $R_i^{A_0}$ is irreflexive and symmetric. An *ordered hypergraph* is an L -structure

$$A = \langle A_0, <^A \rangle,$$

where $L = L_0 \cup \{<\}$, A_0 is a hypergraph of type L_0 , and $<^A$ is a linear ordering of $\text{dom}(A)$.

We first consider the class of all finite ordered hypergraphs of a fixed type. We then record a more general family obtained by forbidding finite irreducible substructures.

6.2.1 Ordered hypergraphs

Let

$$\vec{\mathcal{H}}_{L_0}$$

denote the class of all finite ordered hypergraphs of type L_0 . This is a Fraïssé order class. Indeed, heredity is immediate, and joint embedding and amalgamation are obtained by taking disjoint unions over the underlying ordered hypergraphs.

The Ramsey property for such classes was proved by Nešetřil and Rödl, extending the graph case. Hence Chapter 5 applies exactly as before.

Theorem 6.2.1. *Let*

$$F = \text{Flim}(\vec{\mathcal{H}}_{L_0}).$$

Then $\text{Aut}(F)$ is extremely amenable.

Proof. The class $\vec{\mathcal{H}}_{L_0}$ is a Fraïssé order class with the Ramsey property, so the conclusion follows from Corollary 5.3.2. $\dashv$

Thus every fixed finite hypergraph type yields an extremely amenable automorphism group once a linear ordering is added.

6.2.2 Forbidden irreducible hypergraphs

The previous example admits a substantial generalization. Let $\mathcal{A}$ be a family of finite hypergraphs of type L_0 . We write

$$\text{Forb}(\mathcal{A})$$

for the class of all finite hypergraphs of type L_0 that admit no embedding of a member of $\mathcal{A}$.

To obtain a Fraïssé class in this setting, one usually assumes that the forbidden structures are *irreducible*. Recall that a finite hypergraph A is called *irreducible* if any two distinct

vertices of A lie together in some relation of A . Equivalently, A cannot be written as a free amalgam of two proper substructures.

Let

$$\vec{\text{Forb}}(\mathcal{A})$$

denote the class of all finite linearly ordered members of $\text{Forb}(\mathcal{A})$.

Theorem 6.2.2. *Assume that $\mathcal{A}$ is a family of finite irreducible hypergraphs of type L_0 . Then $\vec{\text{Forb}}(\mathcal{A})$ is a Fraïssé order class with the Ramsey property. Consequently, if*

$$F = \text{Flim}(\vec{\text{Forb}}(\mathcal{A})),$$

then $\text{Aut}(F)$ is extremely amenable.

Proof. It is a standard result of Nešetřil and Rödl, in the form generalized by Abramson and Harrington, that $\vec{\text{Forb}}(\mathcal{A})$ is a Ramsey class whenever the forbidden structures are finite and irreducible. The hereditary property is immediate, and the Fraïssé properties are standard for this class. Therefore $\vec{\text{Forb}}(\mathcal{A})$ is a Fraïssé order class with the Ramsey property. The conclusion then follows from Corollary 5.3.2. $\dashv$

This theorem contains the ordered graph examples from the previous section as special cases. Indeed, if $L_0 = \{E\}$ and $\mathcal{A} = \{K_n\}$, then

$$\vec{\text{Forb}}(\mathcal{A}) = \vec{\text{Forb}}(K_n).$$

The hypergraph setting is therefore not merely parallel to the graph case, but a genuine extension of it.

6.3 Vector spaces over finite fields

We now turn to finite-dimensional vector spaces over a fixed finite field. This provides a different kind of example, in which the ordering is not arbitrary but is induced by an ordered basis.

Fix a finite field F . Let $\mathcal{V}_F$ denote the class of all finite-dimensional vector spaces over F , regarded as structures in the language of vector spaces over F . This is a Fraïssé class, and its Fraïssé limit is the countably infinite-dimensional vector space over F .

To obtain an ordered expansion, we use the anti-lexicographic ordering associated with an ordered basis. Let V be a finite-dimensional vector space over F , and let

$$b_1 < \cdots < b_n$$

be an ordered basis of V . Every element $v \in V$ may be written uniquely in the form

$$v = \alpha_1 b_1 + \cdots + \alpha_n b_n, \quad \alpha_i \in F.$$

Fix once and for all a linear ordering of the field F . This induces a linear ordering on V by declaring that

$$v < w$$

if, at the largest index i for which the coefficients of v and w differ, the coefficient of v is smaller than the coefficient of w . This is the anti-lexicographic ordering determined by the ordered basis $b_1 < \cdots < b_n$.

An ordering of a finite-dimensional vector space obtained in this way will be called a *natural ordering*. Let

$$\mathcal{OV}_F$$

denote the class of all finite-dimensional vector spaces over F equipped with a natural ordering.

The class $\mathcal{OV}_F$ is a Fraïssé order class. Indeed, heredity is immediate, and the amalgamation property follows from the corresponding property for finite-dimensional vector spaces together with the fact that natural orderings are determined by ordered bases.

The Ramsey property for $\mathcal{OV}_F$ is a consequence of the Graham–Leeb–Rothschild theorem; see also Thomas’s formulation for naturally ordered finite vector spaces. Hence Chapter 5 applies exactly as in the previous examples.

Theorem 6.3.1. *Let*

$$F_F = \text{Flim}(\mathcal{OV}_F).$$

Then $\text{Aut}(F_F)$ is extremely amenable.

Proof. The class $\mathcal{OV}_F$ is a Fraïssé order class with the Ramsey property. Therefore Corollary 5.3.2 yields that

$$\text{Aut}(F_F)$$

is extremely amenable. ◻

Thus naturally ordered finite-dimensional vector spaces over a finite field give another family of extremely amenable automorphism groups. Unlike the graph and hypergraph examples, the ordering here is tied to the underlying algebraic structure, and this compatibility is exactly what makes the Ramsey property available.

6.4 Boolean algebras

We next consider finite Boolean algebras. Let

$$L_0 = \{0, 1, \neg, \wedge, \vee\},$$

where 0 and 1 are constant symbols, $\neg$ is unary, and $\wedge, \vee$ are binary. Let

$$\mathcal{BA}$$

denote the class of all finite Boolean algebras in this language. This is a Fraïssé class, and its Fraïssé limit is the countable atomless Boolean algebra.

To obtain an ordered expansion, we define a natural ordering on each finite Boolean algebra in terms of its atoms. Let B be a finite Boolean algebra, and let

$$a_1 < \cdots < a_n$$

be a linear ordering of the set of atoms of B . Every element $x \in B$ has a unique representation

$$x = \delta_1 a_1 \vee \cdots \vee \delta_n a_n, \quad \delta_i \in \{0, 1\},$$

where $\delta_i a_i$ means a_i if $\delta_i = 1$ and 0 if $\delta_i = 0$. This ordering of the atoms induces an anti-lexicographic ordering on B : for

$$x = \delta_1 a_1 \vee \cdots \vee \delta_n a_n \quad \text{and} \quad y = \epsilon_1 a_1 \vee \cdots \vee \epsilon_n a_n,$$

we declare that

$$x < y$$

if at the largest index i for which $\delta_i \neq \epsilon_i$, one has

$$\delta_i < \epsilon_i.$$

An ordering obtained in this way will be called a *natural ordering* of B .

Let

$$\mathcal{OBA}$$

denote the class of all finite Boolean algebras equipped with a natural ordering.

Proposition 6.4.1. *The class $\mathcal{OBA}$ is a Fraïssé order class.*

Proof. Heredity is immediate. If B is a finite Boolean algebra with a natural ordering and $C \leq B$ is a subalgebra, then the atoms of C are unions of consecutive blocks of atoms of B , and the induced ordering on C is again natural.

Joint embedding is clear. For amalgamation, one uses the atom structure of finite Boolean algebras: embeddings of finite Boolean algebras are determined by the images of atoms, and an amalgam may be constructed by refining the atom decompositions of the two larger algebras over the common subalgebra. Ordering the new atoms compatibly with the given orderings yields a natural ordering on the amalgam. Hence $\mathcal{OBA}$ is a Fraïssé order class. $\dashv$

The Ramsey property for naturally ordered finite Boolean algebras is equivalent to the Dual Ramsey Theorem of Graham and Rothschild [33]. Therefore the criterion from Chapter 5 applies once again.

Theorem 6.4.2. *Let*

$$B_\infty^* = \text{Flim}(\mathcal{OBA})$$

be the countable atomless Boolean algebra with its canonical natural ordering. Then

$$\text{Aut}(B_\infty^*)$$

is extremely amenable.

Proof. By Proposition 6.4.1, the class $\mathcal{OBA}$ is a Fraïssé order class, and by the Dual Ramsey Theorem it has the Ramsey property. Corollary 5.3.2 therefore gives that

$$\text{Aut}(B_\infty^*)$$

is extremely amenable. $\dashv$

This example is parallel to the vector-space case. The ordering is not chosen arbitrarily, but is determined by an ordering of the atoms, and this compatibility is what places the class within the Ramsey framework.

6.5 Metric spaces

We conclude with metric spaces, which provide one of the most striking applications of the correspondence developed in Chapter 5.

Let

$$L_0 = \{R_q : q \in \mathbb{Q}_{>0}\},$$

where each R_q is a binary relation symbol. A metric space (X, d) determines an L_0 -structure

$$\mathbf{X} = \langle X, (R_q^{\mathbf{X}})_{q \in \mathbb{Q}_{>0}} \rangle$$

by setting

$$R_q^{\mathbf{X}}(x, y) \iff d(x, y) < q.$$

In this way finite metric spaces with rational distances may be viewed as finite relational structures.

Let

$$\mathcal{M}_{\mathbb{Q}}$$

denote the class of all finite metric spaces with rational distances. This is a Fraïssé class; its Fraïssé limit will be denoted by

$$U_{\mathbb{Q}}$$

and called the *rational Urysohn space*. Let

$$\mathcal{OM}_{\mathbb{Q}}$$

be the class of all finite linearly ordered metric spaces with rational distances. Since $\mathcal{M}_{\mathbb{Q}}$ has strong amalgamation and strong joint embedding, it follows that $\mathcal{OM}_{\mathbb{Q}}$ is a Fraïssé order class.

The Fraïssé limit

$$U_{\mathbb{Q}}^< = \text{Flim}(\mathcal{OM}_{\mathbb{Q}})$$

will be called the *ordered rational Urysohn space*. A result of Nešetřil implies that $\mathcal{OM}_{\mathbb{Q}}$ has the Ramsey property. Hence the criterion from Chapter 5 yields the following.

Theorem 6.5.1. *The automorphism group of the ordered rational Urysohn space is extremely amenable.*

Proof. The class $\mathcal{OM}_{\mathbb{Q}}$ is a Fraïssé order class with the Ramsey property, so the conclusion follows from Corollary 5.3.2. $\dashv$

We now explain how this yields Pestov's theorem on the full Urysohn space.

Let

$$\mathbb{U}$$

denote the Urysohn space, that is, the unique complete separable metric space that is universal for separable metric spaces and ultrahomogeneous for isometries. Urysohn already showed that the completion of $U_{\mathbb{Q}}$ is isometric to $\mathbb{U}$, so we may regard $U_{\mathbb{Q}}$ as a dense subspace of $\mathbb{U}$. Uspenskij proved that $\text{Iso}(\mathbb{U})$, equipped with the pointwise convergence topology, is a universal Polish group. Pestov showed that this group is extremely amenable.

Theorem 6.5.2 (Pestov). *The topological group $\text{Iso}(\mathbb{U})$ is extremely amenable.*

Proof. We first record a simple general fact.

Lemma 6.5.3. *Let $\pi: G \rightarrow H$ be a continuous homomorphism of topological groups with dense range. If G is extremely amenable, then H is extremely amenable.*

Proof. Let X be a compact H -flow. Via π , this becomes a G -flow, so by extreme amenability there is $x \in X$ fixed by every element of G . Since $\pi(G)$ is dense in H and the action is continuous, x is fixed by every element of H . $\dashv$

Let

$$U_{\mathbb{Q}}^{\leq} = \langle U_{\mathbb{Q}}, \prec \rangle$$

be the ordered rational Urysohn space. Every automorphism of $U_{\mathbb{Q}}^{\leq}$ is in particular an isometry of $U_{\mathbb{Q}}$, and therefore extends uniquely to an isometry of the completion $\mathbb{U}$. Thus we obtain a continuous homomorphism

$$\Phi: \text{Aut}(U_{\mathbb{Q}}^{\leq}) \rightarrow \text{Iso}(\mathbb{U}).$$

By Theorem 6.5.1 and Lemma 6.5.3, it is enough to show that $\Phi(\text{Aut}(U_{\mathbb{Q}}^{\leq}))$ is dense in $\text{Iso}(\mathbb{U})$.

Fix

$$h \in \text{Iso}(\mathbb{U}), \quad x_1, \dots, x_n \in \mathbb{U}, \quad \epsilon > 0.$$

We must find

$$g \in \text{Aut}(U_{\mathbb{Q}}^{\leq})$$

such that

$$d(\Phi(g)(x_i), h(x_i)) < \epsilon \quad \text{for } i = 1, \dots, n.$$

For this we use the following approximation lemma.

Lemma 6.5.4. *Let $x_1, \dots, x_n, y_1, \dots, y_n \in \mathbb{U}$ be such that*

$$x_i \mapsto y_i \quad (i = 1, \dots, n)$$

is an isometry between the two finite subspaces they generate. Then for every $\epsilon > 0$ there exist points

$$x'_1, \dots, x'_n, y'_1, \dots, y'_n \in U_{\mathbb{Q}}$$

such that:

- (i) *the map $x'_i \mapsto y'_i$ is an isometry;*
- (ii) *this isometry is order-preserving with respect to $\prec$;*
- (iii)

$$d(x_i, x'_i) < \epsilon \quad \text{and} \quad d(y_i, y'_i) < \epsilon \quad (i = 1, \dots, n).$$

Proof. This is proved by induction on n , using the density of $U_{\mathbb{Q}}$ in $\mathbb{U}$ together with the one-point extension property of the rational Urysohn space. At each stage one chooses rational points sufficiently close to the given ones and realizes the required rational distance pattern so that the resulting partial isometry is order-preserving. $\dashv$

Apply Lemma 6.5.4 to the tuples

$$x_1, \dots, x_n \quad \text{and} \quad h(x_1), \dots, h(x_n).$$

We obtain points

$$x'_1, \dots, x'_n, y'_1, \dots, y'_n \in U_{\mathbb{Q}}$$

such that the map

$$x'_i \mapsto y'_i$$

is an order-preserving isometry and

$$d(x_i, x'_i) < \epsilon/2, \quad d(h(x_i), y'_i) < \epsilon/2 \quad (i = 1, \dots, n).$$

Since $U_{\mathbb{Q}}^{\leq}$ is ultrahomogeneous, this finite order-preserving isometry extends to some

$$g \in \text{Aut}(U_{\mathbb{Q}}^{\leq}).$$

Then $\Phi(g)$ extends the same map on $U_{\mathbb{Q}}$, so for each i ,

$$\begin{aligned} d(\Phi(g)(x_i), h(x_i)) &\leq d(\Phi(g)(x_i), \Phi(g)(x'_i)) + d(\Phi(g)(x'_i), h(x_i)) \\ &= d(x_i, x'_i) + d(y'_i, h(x_i)) \\ &< \epsilon. \end{aligned}$$

Thus $\Phi(g)$ belongs to the prescribed neighbourhood of h , and the range of Φ is dense in $\text{Iso}(\mathbb{U})$. By Lemma 6.5.3, $\text{Iso}(\mathbb{U})$ is extremely amenable. $\dashv$

The metric-space case is the culmination of this chapter. It shows that the criterion from Chapter 5 applies not only to combinatorial structures such as graphs, hypergraphs, vector spaces, and Boolean algebras, but also to metric structures, and it leads to one of the most important examples of an extremely amenable Polish group.

The examples in this chapter show that the criterion from Chapter 5 applies to a broad range of Fraïssé order classes. In each case, the Ramsey property of a suitable ordered class yields the extreme amenability of the automorphism group of its Fraïssé limit. In the next chapter we turn from extreme amenability to universal minimal flows and ordered expansions, where the full scope of the Kechris–Pestov–Todorcevic correspondence becomes visible.

Chapter 7

Reasonable Ordered Expansions

In the previous chapters, ordered Fraïssé classes were used to obtain extremely amenable automorphism groups and to prepare the description of universal minimal flows. To pass from these ordered expansions back to the underlying unordered structures, one must understand how an ordered class is related to the class obtained by forgetting the linear order. The aim of this chapter is to isolate the conditions under which this passage behaves well.

We first introduce the notion of a reasonable ordered expansion and show that, for a Fraïssé order class $\mathcal{K}$, reasonableness is exactly the condition ensuring that the reduct class $\mathcal{K}_0$ is again a Fraïssé class and that the reduct of the Fraïssé limit of $\mathcal{K}$ is the Fraïssé limit of $\mathcal{K}_0$. We then discuss freely ordered expansions and strong amalgamation, and conclude with a brief remark on order-forgetful expansions, where Ramsey properties pass directly between the ordered and unordered settings.

7.1 Reducts and expansions

Let L be a signature containing a distinguished binary relation symbol $<$, and let

$$L_0 = L \setminus \{<\}.$$

If A is an L -structure, we write

$$A_0 = A \upharpoonright L_0$$

for the reduct of A to the signature L_0 . Thus A_0 is obtained from A by forgetting the interpretation of the symbol $<$. Conversely, if A_0 is an L_0 -structure and $\prec$ is a linear ordering of $\text{dom}(A_0)$, we write

$$\langle A_0, \prec \rangle$$

for the corresponding L -expansion of A_0 in which $<$ is interpreted by $\prec$.

If $\mathcal{K}$ is a class of finite L -structures, we write

$$\mathcal{K}_0 = \{A_0 : A \in \mathcal{K}\}$$

for the class of all reducts of members of $\mathcal{K}$ to the signature L_0 . In the situations that matter for us, $\mathcal{K}$ will be an order class, so each structure in $\mathcal{K}$ carries a linear ordering, while $\mathcal{K}_0$ is the corresponding unordered class obtained by forgetting that ordering.

The basic question is then the following. Suppose that $\mathcal{K}$ is a Fraïssé order class with Fraïssé limit F . Under what conditions is the reduct class $\mathcal{K}_0$ again a Fraïssé class, and when is its Fraïssé limit simply the reduct

$$F_0 = F \upharpoonright L_0?$$

The key condition is the notion of a reasonable ordered expansion, introduced in the next section.

7.2 Reasonable Fraïssé order classes

We now isolate the condition that controls when a Fraïssé order class behaves well after the linear order is forgotten. Intuitively, the issue is whether embeddings in the reduct can be lifted to embeddings in the ordered class by choosing a suitable ordering on the larger structure. This leads to the following definition.

Definition 7.2.1. Let L be a signature with $L \supseteq \{<\}$, and put

$$L_0 = L \setminus \{<\}.$$

Let $\mathcal{K}$ be a class of finite L -structures, and let

$$\mathcal{K}_0 = \{A_0 : A \in \mathcal{K}\}.$$

We say that $\mathcal{K}$ is *reasonable* if whenever $A_0, B_0 \in \mathcal{K}_0$, $\pi: A_0 \rightarrow B_0$ is an embedding, and $\prec$ is a linear ordering of $\text{dom}(A_0)$ such that

$$A = \langle A_0, \prec \rangle \in \mathcal{K},$$

there exists a linear ordering $\prec'$ of $\text{dom}(B_0)$ such that

$$B = \langle B_0, \prec' \rangle \in \mathcal{K}$$

and $\pi: A \rightarrow B$ is an embedding.

In other words, $\mathcal{K}$ is reasonable if every embedding in the reduct can be made compatible with the order after choosing a suitable ordering on the target structure.

The importance of this condition is that it exactly characterizes when the reduct of a Fraïssé order class is again a Fraïssé class, with the expected Fraïssé limit.

Proposition 7.2.2. Let $L \supseteq \{<\}$ be a signature, let

$$L_0 = L \setminus \{<\},$$

and let $\mathcal{K}$ be a Fraïssé order class of finite L -structures. Put

$$\mathcal{K}_0 = \{A_0 : A \in \mathcal{K}\}, \quad F = \text{Flim}(\mathcal{K}), \quad F_0 = F \upharpoonright L_0.$$

Then the following are equivalent.

(i) $\mathcal{K}_0$ is a Fraïssé class and

$$F_0 = \text{Flim}(\mathcal{K}_0).$$

(ii) $\mathcal{K}$ is reasonable.

Proof. Assume first that $\mathcal{K}_0$ is a Fraïssé class and that

$$F_0 = \text{Flim}(\mathcal{K}_0).$$

We show that $\mathcal{K}$ is reasonable.

Let $A_0, B_0 \in \mathcal{K}_0$, let $\pi: A_0 \rightarrow B_0$ be an embedding, and let $\prec$ be a linear ordering of $\text{dom}(A_0)$ such that

$$A = \langle A_0, \prec \rangle \in \mathcal{K}.$$

Since $F = \text{Flim}(\mathcal{K})$, there is an embedding

$$\varphi: A \rightarrow F.$$

Forgetting the order, we may also regard φ as an embedding

$$\varphi: A_0 \rightarrow F_0.$$

Because F_0 is the Fraïssé limit of $\mathcal{K}_0$, it has the extension property. Hence there is an embedding

$$\psi: B_0 \rightarrow F_0$$

such that

$$\psi \circ \pi = \varphi.$$

Now let $\prec'$ be the linear ordering on $\text{dom}(B_0)$ pulled back from F by ψ , that is,

$$x \prec' y \iff \psi(x) <^F \psi(y).$$

If we put

$$B = \langle B_0, \prec' \rangle,$$

then B is isomorphic to the substructure $\psi(B_0) \subseteq F$, so

$$B \in \text{Age}(F) = \mathcal{K}.$$

Moreover, since $\psi \circ \pi = \varphi$ and φ preserves the order $\prec$, the map

$$\pi: A \rightarrow B$$

is an embedding. Thus $\mathcal{K}$ is reasonable.

Conversely, assume that $\mathcal{K}$ is reasonable. We show first that $\mathcal{K}_0$ is a Fraïssé class.

Since

$$\mathcal{K}_0 = \text{Age}(F_0),$$

the class $\mathcal{K}_0$ is hereditary, countable up to isomorphism, and has structures of arbitrarily large cardinality. It therefore remains only to verify the amalgamation property.

Let $A_0, B_0, C_0 \in \mathcal{K}_0$, and let

$$f: A_0 \rightarrow B_0, \quad g: A_0 \rightarrow C_0$$

be embeddings. Choose a linear ordering $\prec$ on $\text{dom}(A_0)$ such that

$$A = \langle A_0, \prec \rangle \in \mathcal{K}.$$

Since $\mathcal{K}$ is reasonable, there exist linear orderings $\prec_B$ and $\prec_C$ on $\text{dom}(B_0)$ and $\text{dom}(C_0)$ such that

$$B = \langle B_0, \prec_B \rangle \in \mathcal{K}, \quad C = \langle C_0, \prec_C \rangle \in \mathcal{K},$$

and both

$$f: A \rightarrow B \quad \text{and} \quad g: A \rightarrow C$$

are embeddings. Since $\mathcal{K}$ is a Fraïssé class, it has the amalgamation property. Hence there exist

$$D \in \mathcal{K}$$

and embeddings

$$r: B \rightarrow D, \quad s: C \rightarrow D$$

such that

$$r \circ f = s \circ g.$$

Passing to reducts, if

$$D_0 = D \upharpoonright L_0,$$

then r and s are also embeddings

$$r: B_0 \rightarrow D_0, \quad s: C_0 \rightarrow D_0,$$

and still satisfy

$$r \circ f = s \circ g.$$

Thus $\mathcal{K}_0$ has the amalgamation property, and therefore $\mathcal{K}_0$ is a Fraïssé class.

It remains to show that

$$F_0 = \text{Flim}(\mathcal{K}_0).$$

Since

$$\text{Age}(F_0) = \mathcal{K}_0,$$

it is enough to verify that F_0 is ultrahomogeneous, or equivalently that it has the extension property.

Let $A_0, B_0 \in \mathcal{K}_0$, let $\pi: A_0 \rightarrow B_0$ be an embedding, and let

$$\varphi: A_0 \rightarrow F_0$$

be an embedding. Pull back the order on F along φ and define a linear ordering $\prec$ on $\text{dom}(A_0)$ by

$$x \prec y \iff \varphi(x) <^F \varphi(y).$$

Then

$$A = \langle A_0, \prec \rangle \in \mathcal{K},$$

and φ is in fact an embedding

$$\varphi: A \rightarrow F.$$

Since $\mathcal{K}$ is reasonable, there exists a linear ordering $\prec'$ on $\text{dom}(B_0)$ such that

$$B = \langle B_0, \prec' \rangle \in \mathcal{K}$$

and

$$\pi: A \rightarrow B$$

is an embedding. Because $F = \text{Flim}(\mathcal{K})$, it has the extension property, so there is an embedding

$$\psi: B \rightarrow F$$

such that

$$\psi \circ \pi = \varphi.$$

Forgetting the order, ψ is also an embedding

$$\psi: B_0 \rightarrow F_0$$

with

$$\psi \circ \pi = \varphi.$$

Thus F_0 has the extension property. It follows that F_0 is ultrahomogeneous and

$$\text{Age}(F_0) = \mathcal{K}_0,$$

so

$$F_0 = \text{Flim}(\mathcal{K}_0).$$

+

The proposition shows that for Fraïssé order classes, reasonableness is exactly the condition that allows one to pass from the ordered class back to its reduct without losing the Fraïssé structure. In the next section we consider an important source of reasonable order classes, namely freely ordered expansions.

7.3 Freely ordered expansions and strong amalgamation

A particularly important source of ordered expansions is obtained by imposing no compatibility condition between the additional linear order and the underlying L_0 -structure.

Definition 7.3.1. Let $\mathcal{K}_0$ be a class of finite L_0 -structures. We define

$$\mathcal{K}_0 * \mathcal{LO} = \{ \langle A_0, \prec \rangle : A_0 \in \mathcal{K}_0 \text{ and } \prec \text{ is a linear ordering of } \text{dom}(A_0) \}.$$

We call $\mathcal{K}_0 * \mathcal{LO}$ the *freely ordered expansion* of $\mathcal{K}_0$.

The point is that in a freely ordered expansion every linear order is allowed. In particular, such classes are automatically reasonable.

Lemma 7.3.2. *Let $\mathcal{K} = \mathcal{K}_0 * \mathcal{LO}$. Then $\mathcal{K}$ is reasonable.*

Proof. Let $A_0, B_0 \in \mathcal{K}_0$, let $\pi: A_0 \rightarrow B_0$ be an embedding, and suppose that

$$A = \langle A_0, \prec \rangle \in \mathcal{K}.$$

Choose any linear ordering $\prec'$ of $\text{dom}(B_0)$ extending the order transported from $\prec$ along π . Then

$$B = \langle B_0, \prec' \rangle \in \mathcal{K},$$

and $\pi: A \rightarrow B$ is an embedding. ⊢

For freely ordered expansions, amalgamation in the ordered class is controlled exactly by strong amalgamation in the reduct.

Proposition 7.3.3. *Let $L \supseteq \{<\}$ be a signature, let*

$$L_0 = L \setminus \{<\},$$

let $\mathcal{K}_0$ be a class of finite L_0 -structures, and put

$$\mathcal{K} = \mathcal{K}_0 * \mathcal{LO}.$$

Then the following are equivalent.

- (i) $\mathcal{K}$ has the amalgamation property.
- (ii) $\mathcal{K}$ has the strong amalgamation property.
- (iii) $\mathcal{K}_0$ has the strong amalgamation property.

Proof. The implication (ii) $\Rightarrow$ (i) is immediate.

We prove (i) $\Rightarrow$ (iii). Let

$$A_0, B_0, C_0 \in \mathcal{K}_0$$

and let

$$f: A_0 \rightarrow B_0, \quad g: A_0 \rightarrow C_0$$

be embeddings. Choose a linear ordering $\prec$ on $\text{dom}(A_0)$. Since

$$\mathcal{K} = \mathcal{K}_0 * \mathcal{LO},$$

we may choose linear orderings $\prec_B$ on $\text{dom}(B_0)$ and $\prec_C$ on $\text{dom}(C_0)$ such that

$$A = \langle A_0, \prec \rangle, \quad B = \langle B_0, \prec_B \rangle, \quad C = \langle C_0, \prec_C \rangle$$

all belong to $\mathcal{K}$, the maps

$$f: A \rightarrow B, \quad g: A \rightarrow C$$

are embeddings, and in addition

$$f(A_0) \prec_B (B_0 \setminus f(A_0)) \quad \text{and} \quad (C_0 \setminus g(A_0)) \prec_C g(A_0).$$

Since $\mathcal{K}$ has the amalgamation property, there exist

$$D \in \mathcal{K}$$

and embeddings

$$r: B \rightarrow D, \quad s: C \rightarrow D$$

such that

$$r \circ f = s \circ g.$$

We claim that

$$r(B_0) \cap s(C_0) = r(f(A_0)).$$

Suppose, toward a contradiction, that there is

$$d \in r(B_0) \cap s(C_0) \setminus r(f(A_0)).$$

Since $d \in r(B_0 \setminus f(A_0))$, we have

$$r(f(A_0)) <^D d.$$

On the other hand, since $d \in s(C_0 \setminus g(A_0))$, we have

$$d <^D s(g(A_0)) = r(f(A_0)).$$

This is impossible. Hence the amalgam is strong in the reduct, and $\mathcal{K}_0$ has the strong amalgamation property.

Finally, we prove (iii) $\Rightarrow$ (ii). Let

$$A, B, C \in \mathcal{K}$$

and let

$$f: A \rightarrow B, \quad g: A \rightarrow C$$

be embeddings. Forgetting the order, we obtain embeddings

$$f: A_0 \rightarrow B_0, \quad g: A_0 \rightarrow C_0.$$

By strong amalgamation in $\mathcal{K}_0$, there exist

$$D_0 \in \mathcal{K}_0$$

and embeddings

$$r: B_0 \rightarrow D_0, \quad s: C_0 \rightarrow D_0$$

such that

$$r \circ f = s \circ g \quad \text{and} \quad r(B_0) \cap s(C_0) = r(f(A_0)).$$

Since the images of B_0 and C_0 intersect only in the common copy of A_0 , and the orders on B and C agree there, the two induced linear orders on

$$r(B_0) \subseteq D_0 \quad \text{and} \quad s(C_0) \subseteq D_0$$

can be amalgamated to a linear ordering $\prec$ of $\text{dom}(D_0)$. If we now put

$$D = \langle D_0, \prec \rangle,$$

then

$$D \in \mathcal{K}$$

because $\mathcal{K} = \mathcal{K}_0 * \mathcal{LO}$, and the maps

$$r: B \rightarrow D, \quad s: C \rightarrow D$$

are embeddings witnessing strong amalgamation in $\mathcal{K}$. +

The same argument gives the corresponding statement for strong joint embedding.

Corollary 7.3.4. *Let L , L_0 , $\mathcal{K}_0$, and $\mathcal{K}$ be as in Proposition 7.3.3. Then $\mathcal{K}$ has the strong joint embedding property if and only if $\mathcal{K}_0$ has the strong joint embedding property.*

Proof. Assume first that $\mathcal{K}$ has the strong joint embedding property. Let

$$B_0, C_0 \in \mathcal{K}_0.$$

Choose arbitrary linear orderings on their domains and form

$$B = \langle B_0, \prec_B \rangle, \quad C = \langle C_0, \prec_C \rangle$$

in $\mathcal{K}$. By strong joint embedding in $\mathcal{K}$, there exist

$$D \in \mathcal{K}$$

and embeddings

$$r: B \rightarrow D, \quad s: C \rightarrow D$$

with disjoint images. Passing to reducts shows that $\mathcal{K}_0$ has the strong joint embedding property.

Conversely, assume that $\mathcal{K}_0$ has the strong joint embedding property. Let

$$B, C \in \mathcal{K}.$$

Then there exist

$$D_0 \in \mathcal{K}_0$$

and embeddings

$$r: B_0 \rightarrow D_0, \quad s: C_0 \rightarrow D_0$$

with disjoint images. Since the images are disjoint, the linear orders on B and C may be combined into a linear ordering $\prec$ of $\text{dom}(D_0)$. Setting

$$D = \langle D_0, \prec \rangle,$$

we have

$$D \in \mathcal{K},$$

and the embeddings

$$r: B \rightarrow D, \quad s: C \rightarrow D$$

have disjoint images. Thus $\mathcal{K}$ has the strong joint embedding property. $\dashv$

In particular, if $\mathcal{K}_0$ is a Fraïssé class with the strong amalgamation property, then

$$\mathcal{K}_0 * \mathcal{LO}$$

is a reasonable Fraïssé order class. Many of the ordered graph, hypergraph, and metric-space classes considered earlier arise in exactly this way.

7.4 Order-forgetful expansions and Ramsey transfer

We conclude with a special situation in which the passage between an ordered class and its reduct is particularly simple. Roughly speaking, this happens when the additional order does not create new isomorphism types beyond those already present in the reduct.

Definition 7.4.1. Let $L \supseteq \{<\}$ be a signature, let

$$L_0 = L \setminus \{<\},$$

and let $\mathcal{K}$ be a class of finite L -structures. We say that $\mathcal{K}$ is *order forgetful* if whenever

$$A, B \in \mathcal{K}$$

satisfy

$$A_0 \cong B_0,$$

it follows that

$$A \cong B.$$

Equivalently, for every isomorphism type in the reduct class $\mathcal{K}_0$, there is at most one isomorphism type in $\mathcal{K}$ lying above it.

Thus, in an order-forgetful class, the linear order does not contribute additional isomorphism-type information. A basic example is provided by naturally ordered finite-dimensional vector spaces over a fixed finite field: once the underlying vector space is given, any two natural orderings yield isomorphic ordered structures.

For hereditary order-forgetful classes, the Ramsey property passes directly between the ordered and unordered settings.

Proposition 7.4.2. *Let $L \supseteq \{<\}$ be a signature, let*

$$L_0 = L \setminus \{<\},$$

and let $\mathcal{K}$ be a hereditary order-forgetful class of finite L -structures. Put

$$\mathcal{K}_0 = \{A_0 : A \in \mathcal{K}\}.$$

Then $\mathcal{K}$ has the Ramsey property if and only if $\mathcal{K}_0$ has the Ramsey property.

Proof. Assume first that $\mathcal{K}$ has the Ramsey property. Let

$$A_0 \leq B_0$$

be structures in $\mathcal{K}_0$, and let $k \geq 2$. Choose an expansion

$$B \in \mathcal{K}$$

with reduct B_0 . Since $A_0 \leq B_0$, there is a copy $A'_0 \subseteq B_0$ of A_0 . Let A' be the ordered structure induced on A'_0 by the order of B . Because $\mathcal{K}$ is hereditary, we have

$$A' \in \mathcal{K}.$$

Since $\mathcal{K}$ is order forgetful and $A'_0 \cong A_0$, every expansion of A_0 in $\mathcal{K}$ is isomorphic to A' . In particular, if $A \in \mathcal{K}$ is any expansion of A_0 , then

$$A \cong A' \leq B.$$

Hence, replacing A by an isomorphic copy if necessary, we may apply the Ramsey property of $\mathcal{K}$ to the pair $A \leq B$. We claim that

$$C_0 \rightarrow (B_0)_k^{A_0}.$$

Let

$$c_0: \binom{C_0}{A_0} \rightarrow \{1, \dots, k\}$$

be a coloring. Every copy $A'_0 \subseteq C_0$ inherits from C a linear ordering, and since $\mathcal{K}$ is hereditary, the induced ordered structure on A'_0 belongs to $\mathcal{K}$. Because $\mathcal{K}$ is order forgetful and $A'_0 \cong A_0$, this induced ordered structure is isomorphic to A . Hence c_0 induces a coloring

$$c: \binom{C}{A} \rightarrow \{1, \dots, k\}$$

by setting

$$c(A') = c_0(A'_0),$$

where A'_0 is the underlying reduct of the copy A' .

Choose

$$B' \in \binom{C}{B}$$

homogeneous for c . Then the reduct B'_0 is a copy of B_0 in C_0 , and every copy of A_0 inside B'_0 inherits an ordered structure isomorphic to A . Since B' is homogeneous for c , it follows that B'_0 is homogeneous for c_0 . Thus

$$C_0 \rightarrow (B_0)_k^{A_0},$$

and $\mathcal{K}_0$ has the Ramsey property.

Conversely, assume that $\mathcal{K}_0$ has the Ramsey property. Let

$$A \leq B$$

be structures in $\mathcal{K}$, and let $k \geq 2$. Since $\mathcal{K}_0$ has the Ramsey property, there exists

$$C_0 \in \mathcal{K}_0$$

such that

$$C_0 \rightarrow (B_0)_k^{A_0}.$$

Choose any expansion

$$C \in \mathcal{K}$$

with reduct C_0 .

We claim that

$$C \rightarrow (B)_k^A.$$

Let

$$c: \binom{C}{A} \rightarrow \{1, \dots, k\}$$

be a coloring. Define

$$c_0: \binom{C_0}{A_0} \rightarrow \{1, \dots, k\}$$

by

$$c_0(A'_0) = c(A'),$$

where A' is the ordered structure induced on A'_0 by the order of C . This is well defined because $\mathcal{K}$ is hereditary, and A' is isomorphic to A by order forgetfulness.

Choose

$$B'_0 \in \binom{C_0}{B_0}$$

homogeneous for c_0 . Let B' be the ordered structure induced on B'_0 by the order of C . Again, since $\mathcal{K}$ is hereditary and order forgetful, we have

$$B' \cong B.$$

Moreover, every copy of A inside B' has the same c -color, because the corresponding copies of A_0 inside B'_0 all have the same c_0 -color. Hence B' is homogeneous for c , and therefore

$$C \rightarrow (B)_k^A.$$

Thus $\mathcal{K}$ has the Ramsey property. ◻

The results of this chapter provide the structural link between an ordered Fraïssé class and its reduct. In the next chapter we use this link to study spaces of admissible orderings and to formulate the additional conditions under which ordered expansions determine the universal minimal flow.

Chapter 8

Universal Minimal Flows and the Ordering Property

In the previous chapter we studied reasonable ordered expansions and the passage from an ordered Fraïssé class to its reduct. We now return to the dynamical side of the Kechris–Pestov–Todorcevic correspondence. Starting from a reasonable Fraïssé order class $\mathcal{K}$, we associate to it a compact flow $X_{\mathcal{K}}$ of admissible orderings on the Fraïssé limit of the reduct class $\mathcal{K}_0$. The aim of this chapter is to show that the ordering property characterizes the minimality of this flow, and that together with the Ramsey property it yields the universal minimal flow.

8.1 Admissible orderings and the flow $X_{\mathcal{K}}$

Let $L \supseteq \{<\}$ be a signature, and put

$$L_0 = L \setminus \{<\}.$$

Let $\mathcal{K}$ be a reasonable Fraïssé order class in the language L , and let

$$\mathcal{K}_0 = \mathcal{K}|_{L_0}$$

be its reduct to the language L_0 . By the results of Chapter 7, the class $\mathcal{K}_0$ is a Fraïssé class, and if

$$\mathbf{F} = \text{Flim}(\mathcal{K}), \quad \mathbf{F}_0 = \text{Flim}(\mathcal{K}_0),$$

then

$$\mathbf{F}_0 = \mathbf{F}|_{L_0}.$$

Let

$$<^{\mathbf{F}} = \prec_0,$$

and put

$$G_0 = \text{Aut}(\mathbf{F}_0).$$

Let $\text{LO}(F_0)$ denote the compact space of all linear orderings on the underlying set F_0 of $\mathbf{F}_0$, equipped with the logic action of G_0 . We define

$$X_{\mathcal{K}} = \overline{G_0 \cdot \prec_0} \subseteq \text{LO}(F_0).$$

Thus $X_{\mathcal{K}}$ is the orbit closure of the distinguished ordering $\prec_0$ induced from the Fraïssé limit $\mathbf{F}$.

The first point is that membership in $X_{\mathcal{K}}$ can be recognized purely in terms of finite substructures.

Proposition 8.1.1. *Let $\prec$ be a linear ordering on F_0 . Then $\prec \in X_{\mathcal{K}}$ if and only if for every finite substructure $\mathbf{B}_0$ of $\mathbf{F}_0$, the ordered structure*

$$\langle \mathbf{B}_0, \prec |_{B_0} \rangle$$

belongs to $\mathcal{K}$.

Proof. Assume first that $\prec \in X_{\mathcal{K}}$, and let $\mathbf{B}_0$ be a finite substructure of $\mathbf{F}_0$. Since

$$\prec \in \overline{G_0 \cdot \prec_0},$$

there exists $g \in G_0$ such that

$$\prec |_{B_0} = (g \cdot \prec_0) |_{B_0}.$$

Let

$$\mathbf{A}_0 = g^{-1}(\mathbf{B}_0),$$

which is again a finite substructure of $\mathbf{F}_0$, and let

$$\mathbf{A} = \langle \mathbf{A}_0, \prec_0 |_{A_0} \rangle.$$

Since $\mathbf{A}$ is a finite substructure of $\mathbf{F}$, we have $\mathbf{A} \in \mathcal{K}$. Moreover, the map

$$g|_{A_0}: A_0 \rightarrow B_0$$

is an isomorphism from $\mathbf{A}$ onto

$$\mathbf{B} = \langle \mathbf{B}_0, \prec |_{B_0} \rangle.$$

Hence $\mathbf{B} \in \mathcal{K}$.

Conversely, assume that for every finite substructure $\mathbf{B}_0$ of $\mathbf{F}_0$, the ordered structure

$$\mathbf{B} = \langle \mathbf{B}_0, \prec |_{B_0} \rangle$$

belongs to $\mathcal{K}$. We show that $\prec \in X_{\mathcal{K}}$.

Fix a finite substructure $\mathbf{B}_0$ of $\mathbf{F}_0$. Since $\mathbf{B} \in \mathcal{K} = \text{Age}(\mathbf{F})$, there exists an embedding

$$\pi: \mathbf{B} \rightarrow \mathbf{F}.$$

Let

$$\mathbf{A} = \pi(\mathbf{B}), \quad \mathbf{A}_0 = \mathbf{A}|_{L_0}.$$

Then π is an isomorphism from $\mathbf{B}$ onto $\mathbf{A}$, and in particular an isomorphism from $\mathbf{B}_0$ onto $\mathbf{A}_0$. Since $\mathbf{A}_0$ and $\mathbf{B}_0$ are finite substructures of the ultrahomogeneous structure $\mathbf{F}_0$, there exists

$$g \in \text{Aut}(\mathbf{F}_0) = G_0$$

extending π^{-1} . It follows that

$$\prec|_{B_0} = (g \cdot \prec_0)|_{B_0}.$$

As $\mathbf{B}_0$ was arbitrary, this shows that every finite restriction of $\prec$ agrees with a finite restriction of some translate of $\prec_0$. Therefore

$$\prec \in \overline{G_0 \cdot \prec_0} = X_{\mathcal{K}}.$$

+

Definition 8.1.2. A linear ordering $\prec$ on F_0 is called $\mathcal{K}$ -admissible if

$$\prec \in X_{\mathcal{K}}.$$

Equivalently, $\prec$ is $\mathcal{K}$ -admissible if every finite substructure of $\mathbf{F}_0$, equipped with the induced ordering from $\prec$, belongs to $\mathcal{K}$.

In other words, the points of $X_{\mathcal{K}}$ are exactly the linear orderings on F_0 whose finite restrictions are compatible with the class $\mathcal{K}$. Since $X_{\mathcal{K}}$ is a closed G_0 -invariant subspace of the compact space $\text{LO}(F_0)$, it is a compact G_0 -flow. In the next section we determine exactly when this flow is minimal.

8.2 The ordering property and minimality of $X_{\mathcal{K}}$

We now introduce the combinatorial condition that characterizes the minimality of the flow $X_{\mathcal{K}}$.

Definition 8.2.1. We say that $\mathcal{K}$ has the *ordering property* if for every

$$\mathbf{A}_0 \in \mathcal{K}_0$$

there exists

$$\mathbf{B}_0 \in \mathcal{K}_0$$

such that whenever

$$\mathbf{A} = \langle \mathbf{A}_0, \prec \rangle \in \mathcal{K} \quad \text{and} \quad \mathbf{B} = \langle \mathbf{B}_0, \prec' \rangle \in \mathcal{K},$$

one has

$$\mathbf{A} \leq \mathbf{B}.$$

Thus the ordering property says that for every finite reduct $\mathbf{A}_0$, there is a larger finite reduct $\mathbf{B}_0$ such that every admissible ordering of $\mathbf{B}_0$ contains every admissible ordering of $\mathbf{A}_0$.

The classes considered in Chapter 6 provide both positive and negative examples. For instance, the freely ordered graph and hypergraph classes satisfy the ordering property, whereas the class of all linearly ordered finite equivalence relations does not.

The key step is the following finite reformulation of the condition

$$\prec_0 \in \overline{G_0 \cdot \prec}.$$

Lemma 8.2.2. *Let $\prec$ be a linear ordering on F_0 . Then the following are equivalent.*

(i)

$$\prec_0 \in \overline{G_0 \cdot \prec}.$$

(ii) *For every*

$$\mathbf{A} \in \mathcal{K},$$

there exists a finite substructure $\mathbf{C}_0$ of $\mathbf{F}_0$ such that

$$\langle \mathbf{C}_0, \prec \upharpoonright C_0 \rangle \in \mathcal{K} \quad \text{and} \quad \mathbf{A} \leq \langle \mathbf{C}_0, \prec \upharpoonright C_0 \rangle.$$

In particular, if $\prec \in X_{\mathcal{K}}$, then condition (ii) says exactly that every member of $\mathcal{K}$ embeds into a finite substructure of the ordered structure

$$\langle F_0, \prec \rangle.$$

Proof. Assume first that

$$\prec_0 \in \overline{G_0 \cdot \prec},$$

and let

$$\mathbf{A} \in \mathcal{K}.$$

Since $\mathcal{K} = \text{Age}(\mathbf{F})$, we may identify $\mathbf{A}$ with a finite substructure of $\mathbf{F}$, so that

$$\mathbf{A} = \langle \mathbf{A}_0, \prec_0 \upharpoonright A_0 \rangle$$

for some finite substructure $\mathbf{A}_0$ of $\mathbf{F}_0$. By the assumption on $\prec$, there exists

$$g \in G_0$$

such that

$$(g \cdot \prec) \upharpoonright A_0 = \prec_0 \upharpoonright A_0.$$

Let

$$\mathbf{C}_0 = g^{-1}(\mathbf{A}_0).$$

Then

$$g: \langle \mathbf{C}_0, \prec \upharpoonright C_0 \rangle \rightarrow \mathbf{A}$$

is an isomorphism. Hence

$$\langle \mathbf{C}_0, \prec | C_0 \rangle \in \mathcal{K} \quad \text{and} \quad \mathbf{A} \leq \langle \mathbf{C}_0, \prec | C_0 \rangle.$$

Conversely, assume (ii). To show that

$$\prec_0 \in \overline{G_0 \cdot \prec},$$

it suffices to verify that for every finite substructure $\mathbf{A}_0$ of $\mathbf{F}_0$, there exists

$$g \in G_0$$

such that

$$(g \cdot \prec) | A_0 = \prec_0 | A_0.$$

Fix such a finite substructure $\mathbf{A}_0$, and let

$$\mathbf{A} = \langle \mathbf{A}_0, \prec_0 | A_0 \rangle \in \mathcal{K}.$$

By (ii), there exists a finite substructure $\mathbf{C}_0$ of $\mathbf{F}_0$ such that

$$\mathbf{C} = \langle \mathbf{C}_0, \prec | C_0 \rangle \in \mathcal{K} \quad \text{and} \quad \mathbf{A} \leq \mathbf{C}.$$

Let

$$\pi: \mathbf{A} \rightarrow \mathbf{C}$$

be an embedding. Forgetting the order, π is an embedding

$$\pi: \mathbf{A}_0 \rightarrow \mathbf{F}_0.$$

Since $\mathbf{F}_0$ is ultrahomogeneous, π^{-1} extends to some

$$g \in G_0.$$

For $x, y \in A_0$ we then have

$$x (g \cdot \prec) y \iff g^{-1}(x) \prec g^{-1}(y) \iff \pi(x) \prec \pi(y) \iff x \prec_0 y,$$

because π is an embedding of ordered structures. Thus

$$(g \cdot \prec) | A_0 = \prec_0 | A_0,$$

as required. +

We can now characterize the minimality of $X_{\mathcal{K}}$.

Theorem 8.2.3. *The flow $X_{\mathcal{K}}$ is minimal if and only if $\mathcal{K}$ has the ordering property.*

Proof. Assume first that $\mathcal{K}$ has the ordering property. Let

$$\prec \in X_{\mathcal{K}}.$$

We show that

$$\prec_0 \in \overline{G_0 \cdot \prec}.$$

By Lemma 8.2.2, it is enough to verify condition (ii) of that lemma.

So let

$$\mathbf{A} \in \mathcal{K}.$$

Write

$$\mathbf{A} = \langle \mathbf{A}_0, \prec_A \rangle.$$

By the ordering property, there exists

$$\mathbf{B}_0 \in \mathcal{K}_0$$

such that whenever

$$\mathbf{B} = \langle \mathbf{B}_0, \prec_B \rangle \in \mathcal{K},$$

one has

$$\mathbf{A} \leq \mathbf{B}.$$

Since

$$\text{Age}(\mathbf{F}_0) = \mathcal{K}_0,$$

we may realize $\mathbf{B}_0$ as a finite substructure of $\mathbf{F}_0$. Because $\prec \in X_{\mathcal{K}}$, Proposition 8.1.1 implies that

$$\langle \mathbf{B}_0, \prec \restriction B_0 \rangle \in \mathcal{K}.$$

Hence

$$\mathbf{A} \leq \langle \mathbf{B}_0, \prec \restriction B_0 \rangle.$$

By Lemma 8.2.2,

$$\prec_0 \in \overline{G_0 \cdot \prec}.$$

Now let

$$\prec \in X_{\mathcal{K}}.$$

The orbit closure $\overline{G_0 \cdot \prec}$ is a closed G_0 -invariant subset of $X_{\mathcal{K}}$ containing $\prec_0$. Since

$$X_{\mathcal{K}} = \overline{G_0 \cdot \prec_0},$$

it follows that

$$X_{\mathcal{K}} \subseteq \overline{G_0 \cdot \prec}.$$

The reverse inclusion is automatic, so

$$\overline{G_0 \cdot \prec} = X_{\mathcal{K}}.$$

Thus every orbit is dense, and $X_{\mathcal{K}}$ is minimal.

Conversely, assume that $X_{\mathcal{K}}$ is minimal. Let

$$\mathbf{A}_0 \in \mathcal{K}_0.$$

We must find

$$\mathbf{B}_0 \in \mathcal{K}_0$$

witnessing the ordering property for $\mathbf{A}_0$.

There are only finitely many linear orderings on the finite set A_0 , hence only finitely many ordered structures of the form

$$\langle \mathbf{A}_0, \prec \rangle$$

that belong to $\mathcal{K}$. List them as

$$\mathbf{A}^1, \dots, \mathbf{A}^m.$$

Fix

$$\prec \in X_{\mathcal{K}}.$$

Since $X_{\mathcal{K}}$ is minimal,

$$\prec_0 \in \overline{G_0 \cdot \prec}.$$

Applying Lemma 8.2.2 to each $\mathbf{A}^j$, we obtain finite substructures

$$\mathbf{C}_{\prec,0}^j \leq \mathbf{F}_0 \quad (j = 1, \dots, m)$$

such that

$$\mathbf{C}_{\prec}^j = \langle \mathbf{C}_{\prec,0}^j, \prec | \mathbf{C}_{\prec,0}^j \rangle \in \mathcal{K}$$

and

$$\mathbf{A}^j \leq \mathbf{C}_{\prec}^j.$$

Let

$$C_{\prec,0} = C_{\prec,0}^1 \cup \dots \cup C_{\prec,0}^m,$$

viewed as the finite substructure of $\mathbf{F}_0$ generated by this union. Since $\prec \in X_{\mathcal{K}}$, Proposition 8.1.1 gives

$$\langle C_{\prec,0}, \prec | C_{\prec,0} \rangle \in \mathcal{K},$$

and by construction every $\mathbf{A}^j$ embeds into this ordered structure.

Now define

$$U_{\prec} = \{ \prec' \in X_{\mathcal{K}} : \prec' | C_{\prec,0} = \prec | C_{\prec,0} \}.$$

This is an open neighbourhood of $\prec$ in $X_{\mathcal{K}}$. As $\prec$ ranges over $X_{\mathcal{K}}$, the sets $U_{\prec}$ form an open cover of the compact space $X_{\mathcal{K}}$. Hence there exist

$$\prec_1, \dots, \prec_n \in X_{\mathcal{K}}$$

such that

$$X_{\mathcal{K}} = U_{\prec_1} \cup \dots \cup U_{\prec_n}.$$

Choose a finite substructure

$$\mathbf{B}_0 \leq \mathbf{F}_0$$

containing each of the finite substructures

$$C_{\prec_1,0}, \dots, C_{\prec_n,0}.$$

We claim that $\mathbf{B}_0$ witnesses the ordering property for $\mathbf{A}_0$.

Let

$$\mathbf{A} = \langle \mathbf{A}_0, \prec_A \rangle \in \mathcal{K} \quad \text{and} \quad \mathbf{B} = \langle \mathbf{B}_0, \prec_B \rangle \in \mathcal{K}.$$

Since $\mathbf{B} \in \text{Age}(\mathbf{F}) = \mathcal{K}$, there exists a finite ordered substructure

$$\tilde{\mathbf{B}} \leq \mathbf{F}$$

isomorphic to $\mathbf{B}$. Let

$$h: \tilde{\mathbf{B}} \rightarrow \mathbf{B}$$

be an isomorphism, and let

$$g \in G_0$$

be an automorphism of $\mathbf{F}_0$ extending

$$h^{-1}: \mathbf{B}_0 \rightarrow \tilde{\mathbf{B}}_0.$$

Then

$$\hat{\prec} = g \cdot \prec_0$$

belongs to $X_{\mathcal{K}}$, and its restriction to B_0 is exactly $\prec_B$.

Since

$$X_{\mathcal{K}} = U_{\prec_1} \cup \dots \cup U_{\prec_n},$$

there exists $i \in \{1, \dots, n\}$ such that

$$\hat{\prec} \in U_{\prec_i}.$$

Hence

$$\hat{\prec}|_{C_{\prec_i,0}} = \prec_i|_{C_{\prec_i,0}}.$$

By construction of $C_{\prec_i,0}$, every ordered expansion of $\mathbf{A}_0$ belonging to $\mathcal{K}$ embeds into

$$\langle C_{\prec_i,0}, \prec_i|_{C_{\prec_i,0}} \rangle.$$

In particular,

$$\mathbf{A} \leq \langle C_{\prec_i,0}, \prec_i|_{C_{\prec_i,0}} \rangle = \langle C_{\prec_i,0}, \hat{\prec}|_{C_{\prec_i,0}} \rangle.$$

Since

$$C_{\prec_i,0} \subseteq B_0$$

and $\hat{\prec}|_{B_0} = \prec_B$, the latter is a substructure of $\mathbf{B}$. Therefore

$$\mathbf{A} \leq \mathbf{B}.$$

This proves that $\mathcal{K}$ has the ordering property. +

The theorem shows that the minimality of the flow $X_{\mathcal{K}}$ is controlled by a purely finite combinatorial condition on the class $\mathcal{K}$. In the next section we add the Ramsey property and obtain the universal minimal flow.

8.3 Ramsey property and universal minimal flows

We now add the Ramsey property and obtain the main dynamical consequence of the Kechris–Pestov–Todorcevic correspondence.

Let

$$G = \text{Aut}(\mathbf{F}).$$

Since $\mathbf{F}$ is an ordered expansion of $\mathbf{F}_0$, we have

$$G \leq G_0,$$

and every element of G fixes the distinguished ordering $\prec_0$.

The first result shows that, under the Ramsey property, the pointed flow

$$(X_K, \prec_0)$$

is universal among G_0 -ambits whose distinguished point is fixed by G .

Lemma 8.3.1. *Assume that K has the Ramsey property. Let (X, x_0) be a G_0 -ambit such that*

$$g \cdot x_0 = x_0 \quad \text{for all } g \in G.$$

Let

$$\Phi = \overline{\{(g \cdot \prec_0, g \cdot x_0) : g \in G_0\}} \subseteq X_K \times X.$$

If

$$(\prec, x), (\prec, x') \in \Phi,$$

then

$$x = x'.$$

Proof. Suppose that

$$(\prec, x), (\prec, x') \in \Phi \quad \text{and} \quad x \neq x'.$$

Choose disjoint open neighbourhoods

$$U \ni x, \quad U' \ni x'.$$

Since X is compact Hausdorff, there exists an open neighbourhood D of the diagonal in $X \times X$ such that

$$D[U] \cap U' = \emptyset,$$

where

$$D[U] = \{y \in X : \text{there exists } z \in U \text{ with } (z, y) \in D\}.$$

By replacing D with

$$D \cap D^{-1},$$

we may assume that D is symmetric.

Because the action of G_0 on the compact space X is continuous, there exists an open neighbourhood V of the identity in G_0 such that

$$(v \cdot z, z) \in D \quad \text{for all } z \in X \text{ and all } v \in V.$$

Since $G_0 \leq S_\infty$ is non-Archimedean, we may choose a finite substructure

$$\mathbf{A}_0 \leq \mathbf{F}_0$$

such that the pointwise stabilizer

$$G_{0,(\mathbf{A}_0)} \subseteq V.$$

Because

$$(\prec, x), (\prec, x') \in \Phi,$$

there exist

$$g, h \in G_0$$

such that

$$g \cdot x_0 \in U, \quad h \cdot x_0 \in U',$$

and

$$(g \cdot \prec_0)|_{A_0} = (h \cdot \prec_0)|_{A_0} = \prec|_{A_0}.$$

Consider the map

$$\theta = h^{-1}g: g^{-1}(A_0) \rightarrow h^{-1}(A_0).$$

Since

$$(g \cdot \prec_0)|_{A_0} = (h \cdot \prec_0)|_{A_0},$$

the map θ is an isomorphism of the finite ordered structures

$$\langle g^{-1}(\mathbf{A}_0), \prec_0 \rangle \quad \text{and} \quad \langle h^{-1}(\mathbf{A}_0), \prec_0 \rangle.$$

Because

$$\text{Age}(\mathbf{F}) = \mathcal{K}$$

and $\mathbf{F}$ is ultrahomogeneous, θ extends to some

$$\gamma \in G = \text{Aut}(\mathbf{F}).$$

Then for every $a \in A_0$,

$$h\gamma g^{-1}(a) = a,$$

so

$$v := h\gamma g^{-1} \in G_{0,(\mathbf{A}_0)} \subseteq V.$$

Since $\gamma \in G$ fixes x_0 , we obtain

$$h \cdot x_0 = v \cdot g \cdot \gamma^{-1} \cdot x_0 = v \cdot g \cdot x_0.$$

Hence

$$(h \cdot x_0, g \cdot x_0) \in D.$$

Because D is symmetric, this implies

$$h \cdot x_0 \in D[U].$$

But

$$g \cdot x_0 \in U \quad \text{and} \quad h \cdot x_0 \in U',$$

so

$$h \cdot x_0 \in D[U] \cap U',$$

contrary to the choice of D . Therefore

$$x = x'.$$

+

Theorem 8.3.2. *Assume that $\mathcal{K}$ has the Ramsey property. Let (X, x_0) be a G_0 -ambit such that*

$$g \cdot x_0 = x_0 \quad \text{for all } g \in G.$$

Then there exists a unique G_0 -map

$$\pi: X_{\mathcal{K}} \rightarrow X$$

such that

$$\pi(\prec_0) = x_0.$$

In particular, $(X_{\mathcal{K}}, \prec_0)$ is universal among G_0 -ambits whose distinguished point is fixed by G .

Proof. Let

$$\Phi = \overline{\{(g \cdot \prec_0, g \cdot x_0) : g \in G_0\}} \subseteq X_{\mathcal{K}} \times X.$$

Since the projection

$$p_1: X_{\mathcal{K}} \times X \rightarrow X_{\mathcal{K}}$$

is a closed map and

$$p_1(\Phi)$$

contains the dense set

$$G_0 \cdot \prec_0,$$

we have

$$p_1(\Phi) = X_{\mathcal{K}}.$$

Thus for every

$$\prec \in X_{\mathcal{K}}$$

there exists

$$x \in X$$

such that

$$(\prec, x) \in \Phi.$$

By Lemma 8.3.1, this x is unique. Hence Φ is the graph of a function

$$\pi: X_{\mathcal{K}} \rightarrow X.$$

The map

$$p_1|_{\Phi}: \Phi \rightarrow X_{\mathcal{K}}$$

is a continuous bijection from a compact space to a Hausdorff space, hence a homeomorphism. Therefore

$$\pi = p_2 \circ (p_1|_{\Phi})^{-1}$$

is continuous.

The set

$$\{(g \cdot \prec_0, g \cdot x_0) : g \in G_0\}$$

is G_0 -invariant, and so is its closure Φ . It follows that π is G_0 -equivariant. Also,

$$(\prec_0, x_0) \in \Phi,$$

so

$$\pi(\prec_0) = x_0.$$

To prove uniqueness, let

$$\pi': X_{\mathcal{K}} \rightarrow X$$

be another G_0 -map with

$$\pi'(\prec_0) = x_0.$$

For every $g \in G_0$ we then have

$$\pi'(g \cdot \prec_0) = g \cdot \pi'(\prec_0) = g \cdot x_0 = \pi(g \cdot \prec_0).$$

Since

$$G_0 \cdot \prec_0$$

is dense in $X_{\mathcal{K}}$ and both maps are continuous, it follows that

$$\pi' = \pi.$$

Finally, because (X, x_0) is an ambit, the orbit

$$G_0 \cdot x_0$$

is dense in X . Since π is continuous and G_0 -equivariant, its image is a closed G_0 -invariant subset of X containing

$$\pi(G_0 \cdot \prec_0) = G_0 \cdot x_0.$$

Hence

$$\pi(X_{\mathcal{K}}) = X.$$

Thus $(X_{\mathcal{K}}, \prec_0)$ is universal among such ambits. $\dashv$

We can now combine this with the results of Chapters 5 and 7 and the minimality criterion from the previous section.

Corollary 8.3.3. *Assume that $\mathcal{K}$ has both the Ramsey property and the ordering property. Then $X_{\mathcal{K}}$ is the universal minimal flow of $G_0 = \text{Aut}(\mathbf{F}_0)$.*

Proof. By Corollary 5.3.2, the Ramsey property of the Fraïssé order class $\mathcal{K}$ implies that

$$G = \text{Aut}(\mathbf{F})$$

is extremely amenable. Let Y be a minimal compact G_0 -flow. Restricting the action to the subgroup G , we obtain a compact G -flow, so extreme amenability yields a point

$$y_0 \in Y$$

fixed by every element of G . Since Y is minimal as a G_0 -flow, the orbit

$$G_0 \cdot y_0$$

is dense in Y , and therefore (Y, y_0) is a G_0 -ambit with G -fixed distinguished point.

By Theorem 8.3.2, there exists a surjective G_0 -map

$$\pi: X_{\mathcal{K}} \rightarrow Y.$$

On the other hand, by Theorem 8.2.3, the ordering property implies that $X_{\mathcal{K}}$ is minimal. Hence every minimal compact G_0 -flow is a factor of $X_{\mathcal{K}}$, which is therefore the universal minimal flow of G_0 . $\dashv$

The preceding theorem shows how the Ramsey property and admissible orderings determine the universal minimal flow of the automorphism group of the reduct. In the following chapters we turn to the converse direction, uniqueness questions for such expansions, and the role of Ramsey degrees in the computation of universal minimal flows.

Chapter 9

Examples of Universal Minimal Flows

In Chapter 8 we proved that if a reasonable Fraïssé order class $\mathcal{K}$ has both the Ramsey property and the ordering property, then the flow $X_{\mathcal{K}}$ of admissible orderings is the universal minimal flow of the automorphism group of the reduct. The aim of this chapter is to apply that theorem to concrete classes and to see how different kinds of universal minimal flows arise in practice.

The examples naturally divide into three types. In some cases every linear ordering is admissible, so the universal minimal flow is the full space of linear orderings. In other cases admissible orderings form a proper closed subspace determined by the underlying structure. Finally, there are examples in which the universal minimal flow is finite. We begin with the first phenomenon.

9.1 Cases where the universal minimal flow is the full ordering space

In the examples considered in this section, the ordered expansion is the free one: no compatibility condition is imposed between the additional linear order and the underlying structure. As a result, every linear ordering is admissible, so the flow $X_{\mathcal{K}}$ coincides with the full ordering space. Once the Ramsey and ordering properties are known, the universal minimal flow follows immediately from Chapter 8.

We write $\text{LO}(F_0)$ for the compact space of all linear orderings on the underlying set of a countable structure F_0 .

Theorem 9.1.1. *The following groups have universal minimal flow equal to the full space of linear orderings on their underlying countable structure.*

- (i) *Let Γ be the random graph. Then the universal minimal flow of $\text{Aut}(\Gamma)$ is $\text{LO}(\Gamma)$.*
- (ii) *For each $n \geq 3$, let Γ_n be the random K_n -free graph. Then the universal minimal flow of $\text{Aut}(\Gamma_n)$ is $\text{LO}(\Gamma_n)$.*
- (iii) *The universal minimal flow of S_{∞} is $\text{LO}(\mathbb{N})$.*

- (iv) Let F be the Fraïssé limit of the class of all finite hypergraphs of a fixed finite relational type L_0 . Then the universal minimal flow of $\text{Aut}(F)$ is $\text{LO}(F)$.
- (v) Let $\mathcal{A}$ be a family of finite irreducible hypergraphs of type L_0 , and let $F_{\mathcal{A}}$ be the Fraïssé limit of $\text{Forb}(\mathcal{A})$. Then the universal minimal flow of $\text{Aut}(F_{\mathcal{A}})$ is $\text{LO}(F_{\mathcal{A}})$.
- (vi) Let $U_{\mathbb{Q}}$ be the rational Urysohn space. Then the universal minimal flow of $\text{Aut}(U_{\mathbb{Q}})$ is $\text{LO}(U_{\mathbb{Q}})$.

Proof. In each case, let $\mathcal{K}_0$ be the relevant Fraïssé class and let

$$\mathcal{K} = \mathcal{K}_0 * \mathcal{LO}$$

be its free ordered expansion. By construction, every linear ordering on a finite member of $\mathcal{K}_0$ gives a member of $\mathcal{K}$, so every linear ordering on the Fraïssé limit of $\mathcal{K}_0$ is $\mathcal{K}$ -admissible. Therefore

$$X_{\mathcal{K}} = \text{LO}(F_0).$$

It remains to verify that $\mathcal{K}$ has the Ramsey property and the ordering property. The Ramsey property was established in the preceding chapters: for graphs and K_n -free graphs by the theorem of Nešetřil and Rödl, for finite hypergraphs and $\mathcal{A}$ -free hypergraphs by the corresponding hypergraph Ramsey theorems, and for finite rational metric spaces by Nešetřil's theorem. The ordering property holds for these free ordered expansions. Hence Chapter 8 applies and gives that the universal minimal flow of the automorphism group of the reduct is exactly $X_{\mathcal{K}}$.

This yields (i), (ii), (iv), (v), and (vi). For (iii), take $\mathcal{K}_0$ to be the class of finite sets in the empty language. Then its Fraïssé limit is a countably infinite set, whose automorphism group is S_{∞} , and again

$$X_{\mathcal{K}} = \text{LO}(\mathbb{N}).$$

Thus the universal minimal flow of S_{∞} is $\text{LO}(\mathbb{N})$. ◄

These examples show that when the ordered expansion is completely free, no information is lost in passing from admissible orderings to all linear orderings. In the next section we consider the opposite situation, where admissibility imposes genuine structural restrictions and the universal minimal flow becomes a proper closed subspace of the full ordering space.

9.2 Structured spaces of admissible orderings

We now consider examples in which not every linear ordering is admissible. In these cases the universal minimal flow is still obtained from Chapter 8, but it is a proper closed subspace of the full ordering space, determined by compatibility conditions between the order and the underlying structure.

Theorem 9.2.1. *The following groups have universal minimal flow equal to a proper space of admissible orderings.*

(i) Let

$$\mathbf{F}_0 = \langle \mathbb{Q}, E \rangle$$

be the countable equivalence relation with infinitely many equivalence classes, each of which is dense in $\mathbb{Q}$. Let $X_{\mathcal{K}}$ be the space of all convex linear orderings on $\mathbb{Q}$, where $\mathcal{K}$ is the Fraïssé order class of all finite convexly ordered equivalence relations. Then the universal minimal flow of

$$\text{Aut}(\mathbf{F}_0)$$

is $X_{\mathcal{K}}$.

(ii) Let V_F be the countably infinite-dimensional vector space over a finite field F , and let $X_{\mathcal{K}}$ be the space of all natural orderings on V_F , where $\mathcal{K}$ is the class of naturally ordered finite-dimensional vector spaces over F . Then the universal minimal flow of

$$\text{Aut}(V_F)$$

is $X_{\mathcal{K}}$.

(iii) Let $\mathbf{B}_{\infty}$ be the countable atomless Boolean algebra, and let $X_{\mathcal{K}}$ be the space of all natural orderings on $\mathbf{B}_{\infty}$, where $\mathcal{K}$ is the class of naturally ordered finite Boolean algebras. Then the universal minimal flow of

$$\text{Aut}(\mathbf{B}_{\infty})$$

is $X_{\mathcal{K}}$.

Proof. In each case, the relevant ordered class $\mathcal{K}$ was introduced in Chapter 6. We verify that it fits the framework of Chapter 8.

For (i), let $\mathcal{K}$ be the class of finite convexly ordered equivalence relations. By Chapter 6, $\mathcal{K}$ is a Fraïssé order class with the Ramsey property, and its reduct $\mathcal{K}_0$ is the class of finite equivalence relations. Moreover, Chapter 6 showed that arbitrary linear orderings do not work here: the admissible orderings are exactly the convex ones. Thus the flow $X_{\mathcal{K}}$ is the space of convex linear orderings on $\mathbb{Q}$. Since $\mathcal{K}$ also has the ordering property, Chapter 8 implies that $X_{\mathcal{K}}$ is the universal minimal flow of $\text{Aut}(\mathbf{F}_0)$.

For (ii), let $\mathcal{K}$ be the class of naturally ordered finite-dimensional vector spaces over F . By Chapter 6, this is a Fraïssé order class with the Ramsey property, and its reduct is the class of finite-dimensional vector spaces over F . The admissible orderings are precisely the natural orderings induced by ordered bases. Hence the corresponding flow $X_{\mathcal{K}}$ is the space of natural orderings on V_F . Since $\mathcal{K}$ has the ordering property, Chapter 8 yields that $X_{\mathcal{K}}$ is the universal minimal flow of $\text{Aut}(V_F)$.

For (iii), let $\mathcal{K}$ be the class of naturally ordered finite Boolean algebras. Again, Chapter 6 shows that $\mathcal{K}$ is a Fraïssé order class with the Ramsey property, with reduct the class of finite Boolean algebras. The admissible orderings are precisely the natural orderings induced by orderings of the atoms. Therefore $X_{\mathcal{K}}$ is the space of natural orderings on $\mathbf{B}_{\infty}$. Since $\mathcal{K}$ also has the ordering property, Chapter 8 implies that $X_{\mathcal{K}}$ is the universal minimal flow of $\text{Aut}(\mathbf{B}_{\infty})$. ◻

The significance of these examples is that the universal minimal flow is no longer the full space of all linear orderings. Instead, admissibility imposes a structural constraint: convexity with respect to the equivalence classes, or naturality with respect to a basis or an atomic decomposition. In the Boolean algebra case, this space of admissible orderings admits a more concrete topological description, which we discuss next.

9.3 Boolean algebras and maximal chains

We now return to the Boolean algebra example and record a more concrete description of the corresponding universal minimal flow. Besides the space of natural orderings on the countable atomless Boolean algebra, there is a second model in terms of maximal chains of closed subsets of the Cantor space. The two descriptions are equivalent.

Let

$$\mathcal{C} = 2^{\mathbb{N}}$$

be the Cantor space, and let

$$\Phi(\mathcal{C})$$

denote the compact space of all maximal chains of closed subsets of $\mathcal{C}$, ordered by inclusion. The group

$$\text{Homeo}(\mathcal{C})$$

acts continuously on $\Phi(\mathcal{C})$ by

$$g \cdot \mathcal{M} = \{g(M) : M \in \mathcal{M}\}.$$

By Stone duality, the countable atomless Boolean algebra $\mathbf{B}_{\infty}$ may be identified with the Boolean algebra of clopen subsets of $\mathcal{C}$, and under this identification the group

$$\text{Aut}(\mathbf{B}_{\infty})$$

is naturally isomorphic to

$$\text{Homeo}(\mathcal{C}).$$

The next theorem identifies the universal minimal flow described in Section 2 with the maximal-chain flow on $\mathcal{C}$.

Theorem 9.3.1. *Under the natural identification*

$$\text{Aut}(\mathbf{B}_{\infty}) \cong \text{Homeo}(2^{\mathbb{N}}),$$

the universal minimal flow of $\text{Aut}(\mathbf{B}_{\infty})$ is isomorphic to

$$\Phi(2^{\mathbb{N}}).$$

Equivalently, the space of natural orderings on $\mathbf{B}_{\infty}$ is $\text{Aut}(\mathbf{B}_{\infty})$ -equivariantly homeomorphic to the space of maximal chains of closed subsets of $2^{\mathbb{N}}$.

Proof. We sketch the construction. By Theorem 9.2.1(iii), the universal minimal flow of $\text{Aut}(\mathbf{B}_\infty)$ is the space $X_\mathcal{K}$ of natural orderings on $\mathbf{B}_\infty$.

Given a maximal chain

$$\mathcal{M} \in \Phi(2^\mathbb{N}),$$

one obtains a linear ordering on the atoms of each finite subalgebra of $\mathbf{B}_\infty$ by comparing the unique points at which the members of $\mathcal{M}$ first separate those atoms. Passing to the induced anti-lexicographic orderings, this defines a natural ordering on every finite subalgebra, and these finite orderings are coherent. Hence $\mathcal{M}$ determines an element of $X_\mathcal{K}$.

Conversely, let

$$\prec \in X_\mathcal{K}.$$

The ordering $\prec$ induces, on each finite clopen partition of $2^\mathbb{N}$, a linear ordering of its atoms compatible with refinement. From these finite orderings one reconstructs a unique maximal chain of closed subsets of $2^\mathbb{N}$ by taking the nested family of closed sets determined by initial segments of the induced orderings. This produces an element of $\Phi(2^\mathbb{N})$.

These two constructions are inverse to one another, and they are continuous and $\text{Aut}(\mathbf{B}_\infty)$ -equivariant. Therefore

$$X_\mathcal{K} \cong \Phi(2^\mathbb{N})$$

as $\text{Aut}(\mathbf{B}_\infty)$ -flows. Since $X_\mathcal{K}$ is the universal minimal flow, the same is true of $\Phi(2^\mathbb{N})$. $\dashv$

This identification is useful because it shows that the same universal minimal flow may appear in two rather different forms: combinatorially, as a space of natural orderings on a Fraïssé limit, and topologically, as a space of maximal chains in the Cantor space. In the next section we consider a different phenomenon, where the universal minimal flow is finite.

9.4 A finite universal minimal flow

We conclude with an example in which the universal minimal flow is finite. This shows that, although many of the flows considered so far are infinite compact spaces of orderings, the universal minimal flow can also be a finite homogeneous space.

Fix $n \geq 1$, and let

$$\langle \mathbb{Q}, <, E_n \rangle$$

be the structure consisting of the rationals with their usual ordering together with an equivalence relation E_n having exactly n classes, each of which is dense in $\mathbb{Q}$. Let

$$A_1, \dots, A_n$$

be an enumeration of the E_n -classes, and put

$$\mathbf{F}_n = \langle \mathbb{Q}, <, A_1, \dots, A_n \rangle.$$

We write

$$G_n = \text{Aut}(\langle \mathbb{Q}, <, E_n \rangle) \quad \text{and} \quad H_n = \text{Aut}(\mathbf{F}_n).$$

Thus G_n preserves the equivalence relation and may permute its n dense classes, while H_n preserves each class individually.

Theorem 9.4.1. *For each $n \geq 1$, the subgroup H_n is a clopen normal subgroup of finite index in G_n , and the natural action of G_n on*

$$G_n/H_n$$

is the universal minimal flow of G_n .

Proof. Each element of G_n permutes the n equivalence classes of E_n , so there is a natural continuous homomorphism

$$\pi: G_n \rightarrow S_n$$

sending an automorphism to the permutation it induces on

$$\{A_1, \dots, A_n\}.$$

Its kernel is exactly H_n . Hence H_n is a normal subgroup of G_n of finite index at most $n!$. Since the quotient is finite and discrete, H_n is clopen.

We now show that H_n is extremely amenable. Let $\mathcal{OP}_n$ be the class of all finite structures of the form

$$\langle A, <^A, P_1^A, \dots, P_n^A \rangle,$$

where $<^A$ is a linear ordering of A and

$$P_1^A, \dots, P_n^A$$

form a partition of A into pairwise disjoint unary predicates. This is a Fraïssé order class, and its Fraïssé limit is precisely

$$\mathbf{F}_n = \langle \mathbb{Q}, <, A_1, \dots, A_n \rangle.$$

Moreover, $\mathcal{OP}_n$ has the Ramsey property; this is a special case of the general Ramsey theorem for finite ordered relational structures. Therefore, by Corollary 5.3.2,

$$H_n = \text{Aut}(\mathbf{F}_n)$$

is extremely amenable.

Let now Y be a minimal compact G_n -flow. Restricting the action to the subgroup H_n , extreme amenability gives a point

$$y_0 \in Y$$

fixed by every element of H_n . The orbit map

$$g \mapsto g \cdot y_0$$

is therefore constant on left cosets of H_n , so it factors through a continuous G_n -map

$$\varphi: G_n/H_n \rightarrow Y.$$

Since Y is minimal, the orbit

$$G_n \cdot y_0 = \varphi(G_n/H_n)$$

is dense in Y . But G_n/H_n is finite, so its image is finite and hence closed. Therefore

$$\varphi(G_n/H_n) = Y,$$

and φ is surjective. Thus every minimal compact G_n -flow is a factor of G_n/H_n .

Finally, the action of G_n on G_n/H_n is transitive, hence minimal. It follows that

$$G_n/H_n$$

is the universal minimal flow of G_n . +

For $n = 1$, the quotient G_1/H_1 is trivial, so this recovers the extreme amenability of $\text{Aut}(\mathbb{Q}, <)$. For $n \geq 2$, the universal minimal flow is finite and nontrivial. This contrasts sharply with the earlier examples, where the universal minimal flow was an infinite compact space of orderings.

Chapter 10

Uniqueness of Ramsey Expansions with the Ordering Property

In Chapter 8 we saw that if a reasonable Fraïssé order expansion $\mathcal{K}$ of a Fraïssé class $\mathcal{K}_0$ has both the Ramsey property and the ordering property, then the flow $X_{\mathcal{K}}$ is the universal minimal flow of $\text{Aut}(\text{Flim}(\mathcal{K}_0))$. Since the universal minimal flow is unique up to isomorphism, one expects that such expansions should themselves be unique in an appropriate sense. The aim of this chapter is to make that statement precise.

We first introduce the notion of simple bi-definability between ordered expansions and prove the main uniqueness theorem. We then discuss some strengthenings in special cases and conclude with a minimality result showing that, up to simple bi-definability, expansions with both the Ramsey and ordering properties are the smallest Ramsey expansions of a given reduct class.

10.1 Simple bi-definability and the main uniqueness theorem

Let L_0 be a relational signature, and let

$$L = L_0 \cup \{<\}, \quad L' = L_0 \cup \{<'\},$$

where $<$ and $<'$ are new binary relation symbols. In this section we compare two Fraïssé order expansions of the same reduct class

$$\mathcal{K}_0.$$

We begin by formalizing the notion that one of the two orderings can be recovered from the other by a formula of particularly simple form.

Definition 10.1.1. A formula in a relational language is called *simple* if it is quantifier-free. Equivalently, it is a finite Boolean combination of atomic formulas.

Let $\mathbf{A} = \langle \mathbf{A}_0, \prec^{\mathbf{A}} \rangle$ be an L -structure and let $\varphi(x, y)$ be a simple formula in the language L . We write

$$\varphi^{\mathbf{A}}(a, b)$$

to mean that $\mathbf{A} \models \varphi(a, b)$.

Definition 10.1.2. Let $\mathcal{K}$ be a class of finite L -structures and let $\mathcal{K}'$ be a class of finite L' -structures. We say that $\mathcal{K}$ and $\mathcal{K}'$ are *simply bi-definable* if there exist simple formulas

$$\varphi(x, y) \text{ in } L \quad \text{and} \quad \varphi'(x, y) \text{ in } L'$$

such that the following hold.

For every

$$\mathbf{A} = \langle \mathbf{A}_0, \prec^A \rangle \in \mathcal{K},$$

if we define a binary relation $<'^A$ on A by

$$a <'^A b \iff \mathbf{A} \models \varphi(a, b),$$

then

$$\langle \mathbf{A}_0, <'^A \rangle \in \mathcal{K}'.$$

Similarly, for every

$$\mathbf{B} = \langle \mathbf{B}_0, \prec'^B \rangle \in \mathcal{K}',$$

if we define a binary relation $<^B$ on B by

$$a <^B b \iff \mathbf{B} \models \varphi'(a, b),$$

then

$$\langle \mathbf{B}_0, <^B \rangle \in \mathcal{K}.$$

Moreover, these two constructions are inverse to one another up to isomorphism.

In other words, $\mathcal{K}$ and $\mathcal{K}'$ are simply bi-definable if the two orderings can be recovered from one another by quantifier-free formulas over the common reduct language.

The relevance of this notion is that it preserves the two combinatorial properties that were central in Chapters 5 and 8.

Proposition 10.1.3. *Let $\mathcal{K}$ and $\mathcal{K}'$ be simply bi-definable classes of finite ordered structures with the same reduct class $\mathcal{K}_0$. Then:*

- (i) $\mathcal{K}$ has the Ramsey property if and only if $\mathcal{K}'$ has the Ramsey property;
- (ii) $\mathcal{K}$ has the ordering property if and only if $\mathcal{K}'$ has the ordering property.

Proof. We prove the forward implications; the converse directions are symmetric.

Assume first that $\mathcal{K}$ has the Ramsey property. Let

$$\mathbf{A}' \leq \mathbf{B}'$$

be structures in $\mathcal{K}'$, and let $k \geq 2$. Using simple bi-definability, let $\mathbf{A}, \mathbf{B} \in \mathcal{K}$ be the corresponding structures on the same underlying sets. Choose

$$\mathbf{C} \in \mathcal{K}$$

such that

$$\mathbf{C} \rightarrow (\mathbf{B})_k^{\mathbf{A}}.$$

Let $\mathbf{C}' \in \mathcal{K}'$ be the structure corresponding to $\mathbf{C}$ under the defining formula φ . Since embeddings in one class are exactly embeddings in the other after translating the order relation by φ and φ' , the partition relation for $\mathbf{C}$ transfers to the corresponding partition relation for $\mathbf{C}'$. Hence

$$\mathbf{C}' \rightarrow (\mathbf{B}')_k^{\mathbf{A}'}$$

Thus $\mathcal{K}'$ has the Ramsey property.

Now assume that $\mathcal{K}$ has the ordering property. Let

$$\mathbf{A}_0 \in \mathcal{K}_0.$$

Choose an expansion

$$\mathbf{A}' = \langle \mathbf{A}_0, \prec' \rangle \in \mathcal{K}',$$

and let $\mathbf{A} \in \mathcal{K}$ be the corresponding structure. By the ordering property for $\mathcal{K}$, there exists

$$\mathbf{B}_0 \in \mathcal{K}_0$$

such that every member of $\mathcal{K}$ with reduct $\mathbf{B}_0$ contains a copy of $\mathbf{A}$. Let

$$\mathbf{B}' = \langle \mathbf{B}_0, \triangleleft \rangle \in \mathcal{K}'$$

be arbitrary, and let $\mathbf{B} \in \mathcal{K}$ be the corresponding structure. Then

$$\mathbf{A} \leq \mathbf{B},$$

and therefore, translating back by simple bi-definability,

$$\mathbf{A}' \leq \mathbf{B}'.$$

Thus $\mathcal{K}'$ has the ordering property. +

We now come to the main uniqueness theorem.

Theorem 10.1.4. *Let $\mathcal{K}$ and $\mathcal{K}'$ be reasonable Fraïssé order classes in the signatures L and L' , and suppose that they have the same reduct*

$$\mathcal{K}_0.$$

Assume that both $\mathcal{K}$ and $\mathcal{K}'$ have the Ramsey property and the ordering property. Then $\mathcal{K}$ and $\mathcal{K}'$ are simply bi-definable.

Proof. Let

$$\mathbf{F} = \langle \mathbf{F}_0, \prec_0 \rangle = \text{Flim}(\mathcal{K}) \quad \text{and} \quad \mathbf{F}' = \langle \mathbf{F}_0, \prec'_0 \rangle = \text{Flim}(\mathcal{K}'),$$

where the two limits are taken on the same reduct $\mathbf{F}_0 = \text{Flim}(\mathcal{K}_0)$, using Chapter 7.

Let

$$G_0 = \text{Aut}(\mathbf{F}_0), \quad X_{\mathcal{K}} = \overline{G_0 \cdot \prec_0}, \quad X_{\mathcal{K}'} = \overline{G_0 \cdot \prec'_0}.$$

By Chapter 8, since both $\mathcal{K}$ and $\mathcal{K}'$ have the Ramsey property and the ordering property, the flows $X_{\mathcal{K}}$ and $X_{\mathcal{K}'}$ are both universal minimal flows of G_0 . Hence there exists a G_0 -flow isomorphism

$$\pi: X_{\mathcal{K}} \rightarrow X_{\mathcal{K}'}.$$

We claim that

$$\pi(G_0 \cdot \prec_0) = G_0 \cdot \prec'_0.$$

Indeed, for a linear ordering $\prec \in X_{\mathcal{K}}$, the condition

$$\prec \in G_0 \cdot \prec_0$$

is equivalent to the statement that the ordered structure

$$\langle \mathbf{F}_0, \prec \rangle$$

is isomorphic to $\mathbf{F}$, or equivalently that it has age $\mathcal{K}$ and the extension property. This witnesses the triangle condition for $\mathcal{K}'$. Applying Corollary 10.2.3 with the roles of $\mathcal{K}$ and $\mathcal{K}'$ interchanged, we obtain

$$\mathcal{K} = \mathcal{K}' \quad \text{or} \quad \mathcal{K} = (\mathcal{K}')^*.$$

Equivalently,

$$\mathcal{K}' = \mathcal{K} \quad \text{or} \quad \mathcal{K}' = \mathcal{K}^*.$$

Similarly, $G_0 \cdot \prec'_0$ is a dense G_δ subset of $X_{\mathcal{K}'}$. Because π is a homeomorphism, it must send the unique dense G_δ orbit of $X_{\mathcal{K}}$ onto the unique dense G_δ orbit of $X_{\mathcal{K}'}$. This proves the claim.

Composing π with a suitable element of G_0 if necessary, we may therefore assume that

$$\pi(\prec_0) = \prec'_0.$$

Let

$$G = \text{Aut}(\mathbf{F}) \quad \text{and} \quad G' = \text{Aut}(\mathbf{F}').$$

Then

$$G = \text{Stab}_{G_0}(\prec_0) \quad \text{and} \quad G' = \text{Stab}_{G_0}(\prec'_0).$$

Since π is G_0 -equivariant and sends $\prec_0$ to $\prec'_0$, it follows that

$$G = G'.$$

We now define $<'$ from $<$ by a simple formula. Since $G = G'$, the relation

$$x <' y$$

is invariant under $\text{Aut}(\mathbf{F})$. Because $\mathbf{F}$ is ultrahomogeneous in the relational language L , every $\text{Aut}(\mathbf{F})$ -invariant binary relation is a union of 2-orbits, and each such orbit is determined by the quantifier-free type of the ordered pair in $\mathbf{F}$. There are only finitely many such 2-orbits, so the relation $<'$ is defined in $\mathbf{F}$ by a simple formula

$$\varphi(x, y)$$

in the language L .

By the same argument, the relation $<$ is defined in $\mathbf{F}'$ by a simple formula

$$\varphi'(x, y)$$

in the language L' .

Finally, let

$$\mathbf{A} = \langle \mathbf{A}_0, \prec^A \rangle \in \mathcal{K}.$$

Since $\mathcal{K} = \text{Age}(\mathbf{F})$, the structure $\mathbf{A}$ embeds into $\mathbf{F}$. Restricting the formula φ to the image of such an embedding defines on A a second linear order $<'^A$, and the resulting structure

$$\langle \mathbf{A}_0, <'^A \rangle$$

belongs to $\text{Age}(\mathbf{F}') = \mathcal{K}'$. The same reasoning applies in the other direction using φ' . Because both formulas come from the same subgroup

$$G = G' \leq G_0,$$

the two constructions are inverse to one another up to isomorphism. Therefore $\mathcal{K}$ and $\mathcal{K}'$ are simply bi-definable. $\dashv$

Remark 10.1.5. If the signatures are finite, the same argument yields first-order bi-definability in the usual sense. For the purposes of the present thesis, however, the quantifier-free version is the natural one and is sufficient.

The theorem shows that a reasonable Ramsey expansion with the ordering property is unique up to a very rigid notion of definitional equivalence. In the next section we record some strengthenings and examples of this phenomenon.

10.2 Strengthenings and examples

We now record two useful strengthenings of Theorem 10.1.4, together with some examples.

Proposition 10.2.1. *In the context of Theorem 10.1.4, suppose that $\mathcal{K}$ and $\mathcal{K}'$ are comparable under inclusion, that is,*

$$\mathcal{K} \subseteq \mathcal{K}' \quad \text{or} \quad \mathcal{K}' \subseteq \mathcal{K}.$$

Then

$$\mathcal{K} = \mathcal{K}'.$$

In particular, if

$$\mathcal{K} = \mathcal{K}_0 * \mathcal{LO},$$

then $\mathcal{K} = \mathcal{K}'$.

Proof. Assume, for example, that

$$\mathcal{K} \subseteq \mathcal{K}'.$$

Suppose toward a contradiction that $\mathcal{K} \neq \mathcal{K}'$. Then there exists

$$\mathbf{A} = \langle \mathbf{A}_0, \prec \rangle \in \mathcal{K}' \setminus \mathcal{K}.$$

Since $\mathcal{K}'$ has the ordering property, there exists

$$\mathbf{B}_0 \in \mathcal{K}_0$$

such that for every expansion

$$\mathbf{B}' = \langle \mathbf{B}_0, \prec' \rangle \in \mathcal{K}'$$

one has

$$\mathbf{A} \leq \mathbf{B}'.$$

Choose any expansion

$$\mathbf{B} = \langle \mathbf{B}_0, \prec' \rangle \in \mathcal{K}.$$

Since $\mathcal{K} \subseteq \mathcal{K}'$, we also have $\mathbf{B} \in \mathcal{K}'$, and therefore

$$\mathbf{A} \leq \mathbf{B}.$$

But $\mathcal{K}$ is hereditary, so this implies

$$\mathbf{A} \in \mathcal{K},$$

a contradiction. Hence $\mathcal{K} = \mathcal{K}'$.

For the final statement, note that if

$$\mathcal{K} = \mathcal{K}_0 * \mathcal{LO},$$

then every expansion in $\mathcal{K}'$ also belongs to $\mathcal{K}$, so

$$\mathcal{K}' \subseteq \mathcal{K}.$$

The first part of the proposition therefore gives $\mathcal{K} = \mathcal{K}'$. –

Thus in all the free-expansion examples from Chapter 9, the expansion with both the Ramsey and ordering properties is uniquely determined.

There is another natural way in which two expansions may differ. If

$$\mathcal{K}$$

is a class of ordered structures, let

$$\mathcal{K}^* = \{ \langle \mathbf{A}_0, \prec^* \rangle : \langle \mathbf{A}_0, \prec \rangle \in \mathcal{K} \},$$

where

$$a \prec^* b \iff b \prec a.$$

Then $\mathcal{K}^*$ is simply bi-definable with $\mathcal{K}$, so by Proposition 10.1.3, the classes $\mathcal{K}$ and $\mathcal{K}^*$ either both have, or both fail to have, the Ramsey and ordering properties. The next condition shows that in some cases these are the only possibilities.

Given

$$\mathbf{A} = \langle \mathbf{A}_0, \prec \rangle \in \mathcal{K} \quad \text{and} \quad a, b \in A_0$$

with

$$a \prec b,$$

we write

$$\text{tp}_{\mathbf{A}}(a, b)$$

for the atomic type of the ordered pair (a, b) in the language L . Equivalently, this is the quantifier-free type of (a, b) in $\mathbf{A}$.

Definition 10.2.2. We say that $\mathcal{K}$ satisfies the *triangle condition* if for any two distinct pair-types

$$\sigma \neq \tau$$

realized in members of $\mathcal{K}$, there exist

$$\mathbf{A} = \langle \mathbf{A}_0, \prec \rangle \in \mathcal{K} \quad \text{and} \quad a \prec b \prec c$$

in A_0 such that either

$$\text{tp}_{\mathbf{A}}(a, b) = \text{tp}_{\mathbf{A}}(b, c) = \sigma \quad \text{and} \quad \text{tp}_{\mathbf{A}}(a, c) = \tau,$$

or

$$\text{tp}_{\mathbf{A}}(a, b) = \text{tp}_{\mathbf{A}}(b, c) = \tau \quad \text{and} \quad \text{tp}_{\mathbf{A}}(a, c) = \sigma.$$

The point is that the triangle condition forces any alternative admissible ordering to agree either with the original ordering or with its reverse.

Corollary 10.2.3. *In the context of Theorem 10.1.4, assume that $\mathcal{K}$ satisfies the triangle condition. Then*

$$\mathcal{K}' = \mathcal{K} \quad \text{or} \quad \mathcal{K}' = \mathcal{K}^*.$$

Proof. By Theorem 10.1.4, the classes $\mathcal{K}$ and $\mathcal{K}'$ are simply bi-definable. In the notation of its proof, we may therefore realize the two Fraïssé limits on the same reduct

$$\mathbf{F}_0$$

and write

$$\mathbf{F} = \langle \mathbf{F}_0, \prec_0 \rangle, \quad \mathbf{F}' = \langle \mathbf{F}_0, \prec'_0 \rangle,$$

with

$$\text{Aut}(\mathbf{F}) = \text{Aut}(\mathbf{F}').$$

It is enough to show that

$$\prec'_0 = \prec_0 \quad \text{or} \quad \prec'_0 = \prec_0^*.$$

Assume this fails. Then there exist

$$a, b, c, d \in F_0$$

such that

$$a \prec_0 b, \quad c \prec_0 d,$$

but

$$a \prec'_0 b, \quad d \prec'_0 c.$$

Let

$$\sigma = \text{tp}_{\mathbf{F}}(a, b) \quad \text{and} \quad \tau = \text{tp}_{\mathbf{F}}(c, d).$$

Then

$$\sigma \neq \tau,$$

since $\prec'_0$ is invariant under $\text{Aut}(\mathbf{F}) = \text{Aut}(\mathbf{F}')$.

By the triangle condition, there exist

$$\mathbf{A} \in \mathcal{K} \quad \text{and} \quad u \prec v \prec w$$

in A_0 such that either

$$\text{tp}_{\mathbf{A}}(u, v) = \text{tp}_{\mathbf{A}}(v, w) = \sigma \quad \text{and} \quad \text{tp}_{\mathbf{A}}(u, w) = \tau,$$

or the same statement with σ and τ interchanged. Realizing $\mathbf{A}$ as a substructure of $\mathbf{F}$, the relation $\prec'_0$ must orient the pairs of type σ as in (a, b) and the pairs of type τ as in (d, c) . In the first case this gives

$$u \prec'_0 v \prec'_0 w \quad \text{and} \quad w \prec'_0 u,$$

a contradiction; in the second case one obtains

$$w \prec'_0 v \prec'_0 u \quad \text{and} \quad u \prec'_0 w,$$

again a contradiction. Therefore

$$\prec'_0 = \prec_0 \quad \text{or} \quad \prec'_0 = \prec_0^*,$$

and hence

$$\mathcal{K}' = \mathcal{K} \quad \text{or} \quad \mathcal{K}' = \mathcal{K}^*.$$

+

We now record two examples.

Example 10.2.4. Let $\mathcal{K}_0$ be the class of all finite equivalence relations, and let $\mathcal{K}$ be the class of all finite convexly ordered equivalence relations. Then $\mathcal{K}$ is the unique reasonable Fraïssé order expansion of $\mathcal{K}_0$ with both the Ramsey property and the ordering property.

Proof. Let $\mathcal{K}'$ be any reasonable Fraïssé order expansion of $\mathcal{K}_0$ with both properties. If

$$\mathcal{K}' \subseteq \mathcal{K},$$

then Proposition 10.2.1 yields

$$\mathcal{K}' = \mathcal{K}.$$

So suppose instead that $\mathcal{K}' \not\subseteq \mathcal{K}$. Then $\mathcal{K}'$ contains some

$$\mathbf{A}' = \langle \mathbf{A}_0, E, \prec' \rangle$$

with

$$a \prec' b \prec' c, \quad a E c, \quad a \not E b, \quad b \not E c,$$

for some $a, b, c \in A_0$. This witnesses the triangle condition for $\mathcal{K}'$, and therefore Corollary 10.2.3 implies that

$$\mathcal{K}' = \mathcal{K} \quad \text{or} \quad \mathcal{K}' = \mathcal{K}^*.$$

But $\mathcal{K}^* = \mathcal{K}$ in this case, since reversing a convex ordering is again convex. Hence

$$\mathcal{K}' = \mathcal{K}.$$

+

Example 10.2.5. Let $\mathcal{K}_0$ be the class of all finite posets, and let $\mathcal{K}$ be the class of all finite structures

$$\langle A, \sqsubset, \prec \rangle$$

where $\langle A, \sqsubset \rangle$ is a poset and $\prec$ is a linear ordering extending $\sqsubset$. Then the only reasonable Fraïssé order expansions of $\mathcal{K}_0$ with both the Ramsey and ordering properties are $\mathcal{K}$ and $\mathcal{K}^*$.

Proof. This class has the Ramsey and ordering properties. It also satisfies the triangle condition: one may realize two distinct pair-types by appropriately choosing a three-element chain

$$a \prec b \prec c$$

and using the compatibility of $\prec$ with the partial order. Corollary 10.2.3 therefore gives the conclusion. +

The situation is not always so rigid. In some classes there are several non-isomorphic expansions satisfying the Ramsey and ordering properties. This occurs, for example, for finite Boolean algebras and for finite-dimensional vector spaces over a finite field. Since the classification of all such expansions would take us too far from the main line of the thesis, we do not pursue it here.

The next section explains the conceptual meaning of the uniqueness theorem from the viewpoint of minimality among Ramsey expansions.

10.3 Minimality among Ramsey expansions

We now explain the sense in which expansions with both the Ramsey property and the ordering property are minimal among Ramsey expansions of a fixed reduct class.

Theorem 10.3.1. *Let $\mathcal{K}_0$ be a Fraïssé class in a relational signature L_0 , let*

$$L = L_0 \cup \{<\},$$

and let $\mathcal{K}, \mathcal{K}'$ be reasonable Fraïssé order classes in the language L expanding $\mathcal{K}_0$. Assume that $\mathcal{K}$ has the Ramsey property and that $\mathcal{K}'$ has both the Ramsey property and the ordering property. Then there exists a reasonable Fraïssé order class $\mathcal{K}''$ in the language L such that

$$\mathcal{K}'' \subseteq \mathcal{K}$$

and $\mathcal{K}''$ is simply bi-definable with $\mathcal{K}'$.

Proof. Let

$$\mathbf{F} = \langle \mathbf{F}_0, \prec_0 \rangle = \text{Flim}(\mathcal{K}), \quad \mathbf{F}' = \langle \mathbf{F}_0, \prec'_0 \rangle = \text{Flim}(\mathcal{K}'),$$

where we identify the two reduct limits using Chapter 7. Put

$$G_0 = \text{Aut}(\mathbf{F}_0), \quad G = \text{Aut}(\mathbf{F}).$$

Let

$$X_{\mathcal{K}} = \overline{G_0 \cdot \prec_0} \quad \text{and} \quad X_{\mathcal{K}'} = \overline{G_0 \cdot \prec'_0}.$$

Since $\mathcal{K}$ has the Ramsey property, Corollary 5.3.2 shows that

$$G = \text{Aut}(\mathbf{F})$$

is extremely amenable. Let Y be any minimal subflow of $X_{\mathcal{K}}$. We claim that Y is the universal minimal flow of G_0 .

Indeed, let Z be any minimal compact G_0 -flow. Restricting the action to the subgroup G , extreme amenability yields a point

$$z_0 \in Z$$

fixed by every element of G . By Theorem 8.3.2, there exists a surjective G_0 -map

$$\pi: X_{\mathcal{K}} \rightarrow Z.$$

Since Y is a minimal subflow of $X_{\mathcal{K}}$, the image $\pi(Y)$ is a nonempty closed G_0 -invariant subset of Z , hence

$$\pi(Y) = Z.$$

Thus every minimal compact G_0 -flow is a factor of Y , so Y is universal minimal.

On the other hand, since $\mathcal{K}'$ has both the Ramsey property and the ordering property, Corollary 8.3.3 implies that

$$X_{\mathcal{K}'}$$

is also the universal minimal flow of G_0 . Therefore there is a G_0 -flow isomorphism

$$\theta: X_{\mathcal{K}'} \rightarrow Y.$$

Replacing θ by $g \cdot \theta$ for a suitable $g \in G_0$ if necessary, we may assume that

$$\theta(\prec'_0) = \prec''_0 \in Y.$$

Since θ is G_0 -equivariant, the stabilizers of $\prec'_0$ and $\prec''_0$ in G_0 coincide. Hence

$$\text{Aut}(\langle \mathbf{F}_0, \prec'_0 \rangle) = \text{Aut}(\langle \mathbf{F}_0, \prec''_0 \rangle).$$

Exactly as in the proof of Theorem 10.1.4, it follows that there are simple formulas defining $\prec'_0$ from $\prec''_0$ and $\prec''_0$ from $\prec'_0$. In particular,

$$\langle \mathbf{F}_0, \prec''_0 \rangle$$

is ultrahomogeneous.

Let

$$\mathcal{K}'' = \text{Age}(\langle \mathbf{F}_0, \prec''_0 \rangle).$$

Then $\mathcal{K}''$ is a reasonable Fraïssé order class expanding $\mathcal{K}_0$, and by construction it is simply bi-definable with $\mathcal{K}'$.

It remains to show that

$$\mathcal{K}'' \subseteq \mathcal{K}.$$

Since

$$\prec''_0 \in Y \subseteq X_{\mathcal{K}},$$

Proposition 8.1.1 shows that every finite substructure of

$$\langle \mathbf{F}_0, \prec''_0 \rangle$$

belongs to $\mathcal{K}$. Hence

$$\text{Age}(\langle \mathbf{F}_0, \prec''_0 \rangle) \subseteq \mathcal{K},$$

that is,

$$\mathcal{K}'' \subseteq \mathcal{K}.$$

This completes the proof. ◻

Thus, up to simple bi-definability, the expansions of a given reduct class that have both the Ramsey property and the ordering property are the smallest among those that have the Ramsey property. The proof also suggests a natural strategy: starting from a Ramsey expansion $\mathcal{K}$, one may try to pass to a minimal subflow of $X_{\mathcal{K}}$ and recover from it a smaller expansion with better dynamical properties.

In the next chapter this idea is developed further through Ramsey degrees, which allow one to treat situations in which no reasonable expansion with the full Ramsey property is available.

Chapter 11

Ramsey Degrees and Converse Results

In the previous chapters we studied classes with the Ramsey property and saw how ordered Ramsey expansions determine the universal minimal flow of the automorphism group of the Fraïssé limit. However, many natural Fraïssé classes do not themselves satisfy the Ramsey property. The purpose of this chapter is to explain that, even in that situation, there is still a finite combinatorial invariant measuring the extent to which the Ramsey property fails.

This invariant is the *Ramsey degree* of a finite structure. Roughly speaking, if $\mathbf{A}_0 \in \mathcal{K}_0$, then the Ramsey degree of $\mathbf{A}_0$ is the least number t such that, after passing to a sufficiently large copy of any ambient structure, every coloring of copies of $\mathbf{A}_0$ can be reduced on some substructure to at most t colors. When $\mathcal{K}_0$ admits a reasonable ordered expansion with the Ramsey and ordering properties, these degrees can be computed explicitly in terms of the admissible orderings on $\mathbf{A}_0$.

The chapter has two aims. First, we define these degrees and show how they arise from ordered Ramsey expansions. Second, we prove converse results showing that the combinatorial bounds given by Ramsey degrees recover both the Ramsey property and, ultimately, the universal minimal flow description from the previous chapter.

11.1 Patterns and Ramsey degrees

Throughout this chapter, L_0 is a signature, $\mathcal{K}_0$ is a Fraïssé class in L_0 , and $\mathcal{K}$ is a hereditary order class in the expanded signature

$$L = L_0 \cup \{<\}$$

such that

$$\mathcal{K}|_{L_0} = \mathcal{K}_0.$$

Thus $\mathcal{K}$ is an order expansion of $\mathcal{K}_0$.

Let $\mathbf{A}_0 \in \mathcal{K}_0$. We write

$$X_{\mathcal{K}}^{\mathbf{A}_0} = \{<: < \text{ is a linear ordering on } A_0 \text{ and } \langle \mathbf{A}_0, < \rangle \in \mathcal{K}\}.$$

So $X_{\mathcal{K}}^{\mathbf{A}_0}$ is the set of all $\mathcal{K}$ -admissible linear orderings on the underlying set of $\mathbf{A}_0$.

The group $\text{Aut}(\mathbf{A}_0)$ acts on $X_{\mathcal{K}}^{\mathbf{A}_0}$ in the natural way: if $g \in \text{Aut}(\mathbf{A}_0)$ and $\prec \in X_{\mathcal{K}}^{\mathbf{A}_0}$, then $g \cdot \prec$ is the linear order on A_0 defined by

$$x (g \cdot \prec) y \iff g^{-1}(x) \prec g^{-1}(y).$$

Since $\mathcal{K}$ is closed under isomorphism, we have $g \cdot \prec \in X_{\mathcal{K}}^{\mathbf{A}_0}$ whenever $\prec \in X_{\mathcal{K}}^{\mathbf{A}_0}$.

Definition 11.1.1. Let $\prec \in X_{\mathcal{K}}^{\mathbf{A}_0}$. The $\mathbf{A}_0$ -pattern of $\prec$ (with respect to $\mathcal{K}$) is its orbit under the action of $\text{Aut}(\mathbf{A}_0)$ on $X_{\mathcal{K}}^{\mathbf{A}_0}$.

Equivalently, two orderings $\prec, \prec' \in X_{\mathcal{K}}^{\mathbf{A}_0}$ have the same pattern if and only if

$$\langle \mathbf{A}_0, \prec \rangle \cong \langle \mathbf{A}_0, \prec' \rangle.$$

Thus patterns record the isomorphism types of admissible ordered expansions of $\mathbf{A}_0$.

We write

$$t_{\mathcal{K}}(\mathbf{A}_0) = |X_{\mathcal{K}}^{\mathbf{A}_0} / \text{Aut}(\mathbf{A}_0)|$$

for the number of $\mathbf{A}_0$ -patterns. Since the action of $\text{Aut}(\mathbf{A}_0)$ on $X_{\mathcal{K}}^{\mathbf{A}_0}$ is free, each orbit has cardinality $|\text{Aut}(\mathbf{A}_0)|$. Hence

$$t_{\mathcal{K}}(\mathbf{A}_0) = \frac{|X_{\mathcal{K}}^{\mathbf{A}_0}|}{|\text{Aut}(\mathbf{A}_0)|}.$$

To state the relevant partition relation, let $\mathbf{A}_0 \leq \mathbf{B}_0 \leq \mathbf{C}_0$ in $\mathcal{K}_0$, and let $k, t \geq 1$. We write

$$\mathbf{C}_0 \rightarrow (\mathbf{B}_0)_{k,t}^{\mathbf{A}_0}$$

to mean that for every coloring

$$c : \binom{\mathbf{C}_0}{\mathbf{A}_0} \rightarrow \{1, \dots, k\},$$

there is some

$$\mathbf{B}'_0 \in \binom{\mathbf{C}_0}{\mathbf{B}_0}$$

such that the restriction of c to

$$\binom{\mathbf{B}'_0}{\mathbf{A}_0}$$

takes at most t values. In particular,

$$\mathbf{C}_0 \rightarrow (\mathbf{B}_0)_{k,1}^{\mathbf{A}_0} \iff \mathbf{C}_0 \rightarrow (\mathbf{B}_0)_k^{\mathbf{A}_0}.$$

For later use, suppose $\mathbf{A}_0 \leq \mathbf{B}_0$ in $\mathcal{K}_0$ and

$$\langle \mathbf{B}_0, <_{\mathbf{B}_0} \rangle \in \mathcal{K}.$$

We then define

$$t_{\mathcal{K}}(\mathbf{A}_0, \mathbf{B}_0, <_{\mathbf{B}_0})$$

to be the number of $\mathbf{A}_0$ -patterns represented by orderings $\prec \in X_{\mathcal{K}}^{\mathbf{A}_0}$ such that

$$\langle \mathbf{A}_0, \prec \rangle \leq \langle \mathbf{B}_0, <_{\mathbf{B}_0} \rangle.$$

In other words, $t_{\mathcal{K}}(\mathbf{A}_0, \mathbf{B}_0, <_{\mathbf{B}_0})$ counts the patterns of copies of $\mathbf{A}_0$ that actually occur inside the ordered structure $\langle \mathbf{B}_0, <_{\mathbf{B}_0} \rangle$. Clearly,

$$t_{\mathcal{K}}(\mathbf{A}_0, \mathbf{B}_0, <_{\mathbf{B}_0}) \leq t_{\mathcal{K}}(\mathbf{A}_0).$$

We can now define the main combinatorial invariant of the chapter.

Definition 11.1.2. Let $\mathbf{A}_0 \in \mathcal{K}_0$. The *Ramsey degree* of $\mathbf{A}_0$ in $\mathcal{K}_0$ is the least integer t , if it exists, such that for every $\mathbf{B}_0 \in \mathcal{K}_0$ with $\mathbf{A}_0 \leq \mathbf{B}_0$ and every integer $k \geq 2$, there is $\mathbf{C}_0 \in \mathcal{K}_0$ with $\mathbf{C}_0 \geq \mathbf{B}_0$ and

$$\mathbf{C}_0 \rightarrow (\mathbf{B}_0)_{k,t}^{\mathbf{A}_0}.$$

If no such integer exists, we put

$$t(\mathbf{A}_0, \mathcal{K}_0) = \infty.$$

Thus $t(\mathbf{A}_0, \mathcal{K}_0)$ measures how far $\mathbf{A}_0$ is from satisfying the usual Ramsey property inside the class $\mathcal{K}_0$. The case $t(\mathbf{A}_0, \mathcal{K}_0) = 1$ is exactly the ordinary Ramsey statement for copies of $\mathbf{A}_0$.

The guiding principle of this chapter is that, when $\mathcal{K}_0$ admits a suitable ordered Ramsey expansion $\mathcal{K}$, the Ramsey degree of $\mathbf{A}_0$ is determined by the finite set of admissible patterns on $\mathbf{A}_0$. In the next section we make this precise.

A basic example is given by the class $\mathcal{G}$ of finite graphs. This class does not have the Ramsey property, but its standard ordered expansion by linear orders does. If $\mathbf{A}_0 \in \mathcal{G}$, then every linear ordering of A_0 is admissible, and two such orderings have the same pattern exactly when they differ by an automorphism of $\mathbf{A}_0$. Thus

$$t(\mathbf{A}_0, \mathcal{G}) = \frac{|A_0|!}{|\text{Aut}(\mathbf{A}_0)|}.$$

This illustrates the general picture: Ramsey degrees arise from counting the admissible order-types of a structure up to automorphism.

11.2 Ramsey degrees from Ramsey expansions

We now show that if the ordered class $\mathcal{K}$ has the Ramsey property, then the number of admissible patterns gives an upper bound on the number of colors that must remain on a large substructure. If $\mathcal{K}$ also has the ordering property, then this bound is optimal. In particular, the Ramsey degree of $\mathbf{A}_0$ is exactly the number of $\mathcal{K}$ -patterns on $\mathbf{A}_0$.

Proposition 11.2.1. *Assume that $\mathcal{K}$ has the Ramsey property. Let $\mathbf{A}_0 \leq \mathbf{B}_0$ in $\mathcal{K}_0$, let*

$$\langle \mathbf{B}_0, <_{\mathbf{B}_0} \rangle \in \mathcal{K},$$

and let $k \geq 2$. Then there is $\mathbf{C}_0 \geq \mathbf{B}_0$ in $\mathcal{K}_0$ such that

$$\mathbf{C}_0 \rightarrow (\mathbf{B}_0)_{k, t_{\mathcal{K}}(\mathbf{A}_0, \mathbf{B}_0, <_{\mathbf{B}_0})}^{\mathbf{A}_0}.$$

In particular, for every hereditary class $\mathcal{K}' \subseteq \mathcal{K}$ with $\mathcal{K}'|L_0 = \mathcal{K}_0$, there is $\mathbf{C}_0 \geq \mathbf{B}_0$ in $\mathcal{K}_0$ such that

$$\mathbf{C}_0 \rightarrow (\mathbf{B}_0)_{k, t_{\mathcal{K}'}(\mathbf{A}_0)}^{\mathbf{A}_0}.$$

Proof. Put

$$t = t_{\mathcal{K}}(\mathbf{A}_0, \mathbf{B}_0, <_{\mathbf{B}_0}).$$

Choose representatives

$$<_1, \dots, <_t$$

for the $\mathbf{A}_0$ -patterns realized inside $\langle \mathbf{B}_0, <_{\mathbf{B}_0} \rangle$.

We define inductively ordered structures

$$\langle \mathbf{D}_i, <_{\mathbf{D}_i} \rangle \in \mathcal{K} \quad (0 \leq i \leq t)$$

as follows. Let

$$\langle \mathbf{D}_0, <_{\mathbf{D}_0} \rangle = \langle \mathbf{B}_0, <_{\mathbf{B}_0} \rangle,$$

and, for each $1 \leq i \leq t$, choose

$$\langle \mathbf{D}_i, <_{\mathbf{D}_i} \rangle \in \mathcal{K}$$

such that

$$\langle \mathbf{D}_i, <_{\mathbf{D}_i} \rangle \rightarrow \left(\langle \mathbf{D}_{i-1}, <_{\mathbf{D}_{i-1}} \rangle \right)_k^{\langle \mathbf{A}_0, <_i \rangle}.$$

Set

$$\mathbf{C}_0 = \mathbf{D}_t.$$

We claim that $\mathbf{C}_0$ has the required property. Let

$$c : \binom{\mathbf{C}_0}{\mathbf{A}_0} \rightarrow \{1, \dots, k\}$$

be any coloring of copies of $\mathbf{A}_0$ in $\mathbf{C}_0$.

We now work downward from $\langle \mathbf{D}_t, <_{\mathbf{D}_t} \rangle$. For the pattern represented by $<_t$, define a coloring

$$c_t : \binom{\langle \mathbf{D}_t, <_{\mathbf{D}_t} \rangle}{\langle \mathbf{A}_0, <_t \rangle} \rightarrow \{1, \dots, k\}$$

by sending a copy

$$\langle \mathbf{A}'_0, <_{\mathbf{D}_t} \mid \mathbf{A}'_0 \rangle \cong \langle \mathbf{A}_0, <_t \rangle$$

to the color $c(\mathbf{A}'_0)$ of its reduct. By the choice of $\langle \mathbf{D}_t, <_{\mathbf{D}_t} \rangle$, there is a copy

$$\langle \mathbf{D}'_{t-1}, <_{\mathbf{D}'_{t-1}} \rangle \cong \langle \mathbf{D}_{t-1}, <_{\mathbf{D}_{t-1}} \rangle$$

inside $\langle \mathbf{D}_t, <_{\mathbf{D}_t} \rangle$ on which every copy of

$$\langle \mathbf{A}_0, <_t \rangle$$

has the same c_t -color, say k_t .

Next define

$$c_{t-1} : \left(\langle \mathbf{D}'_{t-1}, <_{\mathbf{D}'_{t-1}} \rangle \right) \rightarrow \{1, \dots, k\}$$

in the same way, by pulling back the coloring c from reducts. Again by the Ramsey property, there is a copy

$$\langle \mathbf{D}'_{t-2}, <_{\mathbf{D}'_{t-2}} \rangle \cong \langle \mathbf{D}_{t-2}, <_{\mathbf{D}_{t-2}} \rangle$$

inside $\langle \mathbf{D}'_{t-1}, <_{\mathbf{D}'_{t-1}} \rangle$ on which every copy of

$$\langle \mathbf{A}_0, <_{t-1} \rangle$$

has the same color, say k_{t-1} .

Continuing in this way, after t steps we obtain a copy

$$\langle \mathbf{B}'_0, <_{\mathbf{B}'_0} \rangle \cong \langle \mathbf{B}_0, <_{\mathbf{B}_0} \rangle$$

inside $\langle \mathbf{C}_0, <_{\mathbf{D}_t} \rangle$ together with colors

$$k_1, \dots, k_t \in \{1, \dots, k\}$$

such that, for each $i \in \{1, \dots, t\}$, every copy of

$$\langle \mathbf{A}_0, <_i \rangle$$

inside $\langle \mathbf{B}'_0, <_{\mathbf{B}'_0} \rangle$ has color k_i .

Now let

$$\mathbf{A}'_0 \in \binom{\mathbf{B}'_0}{\mathbf{A}_0}.$$

The induced ordering $<_{\mathbf{B}'_0} \upharpoonright \mathbf{A}'_0$ realizes one of the patterns represented by some $<_i$. Hence

$$c(\mathbf{A}'_0) = k_i.$$

Therefore the coloring c takes at most the t values

$$k_1, \dots, k_t$$

on

$$\binom{\mathbf{B}'_0}{\mathbf{A}_0}.$$

This proves

$$\mathbf{C}_0 \rightarrow (\mathbf{B}_0)_{k,t}^{\mathbf{A}_0}.$$

For the final claim, note that

$$t_{\mathcal{K}}(\mathbf{A}_0, \mathbf{B}_0, <_{\mathbf{B}_0}) \leq t_{\mathcal{K}'}(\mathbf{A}_0),$$

since every pattern realized in $\langle \mathbf{B}_0, <_{\mathbf{B}_0} \rangle$ with respect to $\mathcal{K}'$ is also a $\mathcal{K}'$ -pattern on $\mathbf{A}_0$. Hence the same $\mathbf{C}_0$ also satisfies

$$\mathbf{C}_0 \rightarrow (\mathbf{B}_0)_{k, t_{\mathcal{K}'}(\mathbf{A}_0)}^{\mathbf{A}_0}.$$

—

The proof yields a slightly stronger conclusion: one may choose a linear ordering on $\mathbf{C}_0$ such that, on a suitable copy of $\mathbf{B}_0$, the color of a copy of $\mathbf{A}_0$ depends only on the pattern induced by that ordering. We will not isolate this as a separate proposition, since the weaker statement above is all that we need.

Proposition 11.2.2. *Assume that $\mathcal{K}$ has both the Ramsey property and the ordering property. Then for every $\mathbf{A}_0 \in \mathcal{K}_0$ there exists $\mathbf{B}_0 \geq \mathbf{A}_0$ in $\mathcal{K}_0$ such that for every $\mathbf{C}_0 \geq \mathbf{B}_0$ in $\mathcal{K}_0$ there is a coloring*

$$c : \binom{\mathbf{C}_0}{\mathbf{A}_0} \rightarrow \{1, \dots, t_{\mathcal{K}}(\mathbf{A}_0)\}$$

with the following property: for every

$$\mathbf{B}'_0 \in \binom{\mathbf{C}_0}{\mathbf{B}_0},$$

the coloring c takes all the values $1, \dots, t_{\mathcal{K}}(\mathbf{A}_0)$ on

$$\binom{\mathbf{B}'_0}{\mathbf{A}_0}.$$

Proof. By the ordering property, there is $\mathbf{B}_0 \geq \mathbf{A}_0$ in $\mathcal{K}_0$ such that whenever

$$\langle \mathbf{A}_0, \prec \rangle, \langle \mathbf{B}_0, \prec' \rangle \in \mathcal{K},$$

we have

$$\langle \mathbf{A}_0, \prec \rangle \leq \langle \mathbf{B}_0, \prec' \rangle.$$

Equivalently, every admissible $\mathbf{A}_0$ -pattern is realized by some copy of $\mathbf{A}_0$ inside every ordered expansion of $\mathbf{B}_0$ belonging to $\mathcal{K}$.

Now let $\mathbf{C}_0 \geq \mathbf{B}_0$ in $\mathcal{K}_0$, and choose a linear ordering $<_{\mathbf{C}_0}$ on $\mathbf{C}_0$ such that

$$\langle \mathbf{C}_0, <_{\mathbf{C}_0} \rangle \in \mathcal{K}.$$

Enumerate the $\mathbf{A}_0$ -patterns as

$$p_1, \dots, p_{t_{\mathcal{K}}(\mathbf{A}_0)}.$$

Define

$$c : \binom{\mathbf{C}_0}{\mathbf{A}_0} \rightarrow \{1, \dots, t_{\mathcal{K}}(\mathbf{A}_0)\}$$

by declaring that, for

$$\mathbf{A}'_0 \in \binom{\mathbf{C}_0}{\mathbf{A}_0},$$

we put

$$c(\mathbf{A}'_0) = i$$

if the ordered structure

$$\langle \mathbf{A}'_0, <_{\mathbf{C}_0} \upharpoonright \mathbf{A}'_0 \rangle$$

has pattern p_i .

Let

$$\mathbf{B}'_0 \in \binom{\mathbf{C}_0}{\mathbf{B}_0}.$$

Then

$$\langle \mathbf{B}'_0, <_{\mathbf{C}_0} \mid \mathbf{B}'_0 \rangle \in \mathcal{K},$$

and by the choice of $\mathbf{B}'_0$, every admissible $\mathbf{A}_0$ -pattern is realized inside this ordered structure. Therefore the coloring c takes every value

$$1, \dots, t_{\mathcal{K}}(\mathbf{A}_0)$$

on

$$\binom{\mathbf{B}'_0}{\mathbf{A}_0}.$$

⊢

Corollary 11.2.3. *Assume that $\mathcal{K}$ has the Ramsey property and the ordering property. Then for every $\mathbf{A}_0 \in \mathcal{K}_0$,*

$$t(\mathbf{A}_0, \mathcal{K}_0) = t_{\mathcal{K}}(\mathbf{A}_0).$$

Equivalently, $t_{\mathcal{K}}(\mathbf{A}_0)$ is the least integer t such that for every $\mathbf{B}_0 \in \mathcal{K}_0$ with $\mathbf{A}_0 \leq \mathbf{B}_0$ and every $k \geq 2$, there exists $\mathbf{C}_0 \geq \mathbf{B}_0$ in $\mathcal{K}_0$ with

$$\mathbf{C}_0 \rightarrow (\mathbf{B}_0)_{k,t}^{\mathbf{A}_0}.$$

Proof. Proposition 11.2.1 shows that $t_{\mathcal{K}}(\mathbf{A}_0)$ has the required partition property, so

$$t(\mathbf{A}_0, \mathcal{K}_0) \leq t_{\mathcal{K}}(\mathbf{A}_0).$$

On the other hand, Proposition 11.2.2 shows that no smaller number can work. Hence

$$t(\mathbf{A}_0, \mathcal{K}_0) = t_{\mathcal{K}}(\mathbf{A}_0).$$

⊢

In particular, whenever $\mathcal{K}$ has the Ramsey and ordering properties, the number of admissible patterns on $\mathbf{A}_0$ depends only on the reduct class $\mathcal{K}_0$, not on the particular expansion $\mathcal{K}$. Thus ordered Ramsey expansions do not merely imply the finiteness of Ramsey degrees; they compute those degrees exactly.

11.3 Converse criteria

We now prove converses to the results of the previous section. The first shows that, under the ordering property, the partition relation with bound $t_{\mathcal{K}}(\mathbf{A}_0)$ already forces the Ramsey property. We then isolate a local form of the ordering property and show that, assuming the Ramsey property, it is equivalent to the equality

$$t(\mathbf{A}_0, \mathcal{K}_0) = t_{\mathcal{K}}(\mathbf{A}_0).$$

Proposition 11.3.1. *Suppose that $\mathcal{K}$ is reasonable and has the ordering property. Assume moreover that for every $\mathbf{A}_0 \leq \mathbf{B}_0$ in $\mathcal{K}_0$ and every $k \geq 2$, there exists $\mathbf{C}_0 \geq \mathbf{B}_0$ in $\mathcal{K}_0$ such that*

$$\mathbf{C}_0 \rightarrow (\mathbf{B}_0)_{k, t_{\mathcal{K}}(\mathbf{A}_0)}^{\mathbf{A}_0}.$$

Then $\mathcal{K}$ has the Ramsey property.

In particular, any reasonable order class $\mathcal{K}$ with the ordering property that is contained in an expansion of $\mathcal{K}_0$ with the Ramsey property also has the Ramsey property.

Proof. Let

$$\langle \mathbf{A}_0, \prec \rangle, \langle \mathbf{B}_0, \prec' \rangle \in \mathcal{K} \quad \text{with} \quad \langle \mathbf{A}_0, \prec \rangle \leq \langle \mathbf{B}_0, \prec' \rangle.$$

We must find

$$\langle \mathbf{C}_0, \prec'' \rangle \in \mathcal{K}$$

such that

$$\langle \mathbf{C}_0, \prec'' \rangle \rightarrow (\langle \mathbf{B}_0, \prec' \rangle)_2^{\langle \mathbf{A}_0, \prec \rangle}.$$

By the ordering property, there exists $\mathbf{B}_1 \geq \mathbf{B}_0$ in $\mathcal{K}_0$ such that whenever

$$\langle \mathbf{B}_0, \triangleleft_0 \rangle, \langle \mathbf{B}_1, \triangleleft_1 \rangle \in \mathcal{K},$$

we have

$$\langle \mathbf{B}_0, \triangleleft_0 \rangle \leq \langle \mathbf{B}_1, \triangleleft_1 \rangle.$$

Since $\mathcal{K}$ is reasonable and

$$\langle \mathbf{A}_0, \prec \rangle \leq \langle \mathbf{B}_0, \prec' \rangle,$$

it follows that for every ordering $\triangleleft_1$ on B_1 with

$$\langle \mathbf{B}_1, \triangleleft_1 \rangle \in \mathcal{K},$$

we also have

$$\langle \mathbf{A}_0, \prec \rangle \leq \langle \mathbf{B}_1, \triangleleft_1 \rangle.$$

Now apply the hypothesis with

$$\mathbf{A}_0, \mathbf{B}_1, \text{ and } k = t_{\mathcal{K}}(\mathbf{A}_0) + 1.$$

Thus there exists $\mathbf{C}_0 \geq \mathbf{B}_1$ in $\mathcal{K}_0$ such that

$$\mathbf{C}_0 \rightarrow (\mathbf{B}_1)_{t_{\mathcal{K}}(\mathbf{A}_0)+1, t_{\mathcal{K}}(\mathbf{A}_0)}^{\mathbf{A}_0}.$$

Choose an ordering $\prec''$ on C_0 such that

$$\langle \mathbf{C}_0, \prec'' \rangle \in \mathcal{K}.$$

We claim that this ordered structure has the required Ramsey property.

Let

$$c : \left(\begin{array}{c} \langle \mathbf{C}_0, \prec'' \rangle \\ \langle \mathbf{A}_0, \prec \rangle \end{array} \right) \rightarrow \{1, 2\}$$

be a coloring of copies of $\langle \mathbf{A}_0, \prec \rangle$ in $\langle \mathbf{C}_0, \prec'' \rangle$. Enumerate the $\mathbf{A}_0$ -patterns as

$$p_1, \dots, p_{t_{\mathcal{K}}(\mathbf{A}_0)}.$$

Using c , define a coloring

$$c' : \binom{\mathbf{C}_0}{\mathbf{A}_0} \rightarrow \{0, 1, \dots, t_{\mathcal{K}}(\mathbf{A}_0)\}$$

as follows. For

$$\mathbf{A}'_0 \in \binom{\mathbf{C}_0}{\mathbf{A}_0},$$

let $\pi(\mathbf{A}'_0)$ be the pattern of the induced ordering

$$\prec'' \upharpoonright \mathbf{A}'_0$$

on $\mathbf{A}'_0$. Set

$$c'(\mathbf{A}'_0) = \begin{cases} 0, & \text{if } \pi(\mathbf{A}'_0) \text{ is the pattern of } \prec \text{ and } c(\langle \mathbf{A}'_0, \prec'' \upharpoonright \mathbf{A}'_0 \rangle) = 2, \\ i, & \text{if } \pi(\mathbf{A}'_0) = p_i \text{ and either } p_i \neq \pi(\prec) \text{ or } c(\langle \mathbf{A}'_0, \prec'' \upharpoonright \mathbf{A}'_0 \rangle) = 1. \end{cases}$$

This uses at most $t_{\mathcal{K}}(\mathbf{A}_0) + 1$ colors.

Hence there exists

$$\mathbf{B}'_1 \in \binom{\mathbf{C}_0}{\mathbf{B}_1}$$

such that c' takes at most $t_{\mathcal{K}}(\mathbf{A}_0)$ values on

$$\binom{\mathbf{B}'_1}{\mathbf{A}_0}.$$

Since

$$\langle \mathbf{B}'_1, \prec'' \upharpoonright \mathbf{B}'_1 \rangle \in \mathcal{K}$$

and $\langle \mathbf{A}_0, \prec \rangle$ embeds into every ordered expansion of $\mathbf{B}_1$ in $\mathcal{K}$, the pattern of $\prec$ is realized by some copy of $\mathbf{A}_0$ inside $\mathbf{B}'_1$. Moreover, every $\mathbf{A}_0$ -pattern realized in $\langle \mathbf{B}'_1, \prec'' \upharpoonright \mathbf{B}'_1 \rangle$ contributes at least one value of c' on

$$\binom{\mathbf{B}'_1}{\mathbf{A}_0}.$$

Since c' takes at most $t_{\mathcal{K}}(\mathbf{A}_0)$ values there, the value 0 must occur and, for the pattern of $\prec$, no additional value can occur. It follows that all copies of

$$\langle \mathbf{A}_0, \prec \rangle$$

inside

$$\langle \mathbf{B}'_1, \prec'' \upharpoonright \mathbf{B}'_1 \rangle$$

have the same c -color, namely 2.

Finally, by the choice of $\mathbf{B}_1$ we have

$$\langle \mathbf{B}_0, \prec' \rangle \leq \langle \mathbf{B}'_1, \prec'' \mid B'_1 \rangle,$$

so

$$\langle \mathbf{C}_0, \prec'' \rangle \rightarrow (\langle \mathbf{B}_0, \prec' \rangle)_2^{\langle \mathbf{A}_0, \prec \rangle}.$$

Thus $\mathcal{K}$ has the Ramsey property.

For the final assertion, if $\mathcal{K}$ is contained in an expansion of $\mathcal{K}_0$ with the Ramsey property, then Proposition 11.2.1 yields exactly the partition relation assumed above. Hence the first part applies. $\dashv$

To formulate the converse to Corollary 11.2.3, it is useful to isolate a local form of the ordering property.

Definition 11.3.2. Let $\mathbf{A}_0 \in \mathcal{K}_0$. We say that $\mathcal{K}$ has the *ordering property at $\mathbf{A}_0$* if there exists $\mathbf{B}_0 \geq \mathbf{A}_0$ in $\mathcal{K}_0$ such that for every pair of orderings $\prec$ on A_0 and $\prec'$ on B_0 with

$$\langle \mathbf{A}_0, \prec \rangle, \langle \mathbf{B}_0, \prec' \rangle \in \mathcal{K},$$

we have

$$\langle \mathbf{A}_0, \prec \rangle \leq \langle \mathbf{B}_0, \prec' \rangle.$$

Thus $\mathcal{K}$ has the ordering property if and only if it has the ordering property at every $\mathbf{A}_0 \in \mathcal{K}_0$.

Proposition 11.3.3. *Assume that $\mathcal{K}$ has the Ramsey property. Then, for each $\mathbf{A}_0 \in \mathcal{K}_0$, the following are equivalent:*

1. $\mathcal{K}$ has the ordering property at $\mathbf{A}_0$;
2. $t(\mathbf{A}_0, \mathcal{K}_0) = t_{\mathcal{K}}(\mathbf{A}_0)$.

Proof. Assume first that $\mathcal{K}$ has the ordering property at $\mathbf{A}_0$. Then the proof of Proposition 11.2.2 shows that there exists $\mathbf{B}_0 \geq \mathbf{A}_0$ such that every admissible $\mathbf{A}_0$ -pattern is realized inside every ordered expansion of $\mathbf{B}_0$ belonging to $\mathcal{K}$. Combining this with Proposition 11.2.1, exactly as in the proof of Corollary 11.2.3, we obtain

$$t(\mathbf{A}_0, \mathcal{K}_0) = t_{\mathcal{K}}(\mathbf{A}_0).$$

Conversely, suppose that

$$t(\mathbf{A}_0, \mathcal{K}_0) = t_{\mathcal{K}}(\mathbf{A}_0)$$

but that $\mathcal{K}$ does not have the ordering property at $\mathbf{A}_0$. Then for every $\mathbf{B}_0 \geq \mathbf{A}_0$ in $\mathcal{K}_0$ there exists an ordering $\prec'$ on B_0 such that

$$\langle \mathbf{B}_0, \prec' \rangle \in \mathcal{K}$$

and some admissible ordering $\prec$ on A_0 with

$$\langle \mathbf{A}_0, \prec \rangle \in \mathcal{K}$$

fails to embed into $\langle \mathbf{B}_0, \prec' \rangle$. Hence the number of $\mathbf{A}_0$ -patterns realized inside $\langle \mathbf{B}_0, \prec' \rangle$ satisfies

$$t_{\mathcal{K}}(\mathbf{A}_0, \mathbf{B}_0, \prec') \leq t_{\mathcal{K}}(\mathbf{A}_0) - 1.$$

Applying Proposition 11.2.1 with this ordered structure, we obtain, for every $k \geq 2$, some $\mathbf{C}_0 \geq \mathbf{B}_0$ such that

$$\mathbf{C}_0 \rightarrow (\mathbf{B}_0)_{k, t_{\mathcal{K}}(\mathbf{A}_0)-1}^{\mathbf{A}_0}.$$

This implies

$$t(\mathbf{A}_0, \mathcal{K}_0) \leq t_{\mathcal{K}}(\mathbf{A}_0) - 1,$$

contrary to the assumption. Therefore $\mathcal{K}$ has the ordering property at $\mathbf{A}_0$. $\dashv$

Corollary 11.3.4. *Assume that $\mathcal{K}$ has the Ramsey property. Then the following are equivalent:*

1. $\mathcal{K}$ has the ordering property;
2. for every $\mathbf{A}_0 \in \mathcal{K}_0$,

$$t(\mathbf{A}_0, \mathcal{K}_0) = t_{\mathcal{K}}(\mathbf{A}_0).$$

Proof. This follows immediately from Proposition 11.3.3, since the ordering property is equivalent to having the ordering property at each $\mathbf{A}_0 \in \mathcal{K}_0$. $\dashv$

11.4 Recovering Ramsey expansions

We now return to the dynamical viewpoint of the previous chapters. The results of this section show that the combinatorial information encoded by Ramsey degrees is strong enough to recover ordered expansions with the Ramsey and ordering properties, and hence to recover the universal minimal flow description from Chapter 9.

Theorem 11.4.1. *Let $\mathcal{K}_0$ be a Fraïssé class in a signature L_0 , and let $\mathcal{K}$ be a reasonable Fraïssé order class in the expanded signature*

$$L = L_0 \cup \{<\}$$

such that $\mathcal{K}|_{L_0} = \mathcal{K}_0$. Assume that $\mathcal{K}$ has the Ramsey property. Then there exists a reasonable Fraïssé order class

$$\mathcal{K}' \subseteq \mathcal{K}$$

such that $\mathcal{K}'|_{L_0} = \mathcal{K}_0$ and $\mathcal{K}'$ has both the Ramsey property and the ordering property.

Proof. Let $\mathbf{F}_0 = \text{Flim}(\mathcal{K}_0)$ and $G_0 = \text{Aut}(\mathbf{F}_0)$. Consider the G_0 -flow $X_{\mathcal{K}}$ of all $\mathcal{K}$ -admissible linear orderings on F_0 . Choose a minimal subflow

$$X' \subseteq X_{\mathcal{K}},$$

and fix some

$$\prec' \in X'.$$

Let

$$\mathcal{K}' = \text{Age}(\langle \mathbf{F}_0, \prec' \rangle).$$

Then $\mathcal{K}' \subseteq \mathcal{K}$, and since every finite L_0 -substructure of $\mathbf{F}_0$ lies in $\mathcal{K}_0$, we have

$$\mathcal{K}'|_{L_0} = \mathcal{K}_0.$$

Moreover, $\mathcal{K}'$ is hereditary and has the joint embedding property, and it is easy to verify that $\mathcal{K}'$ is reasonable.

We claim that

$$X' = X_{\mathcal{K}'}.$$

Indeed, every finite substructure of $\langle \mathbf{F}_0, \prec' \rangle$ belongs to $\mathcal{K}'$, so every ordering in the orbit closure $\overline{G_0 \cdot \prec'} = X'$ is $\mathcal{K}'$ -admissible. Conversely, if an ordering on F_0 is $\mathcal{K}'$ -admissible, then every finite restriction occurs in $\text{Age}(\langle \mathbf{F}_0, \prec' \rangle)$, hence belongs to the orbit closure of $\prec'$. Thus the two spaces coincide.

Since $X' = X_{\mathcal{K}'}$ is minimal, it follows from Theorem ?? that $\mathcal{K}'$ has the ordering property.

It remains to show that $\mathcal{K}'$ has the Ramsey property. By Proposition 11.3.1, it is enough to verify that for every

$$\mathbf{A}_0 \leq \mathbf{B}_0 \quad \text{in } \mathcal{K}_0$$

and every $k \geq 2$, there exists $\mathbf{C}_0 \geq \mathbf{B}_0$ in $\mathcal{K}_0$ such that

$$\mathbf{C}_0 \rightarrow (\mathbf{B}_0)_{k, t_{\mathcal{K}'}(\mathbf{A}_0)}^{\mathbf{A}_0}.$$

But this follows from Proposition 11.2.1, since $\mathcal{K}$ has the Ramsey property and $\mathcal{K}' \subseteq \mathcal{K}$ is a hereditary expansion of $\mathcal{K}_0$.

Finally, because $\mathcal{K}'$ is reasonable and has the Ramsey and ordering properties, Corollary 11.3.4 implies that the numbers $t_{\mathcal{K}'}(\mathbf{A}_0)$ are the Ramsey degrees in $\mathcal{K}_0$. In particular, Proposition 11.3.1 yields the Ramsey property for $\mathcal{K}'$, and hence $\mathcal{K}'$ is a reasonable Fraïssé order class with the required properties. $\dashv$

The preceding theorem shows that, once one has a reasonable Ramsey expansion, one may pass to a smaller expansion in which minimality is built in. The next result is the converse direction of the universal-minimal-flow theorem proved earlier.

Theorem 11.4.2. *Let $L \supseteq \{<\}$ be a signature, let*

$$L_0 = L \setminus \{<\},$$

and let $\mathcal{K}$ be a reasonable Fraïssé order class in L . Put

$$\mathcal{K}_0 = \mathcal{K}|_{L_0}, \quad \mathbf{F} = \text{Flim}(\mathcal{K}), \quad \mathbf{F}_0 = \mathbf{F}|_{L_0} = \text{Flim}(\mathcal{K}_0),$$

and let

$$G_0 = \text{Aut}(\mathbf{F}_0), \quad G = \text{Aut}(\mathbf{F}).$$

Let $X_{\mathcal{K}}$ be the compact space of all $\mathcal{K}$ -admissible linear orderings on the underlying set F_0 of $\mathbf{F}_0$. Then the following are equivalent:

1. $\mathcal{K}$ has the Ramsey property and the ordering property;
2. $X_{\mathcal{K}}$ is the universal minimal flow of G_0 .

Proof. The implication

$$(1) \Rightarrow (2)$$

is Theorem ??.

Now assume that $X_{\mathcal{K}}$ is the universal minimal flow of G_0 . Since it is in particular minimal, Theorem ?? implies that $\mathcal{K}$ has the ordering property. It therefore remains to prove that $\mathcal{K}$ has the Ramsey property.

By Proposition 11.3.1, it is enough to show that for every

$$\mathbf{A}_0 \leq \mathbf{B}_0 \quad \text{in } \mathcal{K}_0,$$

every $k \geq 2$, and every coloring

$$c : \begin{pmatrix} \mathbf{F}_0 \\ \mathbf{A}_0 \end{pmatrix} \rightarrow \{1, \dots, k\},$$

there exists

$$\mathbf{B}'_0 \in \begin{pmatrix} \mathbf{F}_0 \\ \mathbf{B}_0 \end{pmatrix}$$

such that the restriction of c to

$$\begin{pmatrix} \mathbf{B}'_0 \\ \mathbf{A}_0 \end{pmatrix}$$

takes at most $t_{\mathcal{K}}(\mathbf{A}_0)$ values.

Consider the compact G_0 -flow

$$\{1, \dots, k\} \begin{pmatrix} \mathbf{F}_0 \\ \mathbf{A}_0 \end{pmatrix},$$

where G_0 acts by

$$(g \cdot \gamma)(\mathbf{A}'_0) = \gamma(g^{-1}(\mathbf{A}'_0)) \quad \left(g \in G_0, \gamma \in \{1, \dots, k\} \begin{pmatrix} \mathbf{F}_0 \\ \mathbf{A}_0 \end{pmatrix} \right).$$

Let

$$X = \overline{G_0 \cdot c}$$

be the orbit closure of the coloring c . Since $X_{\mathcal{K}}$ is universal minimal, there is a G_0 -map

$$\pi : X_{\mathcal{K}} \rightarrow X.$$

Let

$$\prec_0$$

be the distinguished linear ordering on F_0 coming from the Fraïssé limit $\mathbf{F} = \langle \mathbf{F}_0, \prec_0 \rangle$, and put

$$c_0 = \pi(\prec_0).$$

Because every element of $G = \text{Aut}(\mathbf{F})$ fixes the ordering $\prec_0$, and π is G_0 -equivariant, every element of G also fixes c_0 . Hence if

$$\mathbf{A}'_0, \mathbf{A}''_0 \in \binom{\mathbf{F}_0}{\mathbf{A}_0}$$

have the same $\mathcal{K}$ -pattern with respect to the ordering $\prec_0$, then some element of G sends $\mathbf{A}'_0$ to $\mathbf{A}''_0$, and therefore

$$c_0(\mathbf{A}'_0) = c_0(\mathbf{A}''_0).$$

Thus the value of c_0 on a copy of $\mathbf{A}_0$ depends only on its pattern. It follows that c_0 takes at most

$$t_{\mathcal{K}}(\mathbf{A}_0)$$

values.

Now fix any

$$\mathbf{B}''_0 \in \binom{\mathbf{F}_0}{\mathbf{B}_0}.$$

Since $c_0 \in \overline{G_0 \cdot c}$, there exists $g \in G_0$ such that

$$(g \cdot c) \Big| \binom{\mathbf{B}''_0}{\mathbf{A}_0} = c_0 \Big| \binom{\mathbf{B}''_0}{\mathbf{A}_0}.$$

Let

$$\mathbf{B}'_0 = g^{-1}(\mathbf{B}''_0).$$

Then the restriction of c to

$$\binom{\mathbf{B}'_0}{\mathbf{A}_0}$$

has the same set of values as the restriction of c_0 to

$$\binom{\mathbf{B}''_0}{\mathbf{A}_0},$$

and therefore takes at most $t_{\mathcal{K}}(\mathbf{A}_0)$ values. This is exactly the partition relation required in Proposition 11.3.1.

Hence $\mathcal{K}$ has the Ramsey property. Since we already know that $\mathcal{K}$ has the ordering property, this proves (2) $\Rightarrow$ (1). $\dashv$

Theorems 11.4.1 and 11.4.2 complete the converse direction of the KPT correspondence developed in the previous chapters. Ordered Ramsey expansions not only produce universal minimal flows, but can also be recovered from the dynamical structure of those flows. In this sense, Ramsey degrees measure the residual combinatorial complexity of the reduct class, while the Ramsey and ordering properties together characterize the expansions that realize the universal minimal flow.

Bibliography

- [1] F.G. Abramson and L.A. Harrington, Models without indiscernibles, *J. Symbolic Logic* **43** (1978), 572–600.
- [2] S. Adams and G. Stuck, Splitting of nonnegatively curved leaves in minimal sets of foliations, *Duke Math. J.* (1993), **71** (1), 71–92.
- [3] J. Auslander, *Minimal Flows and Their Extensions*, North Holland, 1988.
- [4] H. Becker and A.S. Kechris, *The Descriptive Set Theory of Polish Group Actions*, London Math. Soc. Lecture Note Series, **232**, Cambridge Univ. Press, 1996.
- [5] Y. Benyamini and J. Lindenstrauss [00], *Geometric Nonlinear Functional Analysis*, Vol. 1, Colloquium Publications **48**, AMS, Providence, RI, xi+488 p., 2000.
- [6] S.K. Berberian, *Lectures in Functional Analysis and Operator Theory*, Springer-Verlag, 1974.
- [7] S.A. Bogatyi [02], Metrically homogeneous spaces, *Russian Math. Surveys* **57**(2) (2002), 221–240.
- [8] P.J. Cameron, *Oligomorphic Permutation Groups*, London Math. Society Lecture Note Series, **152**, 1990.
- [9] G.L. Cherlin [98], *The Classification of Countable Homogeneous Directed Graphs and Countable Homogeneous n -tournaments*, Memoirs of the Amer. Math. Soc., **131**, No. 621, 1998.
- [10] C. Constantinescu, *C^* -algebras, Vol. 2: Banach Algebras and Compact Operators*, North Holland, 2001.
- [11] D. Devlin, *Some partition theorems and ultrafilters on ω* , Ph.D. Thesis, Dartmouth College, 1979.
- [12] J. de Vries, *Elements of Topological Dynamics*, Kluwer, 1993.
- [13] J. Dieudonné, Sur la completion des groupes topologiques, *C. R. Acad. Sci. Paris* **218** (1944), 774–776.
- [14] R. Ellis, Universal minimal sets, *Proc. Amer. Math. Soc.* **11** (1960), 540–543.

- [15] R. Ellis, *Lectures on Topological Dynamics*, W.A. Benjamin, 1969.
- [16] R. Engelking, *General Topology*, Math. Monographs, **60**, PWN - Polish Scient. Publishers, Warsaw, 1977.
- [17] P. Erdős and A. Hajnal, Unsolved and solved problems in set theory, *Proc. Symp. Pure Math.* **25** (1974), 269–287.
- [18] P. Erdős, A. Hajnal and R. Rado, Partition relations for cardinal numbers, *Acta Math. Acad. Sci. Hung.*, **16**, (1965), 93–196.
- [19] P. Erdős and R. Rado, A partition calculus in set theory, *Bull. Amer. Math. Soc.*, **62** (1956), 427–489.
- [20] W.L. Fouché, Symmetry and the Ramsey degree of posets, *Discrete Math.* **167/168** (1997), 309–315.
- [21] W.L. Fouché, Symmetries in Ramsey theory, *East-West J. Math.*, **1** (1998), 43–60.
- [22] W.L. Fouché, Symmetry and the Ramsey degrees of finite relational structures, *J. Comb. Theory Ser. A* **85(2)** (1999), 135–147.
- [23] W.L. Fouché, Symmetries and Ramsey properties of trees, *Discrete Math.* **197/198** (1999), 325–330.
- [24] R. Fraïssé, Sur l’extension aux relations de quelques propriétés des ordres, *Ann. Sci. École Norm. Sup.* **71** (1954), 363–388.
- [25] F. Galvin and K. Prikry, Borel sets and Ramsey’s theorem, *J. Symb. Logic* **38** (1973), 193–198.
- [26] T. Giordano and V. Pestov, Some extremely amenable groups, *C.R. Math. Acad. Sci. Paris* **334 (4)** (2002), 273–278.
- [27] T. Giordano and V. Pestov, Some extremely amenable groups related to operator algebras and ergodic theory, ArXiv e-print math.OA/0405288, May 2004, 37 pp.
- [28] E. Glasner, On minimal actions of Polish groups, *Top. Appl.* **85** (1998), 119–125.
- [29] E. Glasner, Structure theory as a tool in topological dynamics, *Descriptive Set Theory and Dynamical Systems*, M. Foreman et al. (Eds.), London Math. Society Lecture Note Series, **277**, 173–209, Cambridge Univ. Press, 2000.
- [30] E. Glasner and B. Weiss, Minimal actions of the group $S(\mathbb{Z})$ of permutations of the integers, *Geom. and Funct. Anal.* **12** (2002), 964–988.
- [31] E. Glasner and B. Weiss, The universal minimal system for the group of homeomorphisms of the Cantor set, *Fundam. Math.* **176** (2003), 277–289.
- [32] R.L. Graham, K. Leeb, and B.L. Rothschild, Ramsey’s theorem for a class of categories, *Adv. in Math* **8** (1972), 417–433.

- [33] R.L. Graham and B.L. Rothschild, Ramsey's theorem for n -parameter sets, *Trans. Amer. Math. Soc.* **159** (1971), 257–292.
- [34] R.L. Graham, B.L. Rothschild, and J.H. Spencer, *Ramsey Theory*, 2nd Edition, Wiley, 1990.
- [35] E. Granirer, Extremely amenable semigroups I,II, *Math. Scand.* **17** (1965), 177–179; **20** (1967), 93–113.
- [36] E. Granirer and A.T. Lau, Invariant means on locally compact groups, *Ill. J. Math.* **15** (1971), 249–257.
- [37] M. Gromov, Filling Riemannian manifolds, *J. Diff. Geometry* **18** (1983), 1–147.
- [38] M. Gromov and V.D. Milman, A topological application of the isoperimetric inequality, *Amer. J. Math.* **105** (1983), 843–854.
- [39] W. Herer and J.P.R. Christensen, On the existence of pathological submeasures and the construction of exotic topological groups, *Math. Ann.* **213** (1975), 203–210.
- [40] W. Hodges, *Model Theory*, Cambridge Univ. Press, 1993.
- [41] D.E. Knuth and T.J. Buckholtz, Computation of Tangent, Euler, and Bernoulli numbers, *Math. of Computation* **21** (**100**) (1967), 663–688.
- [42] A.H. Lachlan and R. Woodrow, Countable ultrahomogeneous undirected graphs, *Trans. Amer. Math. Soc.* **262** (1980), 51–94.
- [43] A.T.-M. Lau, P. Milnes, and J. Pym, On the structure of minimal left ideals in the largest compactification of a locally compact group, *J. London Math. Soc* **59** (**2**) (1999), 133–152.
- [44] M. Ledoux, *The concentration of measure phenomenon*. Math. Surveys and Monographs, **89**, Amer. Math. Soc., 2001.
- [45] K. Milliken, A Ramsey theorem for trees, *J. Comb. Theory* **26** (1979), 215–237.
- [46] V.D. Milman, Infinite-dimensional geometry of the unit sphere in Banach space, *Sov. Math., Dokl.* **8** (1967), 1440–1444.
- [47] V.D. Milman, The heritage of P. Lévy in geometrical functional analysis, *Astérisque* **157–158** (1988), 273–301.
- [48] V.D. Milman and G. Schechtman, *Asymptotic theory of finite-dimensional normed spaces (with an Appendix by M. Gromov)*, Lecture Notes in Math., **1200**, Springer, 1986.
- [49] T. Mitchell, Fixed points and multiplicative left invariant means, *Trans. Am. Math. Soc.* **122** (1966), 195–202.

- [50] J. Nešetřil, For graphs there are only four types of hereditary Ramsey classes, *J. Comb. Theory, Series B* **46** (2) (1989), 127–132.
- [51] J. Nešetřil, Ramsey Theory, *Handbook of Combinatorics*, R. Graham et al. (Eds.), 1331–1403, Elsevier, 1995.
- [52] J. Nešetřil, Private communication, 2003.
- [53] J. Nešetřil, Ramsey classes and homogeneous structures, Preprint, 2003.
- [54] J. Nešetřil, Metric spaces are Ramsey, Preprint, 2004.
- [55] J. Nešetřil, H.J. Prömel, V. Rödl and B. Voigt, Canonical ordering theorems, a first attempt, Proc. of the 10th Winter School on Abstract Analysis (Srní, 1982), *Rend. Circ. Math. Palermo* (2) (1982), 193–197.
- [56] J. Nešetřil and V. Rödl, Partitions of finite relational and set systems, *J. Comb. Theory* **22** (3) (1977), 289–312.
- [57] J. Nešetřil and V. Rödl, On a probabilistic graph-theoretical method, *Proc. Amer. Math. Soc.* **72** (2) (1978), 417–421.
- [58] J. Nešetřil and V. Rödl, Ramsey classes of set systems, *J. Comb. Theory, Series A* **34** (2) (1983), 183–201.
- [59] J. Nešetřil and V. Rödl, Combinatorial partitions of finite posets and lattices – Ramsey lattices, *Alg. Univers.* **19** (1984), 106–119.
- [60] J. Nešetřil and V. Rödl, *Mathematics of Ramsey Theory*, Springer, 1990.
- [61] L. Nguyen Van The, Ramsey degrees of finite ultrametric spaces, preprint, 2004
- [62] E. Odell and T. Schlumprecht, The distortion problem, *Acta Mathematica* **173** (1994), 259–281.
- [63] V. Pestov, Some universal constructions in abstract topological dynamics, *Contemporary Math.* **215** (1998), 83–99.
- [64] V. Pestov, On free actions, minimal flows and a problem by Ellis, *Trans. Amer. Math. Soc.* **350** (10) (1998), 4149–4165.
- [65] V. Pestov, Topological groups: where to from here?, *Topology Proceed.* **24** (1999), 421–502.
- [66] V. Pestov, Ramsey-Milman phenomenon, Urysohn metric spaces, and extremely amenable groups, *Israel J. Math.* **127** (2002), 317–357.
- [67] V. Pestov, Remarks on actions on compacta by some infinite-dimensional groups, *Infinite dimensional Lie groups in geometry and representation theory*, (Washington, DC 2000), World Sci., 145–163, 2002.

- [68] V. Pestov, MM -spaces and group actions, *L'Enseign. Math.* **48** (2002), 209–236.
- [69] M. Pouzet and N. Sauer, Edge partitions of the Rado graph, *Combinatorica* **16**(4) (1996), 505–520.
- [70] H.J. Prömel, Some remarks on natural orders for combinatorial cubes, *Discrete Math.* **73** (1989), 189–198.
- [71] J. Pym, A note on $G^{\mathcal{LUC}}$ and Veech's Theorem, *Semigroup Forum* **59** (1999), 171–174.
- [72] R. Rado, Direct decomposition of partitions, *J. London Math. Soc.* **29** (1954), 71–83.
- [73] W. Roelcke and S. Dierolf, *Uniform Structures on Topological Groups and Their Quotients*, McGraw-Hill, 1981.
- [74] W. Rudin, *Functional Analysis*, McGraw-Hill, 1973.
- [75] N. Sauer, Coloring finite substructures of countable structures, Paul Erdős and his Mathematics, II, *Bolyai Math. Studies*, **11** (2002), 525–553.
- [76] S. Thomas, Groups acting on infinite dimensional projective spaces, *J. London Math. Soc.* **34** (2) (1986), 265–273.
- [77] S. Todorcevic, *Ramsey spaces*, to appear, 2004.
- [78] S. Turek, Universal minimal dynamical system for reals, *Comment. Math. Univ. Carolin.* **36** (2) (1995), 371–375.
- [79] P. Urysohn, Sur un espace métrique universel, *Bull. Sci. Math.* **51** (1927), 43–64, 74–90.
- [80] V. Uspenskij, On the group of isometries of the Urysohn universal metric space, *Comment. Univ. Carolinae* **31**(1) (1990), 181–182.
- [81] V. Uspenskij, On universal minimal compact G -spaces, *Topology Proc.* **25** (2000), 301–308.
- [82] V. Uspenskij, Compactification of topological groups, *Proc. of the 9th Prague Topol. Symp. (Prague 2001)*, 331–346 (electronic), Topol. Atlas, Toronto, 2002.
- [83] W. Veech, Topological dynamics, *Bull. Amer. Math. Soc.* **83** (5) (1977), 775–830.